



Integrated Sources on Thin-Film Lithium Niobate

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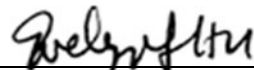
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April 17, 2024

Integrated Sources on Thin-Film Lithium Niobate

A DISSERTATION PRESENTED
BY
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TO
THE JOHN A. PAULSON SCHOOL OF ENGINEERING AND APPLIED SCIENCES
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FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY
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Integrated Sources on Thin-Film Lithium Niobate

ABSTRACT

Photonic integrated circuits have played a crucial role in the miniaturization and efficiency improvement of many optical components. While many material platforms have played host to excellent photonic devices, in recent years, thin-film lithium niobate (TFLN) on insulator has emerged as a promising photonic platform, owing to its large electro-optic effect, low-loss waveguides, and wafer scalability. Devices on TFLN have witnessed remarkable developments over the past decade, most notably in the realization of high-bandwidth, high-efficiency, and low-loss electro-optic modulators. However, the excellent material properties of the TFLN platform extend beyond EO modulation, making it a candidate platform for many high performance chip-scale optical systems.

This dissertation presents work in bringing various functionalities of sources typically realized in bulk systems onto the TFLN platform. Among those included are on-chip pulse and frequency comb generation in lieu of bulk mode-locked lasers, high performance optical isolators using electro-optic modulation in lieu of bulk magneto-optic isolators, and an on-chip, narrow-linewidth extended cavity laser. Included is work involving an emerging technology known as photonic wire bonding, which shows promise as a future integration method to bridge platforms and functionalities.

The first half of the thesis concerns the manipulation of continuous-wave light for frequency comb sources. I start by presenting ultrashort pulse generation using cascaded electro-optic amplitude and phase modulators combined with off-chip dispersion, realized using two designs, the second of which is packaged using the novel photonic wire bonding scheme. I then show how, by interfacing the pulse generation chip with a high- Q resonator with matched free spectral range, broadband spectrum generation can occur with relatively low average on-chip power.

An outstanding challenge for future integrated photonic systems is in the implementation of on-chip gain, which usually involves integration of III-V semiconductor materials to realize integrated lasers and detectors. The second half of the thesis concerns the integration of high-performance lasers onto the TFLN platform. I show how an electro-optic TFLN phase modulator can be used to realize a high-performance, broadband, and low-loss optical isolator, which is a necessary component for any future work with on-chip lasers. Finally, I show the demonstration of a narrow-linewidth, high-power, tunable extended cavity diode laser using InP and TFLN and integrated using photonic wire bonding.

Contents

TITLE PAGE	i
COPYRIGHT	ii
ABSTRACT	iii
TABLE OF CONTENTS	iv
FRONT MATTER	vi
Author List	vi
List of Figures	vii
List of Tables	xix
Dedication	xx
Acknowledgements	xxi
1 THIN-FILM LITHIUM NIOBATE ON INSULATOR	1
1.1 Introduction	2
1.2 Outline of thesis	4
2 ELECTRO-OPTIC PULSE GENERATION ON THIN-FILM LITHIUM NIOBATE	8
2.1 Introduction	9
2.2 Dual drive design and results	12
2.3 Single drive design and results	18
2.4 Conclusion	26
3 SYNCHRONOUS-PUMPED LITHIUM NIOBATE MICRORESONATORS	29
3.1 Introduction	30
3.2 Results	33
3.3 Conclusion	41

4	HIGH PERFORMANCE ELECTRO-OPTIC-BASED OPTICAL ISOLATORS	46
4.1	Introduction	47
4.2	Results	50
4.3	Conclusion	57
5	EXTENDED CAVITY DIODE LASERS ON THIN-FILM LITHIUM NIOBATE	61
5.1	Introduction	62
5.2	Results	67
5.3	Conclusion	71
6	CONCLUSIONS AND OUTLOOK	73
6.1	Summary and conclusion	74
6.2	Outlook	75
	APPENDIX A ELECTRO-OPTIC MODULATION	81
A.1	Basic formalism of the electro-optic effect	81
A.2	Principle of electro-optic phase modulation	83
A.3	V_π measurement	84
	APPENDIX B FABRICATION METHODS	86
B.1	Device fabrication for Chapter 2	86
B.2	Device fabrication for Chapter 3	87
B.3	Device fabrication for Chapter 4	88
B.4	Device fabrication for Chapter 5	88
	REFERENCES	90

Author list

The contents of Chapter 2 are adapted from the contents of the following two publications

- M Yu, DR Barton, **R Cheng**, C Reimer, P Kharel, L He, L Shao, D Zhu, Y Hu, HR Grant, L Johansson, Y Okawachi, A Gaeta, M Zhang, and M Lončar. Integrated femtosecond pulse generator on thin-film lithium niobate. *Nature* **612**, 252–258 (2022).
- **R Cheng**, X Ren, C Reimer, M Yeh, V Rosborough, J Musolf, L Johansson, M Zhang, M Yu, and M Lončar. Single-drive electro-optic frequency comb source on photonic-wire-bonded thin film lithium niobate platform. Forthcoming (2024).

The content in Chapter 3 is included in the publication

- **R Cheng**, M Yu, A Shams-Ansari, Y Hu, C Reimer, M Zhang, and M Lončar. Frequency comb generation via synchronous pumped $\chi^{(3)}$ resonator on thin-film lithium niobate. Forthcoming in *Nature Communications* (2024).

The content in Chapter 4 is included in the publication

- M Yu*, **R Cheng***, C Reimer, L He, K Luke, E Puma, L Shao, A Shams-Ansari, X Ren, HR Grant, L Johansson, M Zhang, and M Lončar. Integrated electro-optic isolator on thin film lithium niobate. *Nature Photonics* **17**, 666–671 (2023).

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- CAA Franken*, **R Cheng***, K Powell, G Kyriazidis, V Rosborough, J Musolf, M Shah, DR Barton, G Hills, L Johansson, KJ Boller, and M Lončar. Photonic wire bonding for narrow-linewidth and high-power lasers on thin-film lithium niobate. Forthcoming (2024).

* equal contribution

List of Figures

2.1	Concept of electrical synthesis of optical pulses via a time lens system.	
	a, Spatial focusing via a spatial lens. A collimated beam passes a spatial lens through its aperture, undergoes diffraction in the free space, and is focused in the focal plane. A perfect lens impacts a quadratic phase profile ($\psi(x) \propto x^2/f$) in the transverse dimension, where f is the focal length of the lens.	
	b, Temporal focusing via an electro-optic time lens (TL). Similar to the spatial focusing in a, a continuous-wave (CW) light passes through a time lens system and is "focused" into a Fourier-transform-limited pulse train. The time lens system is implemented via electro-optic (EO) modulation, consisting of an amplitude modulator (AM), a phase modulator (PM) and a dispersive medium. The AM carves the CW light into a flat-top pulse train, the temporal aperture of the lens. The PM constructs an EO time lens to chirp the initial pulse, where an approximate quadratic phase term ($\psi(t) \sim t^2/f'$) is established at the local minima (or maxima) of a sinusoidal microwave drive, where f' is the focal "length". The dispersive medium induces group delay dispersion (GDD), where different spectral components experience a time delay proportional to the derivative of the spectral phase profile $\phi(\omega) \propto \omega^2 \times \text{GDD}$ with respect to frequency ω . Thus, it retimes all the spectral components to form an ultrashort pulse train. This electrical synthesis of optical pulses generates a flat-top EO comb spectrum with a line spacing equal to the MW drive frequency, with its spectral bandwidth inversely proportional to the pulse duration.	10

2.2	Dual drive design and fabricated chip. a , Schematic of the electro-optic modulation system on TFLN. The recycling phase modulator system is enabled by a low-loss waveguide crossing (grey box). b , SEM and geometry of the waveguide crossing in the phase modulator on a partially etched waveguide with a 300 nm etch depth. The insertion loss of this crossing is extracted to be 0.26 dB per crossing. c , Optical micrograph of the integrated lithium niobate (LN) photonic chip for the time-lens implementation. Zoom-in picture (right) shows the full integrated circuit with a footprint of 25×7 mm. The photonic circuit integrates the following components on a single X-cut 600-nm LN chip: a spot-size converter, a Mach-Zehnder interferometer (MZI) -based AM, a “recycling” PM, and a highly dispersive waveguide.	13
2.3	Frequency-dependent V_π for the dual drive device. a , Measurement scheme for determining the frequency-dependent half-wave voltage of the recycling phase modulator. The amplitude modulator is kept passive while the phase modulation output is monitored at different modulation frequencies using an optical specturm analyzer (OSA). The pump depletion measured on the OSA is fitted to a Bessel function for a given microwave strength. b , Fitted V_π for the recycled phase modulator system measured on the device. The periodicity corresponds to 2.85 GHz spacing of the resonant condition.	14

- 2.4 **Femtosecond pulse generator via an integrated LN-chip-based electro-optic time lens.** **a**, Schematic of the LN chip. In this experiment, the input laser is either a tunable CW laser or a semiconductor-chip-based distributed feedback (DFB) laser. **b**, Characterization setup. The chip output is sent to an optical spectral analyzer for frequency-domain measurement or intensity autocorrelator for time-domain measurement. **c**, The optical microscope picture of the LN photonic chip with edge-coupled DFB laser. The coupling loss is 2 dB with an optimized inverse taper for edge coupling to the DFB laser output (zoom-in). **d**, The output optical spectra and temporal traces at microwave driving frequencies of 10.075 and 30.135 GHz using a tunable CW laser. Two separate EO combs are shown: At 10.075 GHz, a total of 91 comb lines are generated with a 10-dB optical bandwidth of 6.0 nm, and at 30.135 GHz, a total of 67 comb lines are generated with a 10-dB optical bandwidth of 12.6 nm. At 10.075 GHz, the TL focal “length” (referred as focal GDD) is 7.1 ps², corresponding to a PM modulation depth of 11.2 π radians. Aided with the dispersion of single-mode fiber (SMF-28) for full temporal compression, we measured the final pulse duration (full-width-half-maximum (FWHM)) to be 1.140 ps. At 30.135 GHz, the TL focal GDD is 1.1 ps², corresponding to a modulation depth of 7.8 π radians. The final pulses have a measured FWHM of 532 fs. The autocorrelator traces are in good agreement with the simulation (dotted curves). **e**, The optical spectrum with a 30-GHz line spacing and autocorrelator time trace obtained using a DFB laser pump at 1544 nm. The laser wavelength is tuned via changing the injected current, and a 522-fs pulse is measured. OSA: optical spectral analyzer, SMF: single-mode fiber, EDFA: erbium-doped fiber amplifier. 16
- 2.5 **Compact electro-optic comb and pulse generation using a single electrode drive.** **a**, Overview of the electro-optic time lens system for pulse generation. Continuous wave light is converted to a flat-top pulse train via amplitude modulation followed by a cascaded phase modulation to impart a large semi-quadratic phase. The pulse is compressed to its Fourier transform limit by passing through dispersive medium. **b**, Schematic of the thin film lithium niobate photonic device that consists of amplitude modulator followed by a recycling phase modulator. The modulators have common electrode and are driven by the same microwave source. 3-cm long recycling phase modulator is equivalent two two 3-cm long phase modulators. The amplitude modulator segment length L is significantly shorter than the phase modulator length. **c**, Photograph of one of the LN chips next to a quarter for scale. 18

2.6	Frequency-dependent V_π for the single drive device. a , Measurement scheme for determining the frequency-dependent half-wave voltage of the recycling phase modulator. The pump depletion measured on the OSA is fitted to a Bessel function for a given microwave strength. b , Fitted V_π for the recycled phase modulator system measured on the photonic wire bonded device. The periodicity corresponds to 2.1 GHz spacing of the resonant condition.	20
2.7	Optical spectrum characterization of the electro-optic comb generator chips. a , Chips with variable amplitude modulator length L (600, 700, and 800 μm) are tested. A tunable continuous wave laser is coupled into pulse generator chips with different MZI lengths. The generated spectrum is measured with an optical spectrum analyzer (OSA). b , 3-dB optical bandwidth plotted against amplitude modulator length, with the corresponding $1/L$ fit. c , Optical spectrum for the devices. For the 600 μm MZI length, spectral flatness of 3 dB is achieved over 31 comb lines, spanning 7.2 nm bandwidth. For the device with 700 μm length, spectral flatness of 2.2 dB is achieved over 27 comb lines, spanning 6.2 nm bandwidth. For the device with 800 μm MZI length, spectral flatness of 1.7 dB is achieved over 25 comb lines, spanning 5.7 nm bandwidth.	21
2.8	Characterization of optical losses in PWB process. Loss plotted against the wavelength during steps of the photonic wire bonding process, from before wire bonds are cured to after they are cladded. In the measurement, the loss can be improved further by adjusting the input polarization.	23
2.9	Wavelength operation consideration for single-drive device. a , Illustration showing the syncing of the pulse generated by the amplitude modulator with the modulation signal for quadrature points of the amplitude modulator. Because of the single drive operation, the phase modulation signal is automatically in phase with the generated pulse. We note that in the first case, the phase modulator will apply a negative quasi-quadratic phase to the pulse, whereas in the second, the phase modulator applies a positive quasi-quadratic phase. After sending through anomalous dispersion, the first case will translate to a broadening of the pulse (analogous to a concave lens) whereas the second will translate to pulse compression (analogous to a convex lens). b , Passive transmission of the photonic wire bonded device, showing the quadrature points of the unbalanced amplitude modulator. The points of operation for ultrashort pulse generation, where the phase modulation signal applies a positive quadratic phase on the generated pulse, are denoted by the stars.	24

2.10	Photonic wire bonded pulse generation assembly. a , Image of photonic wire bond connecting the output of the fiber array to the input of the lithium niobate waveguide. b , Measurement image. The photonic wire bonded assembly sits on a submount, and GSG probes are touched down to the electrodes to drive the LN chip. c , Setup for the pulse width measurement and corresponding autocorrelator trace for the photonic wire bonded device. The pulse duration using an intensity autocorrelator after compression in 95 m of single-mode fiber. The final pulse duration is 840 fs. d , Stability measurement results conducted over three hours, comparing a photonic wire bonded device with an unpackaged device.	26
3.1	Visualization and concept of pulse and frequency comb generation. a , CW scheme in time and frequency domain; CW light is tuned to the resonant frequency of a microresonator and undergoes four-wave mixing process. The generated frequency comb pulse train has a repetition rate equal to the free spectral range of the microresonator. b , Pulse generation and synchronous pumping scheme in time and frequency domain; a pulse train is formed through amplitude and phase modulation of CW light and compressed in a dispersive medium before resonantly driving the microresonator.	32
3.2	Device images and parameters. a , Optical microscope image of racetrack resonator showing 2 mm footprint size. b , Scanning electron microscope (SEM) image of waveguide sidewalls in the bending region of the racetrack resonator. c , Wide resonance scan (blue) showing FSR of 30.14 GHz for the fundamental TE mode (indicated in red) around 1557 nm. d , Resonance linewidth fitting of fundamental TE mode, showing loaded optical Q -factor of 3.2 million and linewidth of ~60 MHz. e , Dispersion profiles of fabricated geometry with pumping at the resonance frequency.	34
3.3	Effect of annealing on dispersion. Group velocity dispersion (left) and dispersion operator (right) curves of resonator waveguide geometry simulated using thin-film index data collected before and after high temperature furnace annealing. The annealing process pulls the dispersion from the anomalous into normal regime. Index measurements performed by J.A. Woollam.	35
3.4	Full experimental setup. a , Setup schematic for pulse generation and driving across two LN photonic chips. b , Setup for testing spectrum under CW pumping. A circulator is used to monitor the spectrum in the backward direction. EDFA: erbium doped fiber amplifier; OSA: optical spectrum analyzer.	36

3.5	Frequency comb generation in pulse-driven vs. CW-driven case. a, Measurement scheme for on-chip pulse-pumped frequency comb. CW light is coupled onto the first LN chip for pulse generation. The outgoing pulse train is amplified using an EDFA and compressed in single mode fiber. The pulsed light is then coupled to the second LN chip for spectral broadening. b, Pulse-pumped spectrum, generated with 12.5 mW average on-chip power. The spectrum spans 2/5 octave. Red trace: electro-optic frequency comb produced by the pulse generator chip, measured before coupling into the resonator chip. Blue trace: frequency comb spectrum produced via pulse-pumping, measured at the resonator chip output. The power level of the red trace is adjusted to account for chip-chip coupling loss, so that the relative power of the pump comb and broadened comb match appropriately. Inset: visualization of light circulation in microresonator for synchronous driving. SRS is avoided and FWM co-propagates with the pump. c, CW-pumped spectrum, generated with 200 mW on-chip power, monitored in the backward direction. A strong Raman peak is observed at 1720 nm. Inset: visualization of light circulation. SRS threshold is lower than that for FWM, and so both SRS and subsequent comb lines generated through FWM counter-propagate with the pump.	37
3.6	Optical spectrum as a function of repetition rate detuning. Broadened comb spectra collected at 0-3 MHz RF detuning of the pulse source repetition rate from the Kerr resonator's free spectral range with constant average pump power. The spectrum is noticeably affected for >2 MHz detuning.	39
3.7	Pulse-pumped spectrum for different optical pump powers. Generated optical spectra for average on-chip powers ranging from 1.0 to 5.0 mW. The power is adjusted by tuning the current on the EDFA. As expected, we see broader comb spectrum for increasing optical power. .	40

- 3.8 **Outlook of synchronously driven resonators on thin-film LN.** **a,** Schematic for integrated on-chip pulse-driven comb generation, which includes integrated on-chip DFB laser source, amplitude and phase modulators for pulse generation, on-chip dispersion management, and dispersion engineered $\chi^{(2)}$ or $\chi^{(3)}$ resonators. For future integrated schemes, fabrication developments can be made on modulator side (Z-cut modulators) or on the resonator side (dispersion engineering with respect to material birefringence) for compatibility of integration. **b,** Pulse generation at different center wavelengths. The pulse source has flexible wavelength tunability and can be precisely matched to resonator frequency without large need for thermal tuning. **c,** V_π of the recycled phase modulator (layout shown in inset) utilized in the pulse generator design. The waveguide wraps around before passing through modulation again, allowing the phase modulator to be driven with a single GSG electrode and microwave source but imparting a periodic resonance condition on the V_π . The V_π is measured around the resonance condition 30.135 GHz (2.5 V). Within the range that we can fabricate reproducibly, indicated by black arrow, there is < 0.5 dB change in the V_π and minor effect on the pulse generation efficiency. 42
- 4.1 **Concept of an integrated electro-optic isolator.** **a,** The schematic of an integrated laser system consisting of a III-V-based distributed feedback laser (DFB) followed by an electro-optic isolator on thin film lithium niobate (LN). The microwave signal is counter-propagating with respect to the DFB laser output. Under single-tone travelling-wave-based microwave drive at f_{RF}, the accumulated optical phase of the laser output after passing a phase modulator of length L is equal to zero when $f_{RF} = mc/(2Ln_{opt})$ where m is an integer number and c and n_{opt} are light velocity and optical group index, respectively. Any reflected laser light which propagates in the backward direction will undergo efficient phase modulation, which results in $V_{\pi,FW}/V_{\pi,BW} \gg 1$. FW: forward; BW: backward. **b,** The illustration of the travelling-wave EO isolator in frequency domain. The laser output at frequency f_0 is fully transmitted since there is no added phase modulation. All the energy of the reflection signal at frequency f_0 is distributed to the optical sidebands at frequency equal to $f_0 + nf_{RF}$, where n is an integer number. The energy at frequency f_0 can be fully depleted when the modulation depth $\beta \sim 0.765\pi$ satisfying $J_0(\beta) = 0$. **c,** The optical micrograph of the EO isolator chip on thin film LN, which consists of four individual devices with varying modulation length L of 1.25, 1.5, 1.75 and 2 cm. 48

4.2	Coupling between DFB laser and LN chip. a , Quantum well waveguide structure and (inset) simulated output mode of the III-V DFB laser: $W = 5 \mu\text{m}$, $H_p = 2.5 \mu\text{m}$, $h_p = 500 \text{ nm}$, $H_{QW} = 450 \text{ nm}$, $H_N = 4 \mu\text{m}$. b , Structure of the LN mode converter at the coupling facet: $b = 300 \text{ nm}$, $w = 250 \text{ nm}$. c , Simulation of the coupling efficiency between the DFB laser mode and LN facet. An anti-reflective coating is added to the laser structure, as in experiment, to minimize reflection between the laser and the air gap. The minimum coupling gap achievable in our experiment is $\sim 2 \mu\text{m}$, corresponding to about 2-dB coupling loss (indicated by the star). This loss is further verified experimentally on other LN chips.	50
4.3	Full measurement setup for the LN electro-optic isolator. Forward and backward spectra of the isolator are monitored using a tunable laser and 2x2 switch (forward: $1 \rightarrow 3$, backward: $1 \rightarrow 4$) to couple light onto the chip. Polarization controllers are used in both directions to couple light into the transverse-electric (TE) waveguide mode. Spectra are measured on an OSA (optical spectrum analyzer). For III-V laser operation, the DFB laser is butt-coupled directly to the isolator chip, foregoing the switch. The output is then split between a retroreflector and linewidth measurement to monitor the behavior of the DFB laser with and without isolation. The DUT (device under test) includes both devices with and without the add-drop filter.	51
4.4	Characterization of isolation and optical bandwidth of the electro-optic isolator. a , The optical spectrum of the forward and backward optical signal when the microwave signal is propagating in the backward direction at a frequency of 26.5 GHz and a power of 21 dBm (125 mW). b , The optical spectra (power in dB scale) at two different microwave driving frequencies of 26.5 GHz (left) and 10.4 GHz (right), respectively. The achieved isolation at the input laser frequency is 48 dB at 26.5 GHz driving, and $> 44 \text{ dB}$ at 10.4 GHz driving (instrumental resolution limited), respectively. We have observed sidebands with $> 37 \text{ dB}$ lower power than the pump line in the forward signal in both cases. On-chip insertion loss is 0.5 dB. The yellow curve (laser output) is sitting on top of the red curve (forward signal). c , The isolation ratio as a function of optical wavelength. The isolation ratio remains larger than 37 dB across a tuning range of 1510 – 1630 nm.	52

4.5	Stability measurement of the isolation ratio. a, The measured isolation ratio as a function of microwave power. The simulation is performed using the fundamental Bessel function $J_0(\beta)$ (see Appendix A), where the microwave power is proportional to β^2 . The highest simulated isolation ratio is limited by the finite sampling of simulation points. The experimental data shows excellent agreement with the simulation when the microwave power is detuned more than 0.05 dB ($\sim 1.1\%$) from the optimal power. The maximum isolation ratio is likely limited by polarization purity of the input light. b, The measured isolation ratio over 14 hours at a time interval of 30 seconds. At the beginning of the measurement, the isolation of 46 dB is achieved at an optical wavelength of 1555 nm and similar microwave setting to that in Fig. 4.4a,b (inset: the optical spectrum). To assess the effects of environmental conditions, such as temperature, air fluctuation, and acoustic noise, the isolation ratio is monitored as a function of time while keeping all the operation conditions the same. The isolation gradually decreases to 41 dB after 2 hours and stabilizes to a mean value of 41.3 dB with a standard deviation of 0.56 dB over a 12-hour period. The main reason for drop in isolation is the polarization variation over time, which is confirmed by the fact that the isolation ratio can be adjusted back to 46 dB after manually adjusting the polarization controller in the setup.	53
4.6	Integration of a tunable add-drop ring filter on one thin film lithium niobate chip. a, The optical spectra of the forward and backward signals. Laser output spectrum is taken when the microwave drive is turned off. The inset shows an add-drop-ring filter with an integrated heater, followed by a travelling-wave EO isolator. In the forward direction, the laser inputs at Port 1, travels to Port 2 and exits from the phase modulator. In the backward direction, the laser inputs from the modulator, travels to Port 2 and exits from Port 1. With the ring filter, the sideband power is 34 dB lower (0.04%) than the pump power while the isolation ratio remains 48 dB. b, The optical micrograph of the ring filter with an integrated NiCr heater. c, The insertion loss of the micro-ring filter as a function of sideband suppression at an intrinsic quality factor (Q_{in}) of 1×10^6 . The sideband suppression is defined as the power ratio between the first sideband and the pump. We measured an insertion loss of 0.9 dB along with the 34-dB sideband suppression at 26.5 GHz. d, Characterization of the integrated heater. The resonance wavelength is recorded as a function of heater power. We can tune the cavity resonance over one free spectral range of 1.0 nm with 125 mW heater power, based on a fitted tuning efficiency of 8 nm/W.	55

4.7	Characterization of the DFB laser and isolator module. a , Photograph of measurement system. An InP DFB laser is butt-coupled to the LN isolator chip. The laser system output is collected with a lensed fiber. b , Measurement setup. The output of the laser system is split between a reflection port and a self-heterodyne linewidth measurement system to monitor the DFB linewidth. AOM: acousto-optic modulator, RSA: RF spectrum analyzer. c , Linewidth measurements are given for (i) with isolator turned on and reflection turned off, (ii) with isolator turned off and reflection turned on, (iii) with both isolator and reflection turned on. RSA resolution bandwidth: 20 kHz.	58
5.1	Photonic wire bonded TFLN-based ECL. a , Schematic showing the design of the laser, which comprises of a dual channel InP amplifier, an LN feedback circuit and a fiber array, with all components integrated by photonic wire bonding. b , Fully integrated laser with photonic and electrical wire bonds. c , Microscope image of the InP-LN interface, with photonic wire bonds and waveguides angled to mitigate Fresnel reflections. d , Scanning electron microscope image of a photonic wire bond to a TFLN waveguide.	63
5.2	Full design schematic of the extended cavity diode laser. The lithium niobate circuit consists of a Vernier filter and two MZI couplers to maximize the output power. Ti-Pt heaters are included to tune the phase, on each ring to align the Vernier filter, and on the MZI couplers. All of the waveguides extend to the end of the LN chip and are photonic wire bonded to individual fibers in the grating array. The top waveguide is the laser output, and the bottom waveguide is used to characterize the coupling loss. (i)-(v) shows microscope images of each of these components taken before the packaging process.	65

5.3	Loss characterization of the TFLN circuit at different points in the fabrication process. a , Measurement method for characterizing diagnostic ring resonator devices. Grating coupled rings are probed at each stage of fabrication and the resonances are fitted to the transmission function $T(\omega)$. A : background signal; k_i : intrinsic loss rate; k_e : extrinsic loss rate; ω_0 : resonance frequency. b , Stages of the fabrication process during which the intrinsic Q is fitted. (i) The chips first undergo an annealing process as in Ref. ¹ . (ii) The devices are cladded with PECVD silicon dioxide, annealed again, and a 20 nm layer of aluminum oxide is deposited via atomic layer deposition (ALD) to prevent oxide degradation. (iii) Finally, the devices are measured again after 2 weeks of aging in ambient environment. c , Sample transmission spectra taken at points (i)-(iii) with fitted Q . We see a drop in the intrinsic Q after cladding, which is likely due to a combination of weaker confinement and low-quality oxide. After 2 weeks, the intrinsic Q drops further to 1.7 million, indicating that the aluminum oxide layer has slowed down oxide degradation but not completely eliminated it.	66
5.4	Characterization of the InP gain waveguides before photonic wire bonding. a , LI curve measured using an integrating sphere and gain curve measured on an optical spectrum analyzer for amplifier 1 as labeled in Fig. 5.1c. At the maximum current of 800 mA, the total ASE power of the amplifier reaches 73 mW. The peak gain for the amplifier is at 1512 nm. b , LI and gain curve for amplifier 2 as labeled in Fig. 5.1c. At the maximum current of 800 mA, the total ASE power of the amplifier reaches 92 mW. The peak gain for the amplifier is at 1518 nm.	68
5.5	Delayed self heterodyne set-up for measuring the linewidth of our integrated laser. The output of the laser is sent through an optical isolator before being sent into measurement setup. The signal is split into two arms; one arm is modulated at 160 MHz and the other arm has a fixed delay length. The combined signal is measured using a balanced photodetector and optimized using a polarization controller. For the frequency noise measurement, the delay is set to 19.6 m and the time trace is measured using a digital oscilloscope and Fourier transformed. For the Voigt fit for the integrated linewidth, the delay is adjusted to 20 km and the beat is detected on an RSA. AOM: acousto-optic modulator; PD: photodetector; RSA: RF spectrum analyzer.	69

5.6	Laser performance. a , On-chip optical power as a function of current applied to each amplifier. Residual variations to the linear trend are caused by longitudinal mode hops. The highest power of 78 mW corresponds to 200 mA to amplifier 1 and 800 mA to amplifier 2. b , Optical spectrum showing single frequency operation with up to 61 dB side mode suppression at the peak gain wavelength. The laser is tunable over 40.6 nm (shaded region) beyond the 3-dB gain bandwidth. c , Frequency noise power spectral density plot acquired using delayed self-heterodyne measurement with a 19.6 m delay, showing an ultra-narrow 550 Hz intrinsic linewidth. Inset: beat note recorded in the frequency domain confirming the low intrinsic linewidth.	70
6.1	Solutions for on-chip dispersion compensation, adapted from Ref.². Dispersive waveguide approach. a , SEM micrograph of the transition taper to the compression waveguide. The partially etched waveguide with a top width of 800 nm is adiabatically tapered to a top width of 300 nm with a 1 μ m slab width. The fully etched slab waveguide is then tapered to a final width of 500 nm. Scale bar, 20 μ m. b , The dispersion parameter D of the integrated lithium niobate waveguides. The compression waveguide is dispersion-engineered to have a large dispersion value of $-2.15 \text{ ps nm}^{-1}\text{m}^{-1}$ at 1557 nm. Its absolute value is a factor of 10 and 120 larger than the $ D $ of the partially etched waveguide and single-mode fiber. The total required length to reach full pulse compression at 30 GHz is 0.49 m. c , On-chip pulse compression using a 9.56-cm dispersive waveguide down to 8.06 ps. Chirped waveguide grating approach. d , The grating period is linearly chirped from $\Lambda_1 = 406.5 \text{ nm}$ to $\Lambda_2 = 414.5 \text{ nm}$ along the grating length L . The Bragg grating device is air-cladded with a fin width w_1 of 200 nm and a waveguide width w_2 of 1.1 μ m. e , Group delay dispersion of three different grating lengths, extracted from the wavelength-dependent fringe periods in its corresponding reflection spectrum. Dashed lines represent fits. f , Optical image of the TFLN chip, which consists of the time lens system from Chapter 2 followed by a Y-splitter and a chirped Bragg grating waveguide. g , Magnified view of the chirped Bragg grating. A window is opened to remove the cladding (SEM graph, bottom). Scale bar, 1 μ m. h , Autocorrelator trace showing the pulse is fully compressed to its Fourier transform limit of 500 fs using a 1.75-mm integrated lithium niobate Bragg grating.	78
A.1	Plot of the Bessel functions of the first kind J_n for indices $n = 0, 1, 2, 3, 4$.	84

List of Tables

3.1	Performance comparison of our pulse-pumped racetrack with other LN Kerr combs †: extracted.	45
4.1	Performance comparison with other on-chip isolator approaches. EO: electro-optic; MO: magneto-optic; AO: acousto-optic; WB: wafer bonding; PLD: pulsed laser deposition. *: including 3.7 dB loss per facet; †: optical power; ^: power dependent; : only backward transmission tested.	60

TO MY FAMILY.

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1

Thin-film lithium niobate on
insulator

1.1 INTRODUCTION

The development of the optical fiber, which guides light through a high index glass core surrounded by a lower index cladding material, has revolutionized high-speed data transmission. Telecom fiber networks provide a clear advantage over data transmission through electrical cables, as information can be sent over much longer distances with much higher bandwidth. Modern day communication is enabled through the routing of large undersea fiber-optic cables comprised of bundles and bundles of optical fibers, sending streams of data over large distances.

In recent decades, the desire for compact and high-efficiency optical components for data centers in the telecommunications market has spurred the field of integrated photonics, enabled by the optical waveguide. Functioning similarly to an optical fiber at a much smaller scale due to larger index contrast between core and cladding, optical waveguides confine and guide light in a high index material structure. The strong light-matter interaction resulting tight confinement in optical waveguides enables much higher effective nonlinearity, and as a result, device efficiency. Silicon, which boasts a large refractive index (3.47 at 1550 nm) and was previously already compatible with integrated complementary metal-oxide-semiconductor (CMOS) fabrication, was a natural candidate for an integrated photonics material platform. To this day, silicon photonics remains an active area of research for many research groups and companies, and fabrication of silicon photonic chips is mature to the foundry level.

Despite its nice properties and strong industry support, the fundamental material limitations of the silicon on insulator (SOI) platform has necessitated reliance on other photonic platforms for different applications. For example, two-photon absorption limits SOI for broadband and high-power applications which silicon nitride (SiN) is bet-

ter suited for, and on-chip lasers and amplification are enabled by III-V direct bandgap semiconductor platforms like indium phosphide (InP). These three platforms make up a large portion of the current integrated photonics market. However, electro-optic modulators with large modulation bandwidth, high efficiency, and low optical loss^{3,4} have not yet been demonstrated on these platforms, due to the material limitations of Si and InP.

In telecommunications and datacommunications, the role of electro-optic modulation has traditionally been fulfilled using bulk lithium niobate modulators⁵. However, due the low index contrast of bulk LN waveguides realized through Ti-indiffusion and proton exchange, the optical mode of these waveguides span tens of microns. As a result, the signal and ground electrodes of these modulators are patterned far from each other, resulting in a weak electric field for a given applied voltage. These bulk modulators do not fully take advantage of the large EO coefficient in LN. Significant progress was made in overcoming fabrication of thin-film lithium niobate (TFLN) on insulator, both in the fabrication of high-quality thin-film wafers and in the fabrication of low-loss ridge waveguides. Importantly, compact and low-loss propagation was possible, allowing electrodes to be patterned much closer for higher EO efficiency, which enabled the realization of EO amplitude and phase modulators with record-breaking performance⁶.

Since the initial breakthrough in TFLN fabrication technology, many devices have been demonstrated on the platform beyond traditional electro-optic amplitude and phase modulation, including but not limited to electro-optic frequency comb devices, frequency shifters, Kerr frequency combs, waveguides for broadband supercontinuum generation, periodically poled waveguides for efficient frequency conversion, acousto-optic modulators⁷. Concurrently, the field of integrated photonics has expanded to applications beyond telecommunications due to the high efficiency and compact form

factor, with goals of realizing systems traditionally implemented using bulk optics onto a single chip. However, crucial components are missing for fully integrated systems on TFLN. For example, photonic applications relying on pulsed sources typically use off-chip mode-locked laser frequency comb sources, which can be bulky and expensive. Additionally, getting light onto the platform itself is a challenge. Efficient on-chip active gain, which could enable integration of lasers, is largely missing from the TFLN platform. While groups are exploring rare-earth-ion-doping as a possible solution⁸, most research in gain and laser integration involves hybrid or heterogeneous integration of III-V semiconductor materials.

1.2 OUTLINE OF THESIS

This thesis realizes electro-optic manipulation of light using thin-film lithium niobate for frequency comb source applications and chip-scale laser integration. The chapters of the thesis show work that addresses two sides of on-chip integrated sources on TFLN; the first half focuses on utilization of the nonlinear material properties of lithium niobate (here, electro-optic and all-optical nonlinearity) for broadband frequency comb and pulsed laser sources, while the second half is focused on laser integration using III-V gain waveguides and interfacing with on-chip optical isolators.

While typical benchtop frequency comb sources are generated using a mode-locked laser cavity with a fixed repetition rate, electro-optic phase and amplitude modulators have been in use since the 1960s for generating frequency combs and pulses with tunable repetition rate and center wavelength. In **Chapter 2**, we show that the same system can be implemented using integrated photonics components with much higher efficiency and improved performance. Notably, while implementations of time lens sys-

tems with benchtop electro-optic phase modulators require the cascading of numerous phase modulators together to reach high modulation index and broadband frequency comb span, the modulation strength required for ultrafast (sub-ps) pulses can be achieved on-chip using a recycling phase modulator system driven by a single microwave source. Using an amplitude and phase modulator system combined with off-chip dispersion, we measure femtosecond pulse trains (530 fs pulse duration) with a 30 GHz repetition rate and flat-top optical spectra with a 10 dB optical bandwidth of 12.6 nm. We also show that the equivalent system can be implemented using only a single electrode drive, and measure ultrafast pulse trains of 840 fs pulse duration with a 28.7 GHz and flat-top frequency combs with a 10 dB optical bandwidth of 8.8 nm. The time-bandwidth product of the two demonstrations are reasonably consistent. The same system is optically packaged using photonic wire bonding for stable operation.

The ability to generate tunable and ultrafast microwave-rate pulses on-chip opens the door for a myriad of applications. A recent regime of interest for microresonator-based Kerr frequency comb generation is the pulsed-pumping operation regime in comparison to more traditional continuous-wave pumping operation. The content of **Chapter 3** expands on the work highlighted in the previous chapter and realizes a synchronous-pumped resonator on Z-cut TFLN whose free-spectral range (FSR) is precisely matched to one of the optimal repetition rate frequencies of the on-chip pulsed source. The resonator chip has an ultra-high loaded Q -factor of 3.2 million and successfully broadens the pump comb to a broadband comb of over 1400 comb lines with 2/5 of an octave above the noise floor of the optical spectrum analyzer with only 12.5 mW of average on-chip power. Not only is the pulsed-pumping scheme a promising approach to broadband frequency combs with lower repetition rate and higher conversion efficiencies, but also shows that synchronous pumping lowers the threshold for four-wave mixing and

mitigates the strong Raman scattering effect normally present in crystalline resonators without the need for FSR or coupling engineering.

Transitioning in the thesis to the integration of on-chip lasers with the thin-film LN platform, **Chapter 4** addresses the need for a low-loss and high-performance on-chip optical isolator. Laser cavities are extremely sensitive to backreflection, and un-isolated reflection in complex systems may lead to unwanted locking to the reflection or to multimode isolation. In this chapter, an electro-optic optical isolator is realized by utilizing the inherent nonreciprocity of a travelling wave modulator. Using a 1.75-cm long phase modulator, we realize a broadband optical isolator with a low 0.5 dB of insertion loss and a high isolation ratio of 48 dB. By employing a tunable add-drop ring filter, the optical sidebands from the modulation in the reflected direction are suppressed to 34 dB below the pump in the forward direction with 0.9 dB of filter insertion loss. The effectiveness of the isolator is tested and verified using a butt-coupled indium phosphide distributed feedback laser.

In **Chapter 5**, we address the issue of on-chip laser integration. A high-performance extended cavity diode laser is demonstrated on the thin-film lithium niobate platform, with the outlook of embedded lasers for future chip-scale systems. By utilizing the flexible and scalable integration method of photonic wire bonding, a laser cavity is formed using a lithium niobate feedback circuit and an indium phosphide reflective semiconductor optical amplifier (RSOA) with two intra-cavity gain sections. The laser outputs up to 78 mW of on-chip laser power with excellent side mode suppression (up to 61 dB), wide tunability (40 nm), and ultra-low linewidth (550 Hz). These metrics outperform all previous lasers demonstrated on the thin-film lithium niobate platform. The work demonstrates a viable on-chip laser solution for future TFLN photonic systems and the use of photonic wire bonding for bridging various functionalities.

Finally, the thesis offers an outlook on the thin-film lithium niobate platform beyond the work presented in this thesis, including the potential for integrating each device component demonstrated in the thesis along with device improvements and developments for a fully chip-scale microwave-rate frequency comb. I discuss some of the current challenges and additional required components needed for such a demonstration.

2

Electro-optic pulse generation on thin-film lithium niobate

2.1 INTRODUCTION

With the rapid advancement of integrated photonic systems and continuous improvement of photonic device performance, there is a growing need for compactness and stability of individual photonic devices. The ability to generate ultrashort, broadband and high-peak-power optical pulses on-chip has long been sought after due to a slew of applications ranging across optical communications⁹, optical atomic clocks¹⁰, LIDAR^{11,12}, spectroscopy^{13,14}, and more¹⁵. Improvements to low-loss nanophotonic waveguides have reduced the pulse energy required for achieving nonlinear spectral broadening across octaves of bandwidth to sub-picojoules via supercontinuum generation¹⁶. However, all demonstrations to date rely on a table-top pulsed laser sources, which increases system complexity, size, and cost, hindering practical applications. On-chip, optical pulses can be generated via microresonator frequency comb sources through coupling a continuous-wave laser into a high-quality-factor microresonator¹⁷. However, microcombs are ultimately limited by their low efficiencies (usually $\sim 2\%$), comb-line power, and high repetition rates, resulting in only tens of femtojoule pulse energies^{17,18,19}. An alternative approach, based on compact and electrically pumped on-chip semiconductor mode-locked lasers^{20,21}, still faces significant challenges including available pulse energy, pulse width, and optical and radio-frequency (RF) noise. Therefore, high-power and controllable integrated pulse sources are still missing, presenting a major road block to achieve fully integrated nonlinear photonic circuits. Additionally, mode-locked lasers are realized using saturable absorption (passive mode-locking) or modulation (active mode-locking) inside a gain cavity, which are lithographically defined for on-chip lasers. Therefore, in both the microresonator frequency comb and the on-chip mode-locked laser case, the generated pulses not only suffer from the challenges described, but are

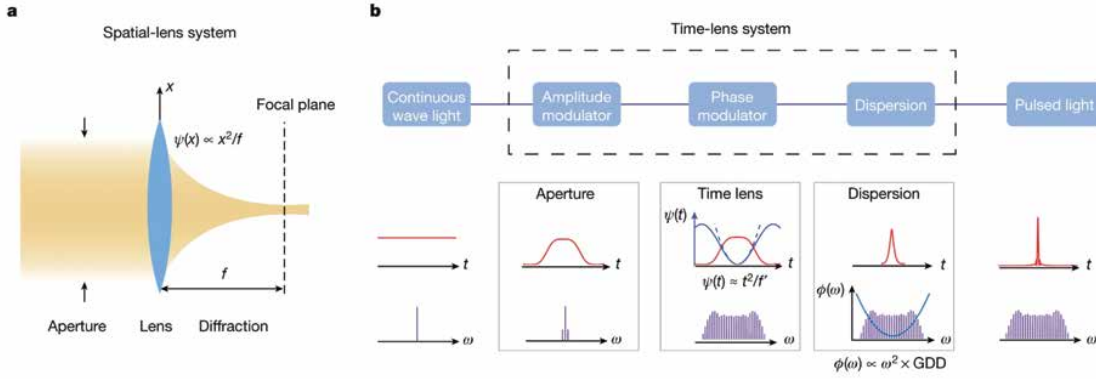


Figure 2.1: Concept of electrical synthesis of optical pulses via a time lens system. **a**, Spatial focusing via a spatial lens. A collimated beam passes a spatial lens through its aperture, undergoes diffraction in the free space, and is focused in the focal plane. A perfect lens impacts a quadratic phase profile ($\psi(x) \propto x^2/f$) in the transverse dimension, where f is the focal length of the lens. **b**, Temporal focusing via an electro-optic time lens (TL). Similar to the spatial focusing in **a**, a continuous-wave (CW) light passes through a time lens system and is "focused" into a Fourier-transform-limited pulse train. The time lens system is implemented via electro-optic (EO) modulation, consisting of an amplitude modulator (AM), a phase modulator (PM) and a dispersive medium. The AM carves the CW light into a flat-top pulse train, the temporal aperture of the lens. The PM constructs an EO time lens to chirp the initial pulse, where an approximate quadratic phase term ($\psi(t) \sim t^2/f'$) is established at the local minima (or maxima) of a sinusoidal microwave drive, where f' is the focal "length". The dispersive medium induces group delay dispersion (GDD), where different spectral components experience a time delay proportional to the derivative of the spectral phase profile $\phi(\omega) \propto \omega^2 \times \text{GDD}$ with respect to frequency ω . Thus, it retimes all the spectral components to form an ultrashort pulse train. This electrical synthesis of optical pulses generates a flat-top EO comb spectrum with a line spacing equal to the MW drive frequency, with its spectral bandwidth inversely proportional to the pulse duration.

also limited in flexibility of repetition rate.

A well-known and widely utilized table-top approach for synthesizing optical pulses uses non-resonant EO modulation of a CW laser light^{22,23}, which delivers flat, high-power and coherent optical frequency combs with up to tens of GHz repetition rates. While their initial proposal dates back to the 1960s²⁴, EO comb generators have attracted significant interest in the past decade due to their flexibility in operating wavelengths and repetition rates, as well as exceptional spectral flatness^{25,26,27}, making them incredibly useful for optical communications and wavelength division multiplexing.

Pulse synthesis using cascaded amplitude and phase modulation can be described as a time lens approach²⁸, or a temporal analogue of a conventional spatial lens (Fig. 2.1a). While a spatial lens system can focus a collimated laser beam to a spot with a small waist size on its focal plane, a time lens system can compress CW light to a short pulse at a proper dispersive focal length. This space-time duality originates from the mathematical analogy between the equation for beam diffraction in the spatial domain and the equation for pulse dispersion in the temporal domain²⁹. Three key components essential to building such a spatial/temporal lens system are: an aperture, a lens that induces a quadratic phase term in space/time, and a diffraction/dispersion medium representing the focal length of the imaging system. In an EO time lens, shown in Fig. 2.1b, an amplitude modulator (AM) carves the initial flat-top pulse, representing a temporal aperture; a phase modulator (PM) induces a (quasi-) quadratic temporal phase; and a dispersive medium imparts a group delay dispersion (GDD) in the frequency domain, which acts as the focal distance and compresses the periodic waveform into ultrashort pulses. Since the EO modulation is periodic in time with a microwave frequency f_{MW} , such a system converts a continuous-wave (CW) single-frequency light to a pulse train with a repetition rate of f_{MW} . In the frequency domain, the output features an optical comb spectrum with a flat-top envelope, which is a result of the time-to-frequency mapping due to the Fourier transform operation of a lens³⁰. An important figure of merit of a time-lens system is the minimal achievable pulse width $\tau \sim 2 \ln(2) V_\pi / (\pi^2 V_{MW} \times f_{MW})$, where V_π is the half-wave voltage of the PM, and V_{MW} is the microwave voltage applied on the PM. Therefore, for ultrashort pulse generation, EO modulators used in time lens must have a low V_π and need to be able to handle large microwave power, in addition to a high EO bandwidth.

Due to the high V_π of commercially available EO modulators, previously demon-

strated systems required 2-4 stages of phase modulators combined with a separate intensity modulator to achieve the optical bandwidth corresponding to picosecond-pulses^{23,26}. Each phase modulator is driven by an RF amplifier with several Watts of power consumption as well as tunable RF phase shifters to align the relative phase between each subsequent modulator. Furthermore, laser amplifiers are needed to compensate for the insertion losses and limited optical power handling capabilities of discrete components used^{31,26}. As a result, the complexity, power-consumption, and high-cost of this approach have limited its adoption in emerging applications.

Here, we address these limitations by fully integrating an EO comb source on an LN photonic chip. The rapid development of the thin-film lithium niobate (LN) platform has resulted in low-loss nanophotonic waveguides (3 dB/m) and state-of-the-art EO efficiency for modulation, storage, and control of optical signals⁷. By utilizing both the low loss and high EO efficiency of the TFLN platform, as well as a loop-back phase modulator design enabled by the integrated photonics platform, we realize flat-top electro-optic combs and, combined with off-chip dispersion in optical fiber, ultrashort pulse generation using vastly fewer off-chip microwave components than this approach has historically used. We present two implementations of the same system. The first uses a dual-drive design, which drives the amplitude and phase modulation components separately. The second, a single-drive design, combines the amplitude and phase modulation into a single transmission line, simplifying operation even further.

2.2 DUAL DRIVE DESIGN AND RESULTS

We first present the results of the dual-drive time lens design, where the amplitude and phase modulator components are driven separately. Figure 1c shows the LN chip, with

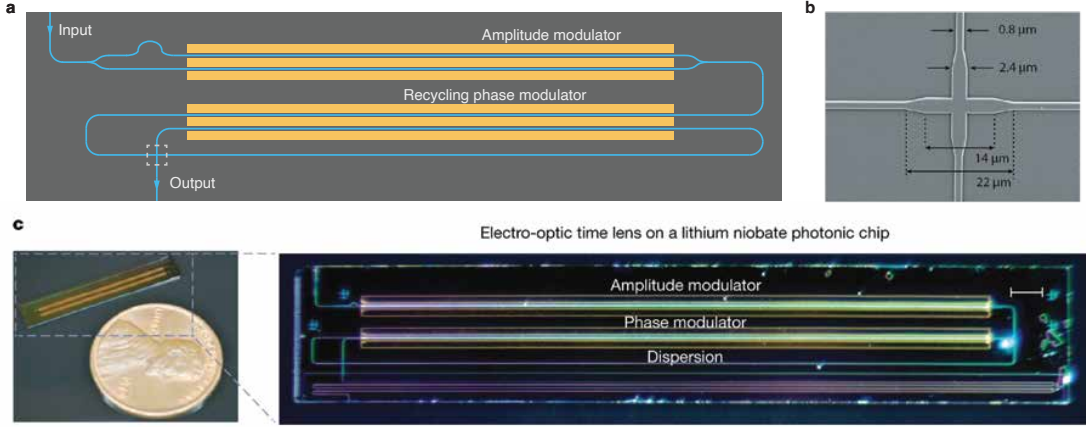


Figure 2.2: Dual drive design and fabricated chip. **a**, Schematic of the electro-optic modulation system on TFLN. The recycling phase modulator system is enabled by a low-loss waveguide crossing (grey box). **b**, SEM and geometry of the waveguide crossing in the phase modulator on a partially etched waveguide with a 300 nm etch depth. The insertion loss of this crossing is extracted to be 0.26 dB per crossing. **c**, Optical micrograph of the integrated lithium niobate (LN) photonic chip for the time-lens implementation. Zoom-in picture (right) shows the full integrated circuit with a footprint of 25×7 mm. The photonic circuit integrates the following components on a single X-cut 600-nm LN chip: a spot-size converter, a Mach-Zehnder interferometer (MZI) -based AM, a “recycling” PM, and a highly dispersive waveguide.

all three building blocks – one AM (in push-pull configuration), one PM, and one dispersive waveguide, integrated into a footprint as small as 25×7 mm. The optical layer is fabricated using a two-layer electron-beam lithography and etching process on an X-cut 600-nm LN substrate. The first layer is used to define the optical slab waveguides for the AM and PM with a top width of 1500 nm and an etch depth of 300 nm. This allows for an efficient modal overlap between the microwave field and the optical field, while maintaining a low optical propagation loss of 0.3 dB/cm. The second layer creates fully etched nanostructures that form the dispersive waveguide for pulse compression as well as spot-size converters³² for efficient coupling to a distributed feedback (DFB) laser chip (2 dB) and a lensed fiber (3 dB). Adiabatic tapering is used to ensure low-loss transitions of < 0.1 dB between optical waveguides on different layers.

Both the AM and PM utilize 2-cm-long traveling-wave coplanar waveguide (CPW)

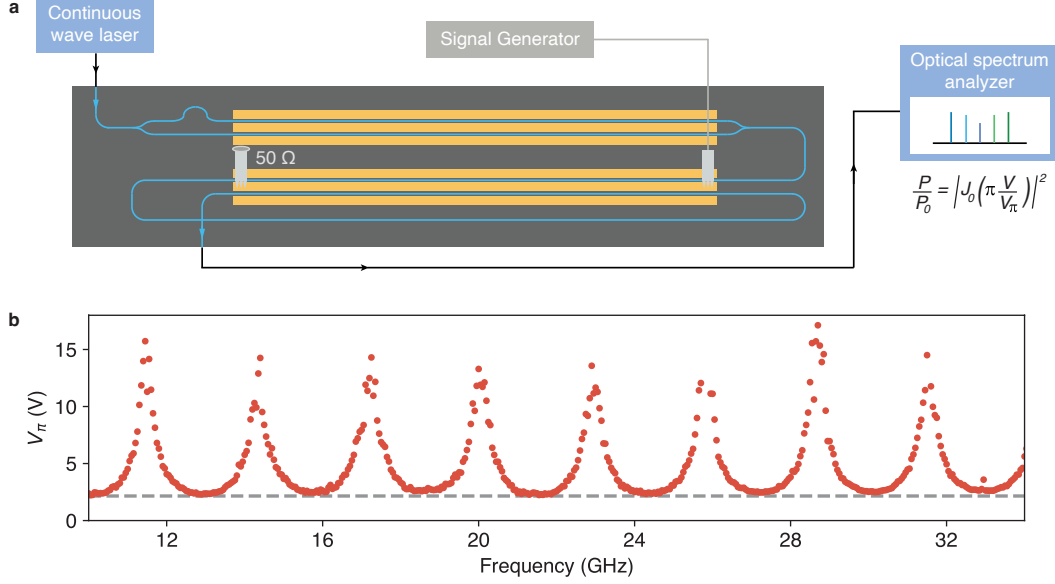


Figure 2.3: Frequency-dependent V_π for the dual drive device. **a**, Measurement scheme for determining the frequency-dependent half-wave voltage of the recycling phase modulator. The amplitude modulator is kept passive while the phase modulation output is monitored at different modulation frequencies using an optical spectrum analyzer (OSA). The pump depletion measured on the OSA is fitted to a Bessel function for a given microwave strength. **b**, Fitted V_π for the recycled phase modulator system measured on the device. The periodicity corresponds to 2.85 GHz spacing of the resonant condition.

electrodes with parameters optimized for RF impedance, RF loss, and microwave-optical velocity matching. Based on an unbalanced Mach-Zehnder interferometer in a push-pull configuration, the measured EO S_{21} parameter of the AM indicates a 3-dB EO bandwidth of 45 GHz and a low DC switching voltage V_π of 1.2 V at DC. To reduce V_π of the PM, we developed a recycling-PM design. Here, after passing one of CPW gaps, the optical waveguide loops back and passes through the other CPW gap. This design maximizes the interaction between optical and microwave fields and effectively reduces V_π by a factor of two. To achieve this, low insertion loss optical waveguide crossings (loss 0.3 dB/per crossing) were developed with minimal cross-talk (Figure 2.2b). The V_π of the recycling-PM is measured to be 2-2.5 V in the RF frequency range from 4 to 39 GHz.

Specifically, the V_π is measured to be 2.2 V and 2.5 V at 10.075 GHz and 30.135 GHz, respectively. We note that in the recycling-PM the lowest V_π is achieved when looped-back optical signal is in phase with microwave drive, which results in resonant behavior with microwave free-spectral range (FSR) of 2.85 GHz and a 3-dB power bandwidth ($\sqrt{2}V_\pi$) of 1.5 GHz (Figure 2.3). The on-chip insertion loss of PM (with waveguide crossing) is 2 dB. The realization of such a looped structure uniquely benefits from the small bending radius and waveguide crossing achievable on a nanophotonic platform which can support tightly confined optical modes.

We show that the time-lens chip allows for generation of pulses in the femtosecond regime, while significantly reducing the RF power consumption and complexity of the control circuitry compared to discrete-component implementation. The experimental setup is shown in Fig. 2.2a. Light is edge-coupled into the LN chip and polarized along the polar axis of LN to excite the transverse electric (TE) mode (Fig. 2.2b). The AM is biased at the quadrature point and driven with a drive voltage V_p equal to the V_π of the AM. The RF signal from a signal generator is split into two: one drives the AM, and the other drives the PM after being amplified and phase-aligned to the AM arm. The output of the chip is sent to an optical spectral analyzer and an intensity autocorrelator for spectral and temporal measurements, respectively (Fig. 2.2c). The system performance is characterized using a single-mode fiber (SMF-28) for dispersion and pulse compression, which has slightly anomalous group velocity dispersion ($GVD < 0$). Figure 2.2d shows the generated optical spectra with a 10-dB optical bandwidth of 6.0 nm and 12.6 nm for two different line spacings (RF drive frequencies) of 10.075 GHz and 30.135 GHz, respectively, obtained using the same chip. A total of 91 comb lines are generated at 10.075 GHz, corresponding to a modulation index of 11.2π radians at an RF power of 38 dBm. A total of 67 comb lines are achieved at 30.135 GHz, corresponding to a

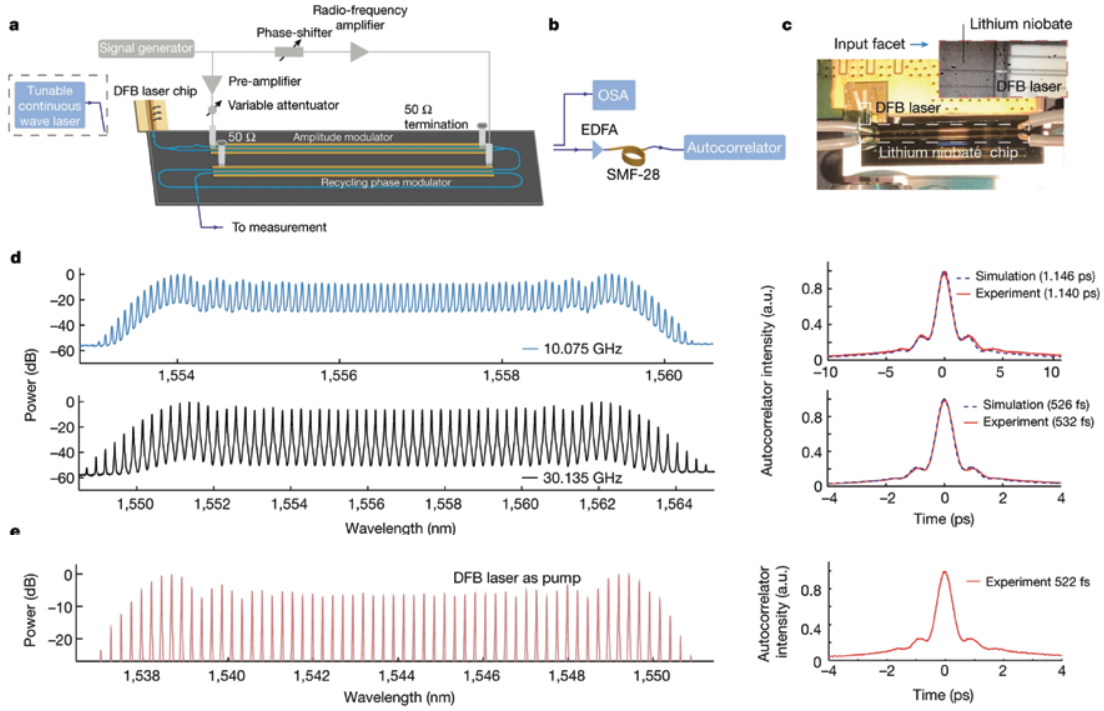


Figure 2.4: Femtosecond pulse generator via an integrated LN-chip-based electro-optic time lens. **a**, Schematic of the LN chip. In this experiment, the input laser is either a tunable CW laser or a semiconductor-chip-based distributed feedback (DFB) laser. **b**, Characterization setup. The chip output is sent to an optical spectral analyzer for frequency-domain measurement or intensity autocorrelator for time-domain measurement. **c**, The optical microscope picture of the LN photonic chip with edge-coupled DFB laser. The coupling loss is 2 dB with an optimized inverse taper for edge coupling to the DFB laser output (zoom-in). **d**, The output optical spectra and temporal traces at microwave driving frequencies of 10.075 and 30.135 GHz using a tunable CW laser. Two separate EO combs are shown: At 10.075 GHz, a total of 91 comb lines are generated with a 10-dB optical bandwidth of 6.0 nm, and at 30.135 GHz, a total of 67 comb lines are generated with a 10-dB optical bandwidth of 12.6 nm. At 10.075 GHz, the TL focal “length” (referred as focal GDD) is 7.1 ps^2 , corresponding to a PM modulation depth of 11.2π radians. Aided with the dispersion of single-mode fiber (SMF-28) for full temporal compression, we measured the final pulse duration (full-width-half-maximum (FWHM)) to be 1.140 ps. At 30.135 GHz, the TL focal GDD is 1.1 ps^2 , corresponding to a modulation depth of 7.8π radians. The final pulses have a measured FWHM of 532 fs. The autocorrelator traces are in good agreement with the simulation (dotted curves). **e**, The optical spectrum with a 30-GHz line spacing and autocorrelator time trace obtained using a DFB laser pump at 1544 nm. The laser wavelength is tuned via changing the injected current, and a 522-fs pulse is measured. OSA: optical spectral analyzer, SMF: single-mode fiber, EDFA: erbium-doped fiber amplifier.

modulation index of 7.8π radian at an RF power of 36 dBm. The modulation index is currently limited by the available power of our RF amplifiers. We have not observed

the damage of our chip due to the applied microwave power. The optical spectra have also shown an excellent spectral power flatness of total 35 and 25 comb lines within < 1 dB power variation. We achieve the temporal focal “length” f' of 7.1 ps² and 1.1 ps² at 10 and 30 GHz, which leads to a measured transform-limited pulse of 1.14-ps and 532-fs duration after pulse compression in an SMF-28 fiber with lengths of 360 and 59 meters, respectively, showing good agreement with simulation (Fig. 2.2d). Both of the EO-based combs and pulses are generated on the same device simply via tuning the microwave driving frequency. The overall insertion loss of the chip is 8 dB including 3-dB on-chip insertion loss and 5-dB facet coupling losses. The insertion loss could be further reduced to a total of 1.3 dB with 0.5 dB/facet and 3 dB/m propagation loss via improved coupler³³ and fabrication³⁴.

In contrast, the state-of-the-art discrete component EO time lens gives 600-fs pulses at both 10 and 30 GHz repetition rate with 3-4 phase modulators and phase shifters with an unavoidably higher optical insertion loss of 12-18 dB and a total RF power consumption of 42 dBm. Operation beyond 30 GHz RF frequency to further reduce the pulse duration is challenging due to the critical voltage-bandwidth trade-off of conventional LN modulators. However, our integrated time lens has not hit the EO bandwidth limit and could generate pulses as short as 200 fs by operating at 45 GHz assuming a 38-dBm RF power, based on $\tau \sim 2 \ln(2) V_{\pi} / (\pi^2 V_{MW} \times f_{MW})$, and are capable of coherent nonlinear broadening to an octave-spanning spectrum³⁵.

The performance of the pulse generator chip is further tested using a butt-coupled indium phosphide DFB laser to optically drive the system (Fig. 2.2a,c). The coupling loss between the DFB laser and the LN chip is 2-dB, limited by the minimal coupling gap achievable on our optical setup. Figure 2.2e shows the results of the DFB laser operation at 30.135 GHz microwave drive frequency, which is in good agreement with simulation

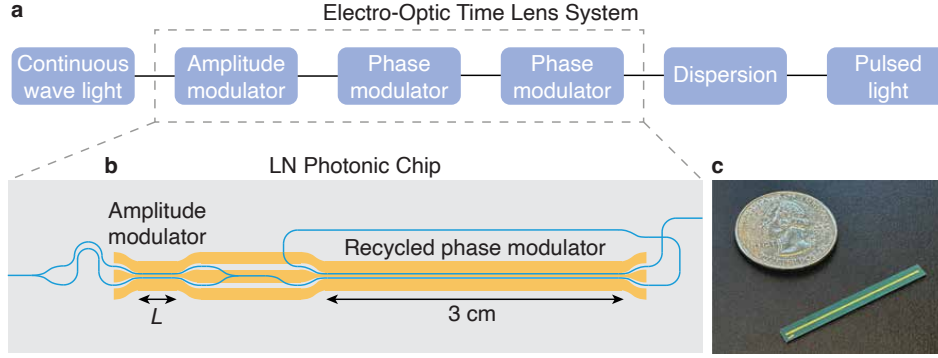


Figure 2.5: Compact electro-optic comb and pulse generation using a single electrode drive. **a**, Overview of the electro-optic time lens system for pulse generation. Continuous wave light is converted to a flat-top pulse train via amplitude modulation followed by a cascaded phase modulation to impart a large semi-quadratic phase. The pulse is compressed to its Fourier transform limit by passing through dispersive medium. **b**, Schematic of the thin film lithium niobate photonic device that consists of amplitude modulator followed by a recycling phase modulator. The modulators have common electrode and are driven by the same microwave source. 3-cm long recycling phase modulator is equivalent two two 3-cm long phase modulators. The amplitude modulator segment length L is significantly shorter than the phase modulator length. **c**, Photograph of one of the LN chips next to a quarter for scale.

and the results from the same comb line spacing when pumped with an off-chip laser.

2.3 SINGLE DRIVE DESIGN AND RESULTS

Next, we take the same system to a further level of compactness by designing the amplitude modulator (AM) and phase modulator (PM) system using a single electrode drive, greatly reducing the overhead on the microwave components and sources needed for EO comb and pulse generation. Notably, the single drive operation requires only one microwave source and eliminates the need for off-chip phase shifters that are typically required. Additionally, we optically package one of the TFLN chips by permanently attaching it to input and output fibers using photonic wire bonding (PWB). We show that this further simplifies the operation of the comb source, improves its stability, and allows the EO comb and pulse generator to function as a plug-and-play module.

The lithium niobate chip is based on the same electro-optic time lens system (Fig. 2.5a) and consists of a travelling-wave Mach-Zehnder interferometer (MZI) amplitude modulator and recycled phase modulator system driven simultaneously by one set of electrodes. As described earlier in the chapter, for these systems, flat-top comb generation is achieved when the amplitude modulator is driven at its half-wave voltage (V_π), carving out an initial train of pulses. The comb bandwidth and pulse duration limit depend on the strength of the phase modulation applied to this train of pulses, requiring the subsequent phase modulator to be driven at many multiples of its V_π for wide span and narrow pulse width.

To achieve this condition, the amplitude modulator portion of the device is made significantly shorter than the phase modulator one (Fig. 2.5b), translating to the condition $V_{\pi,AM} \gg V_{\pi,PM}$, where $V_{\pi,AM}$ and $V_{\pi,PM}$ are the half-wave voltages of the AM and PM, respectively. The length L of the AM is varied over multiple chips (600 μm , 700 μm , and 800 μm) while the recycled PM is 3 cm long, resulting in effective PM length of 6 cm. An image of one of the fabricated chips is given in Fig. 2.5c.

As in the case of the dual drive device, the cascaded phase modulator geometry imposes a resonant condition on the half-wave voltage of the device. The phase modulation V_π is minimized when the modulation signal when passing through the second arm of the phase modulator is in phase with the first, with a measured periodicity of 2.1 GHz (Figure 2.6). The electro-optic comb span is maximized at frequencies where the phase modulator $V_{\pi,PM}$ is minimized at 1.45 V for the device with AM length of 700 μm , corresponding to a voltage-length product of 8.7 V·cm. We assume similar EO performance for the remaining two devices due to an identical PM geometry.

The EO comb devices were optically characterized as in Figure 2.7a. A tunable continuous-wave laser is used to drive the device at the amplitude modulator quadrature

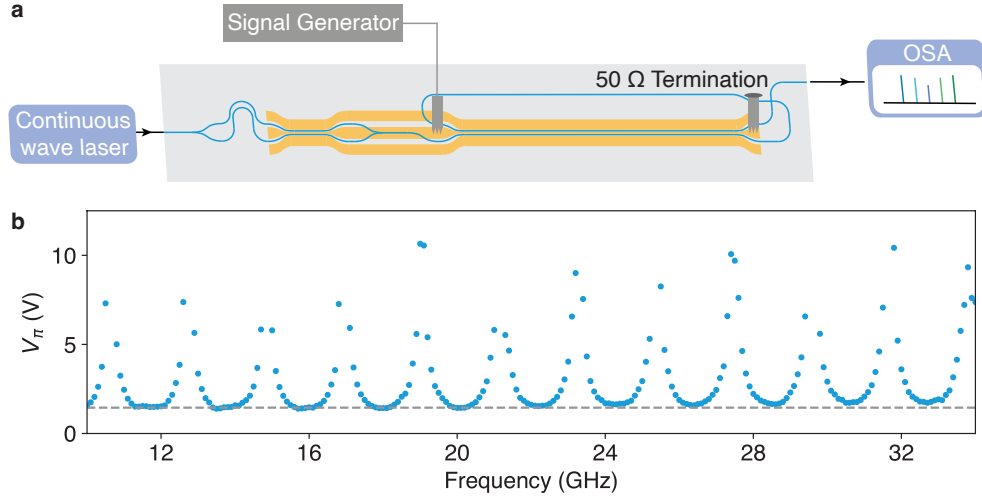


Figure 2.6: Frequency-dependent V_π for the single drive device. **a**, Measurement scheme for determining the frequency-dependent half-wave voltage of the recycling phase modulator. The pump depletion measured on the OSA is fitted to a Bessel function for a given microwave strength. **b**, Fitted V_π for the recycled phase modulator system measured on the photonic wire bonded device. The periodicity corresponds to 2.1 GHz spacing of the resonant condition.

point and a signal generator and amplifier deliver the microwave signal to the electrodes using external GSG probes. The electrode is terminated with a 50Ω load. The EO comb source is driven at 28.7 GHz microwave frequency (one of the minimum $V_{\pi,PM}$ frequencies) and the generated optical spectra are measured on an optical-spectrum analyzer.

The microwave power is selected such that the amplitude modulator is driven at its half-wave voltage $V_{\pi,AM}$, ensuring a flat-top pulse train^{22,2}. For a given electrode geometry, the voltage-length product ($V_\pi \cdot L$), which describes the relationship between the voltage required to modulate at the half wave voltage and the length of the modulation region, should be constant if the electrode loss is consistent across the devices. Under this assumption, the half-wave voltage of the amplitude modulators $V_{\pi,AM}$, and the peak-peak drive voltage V_p of the devices, should scale inversely with their length, $\propto 1/L$.

Similarly, under the assumption that the phase modulator $V_{\pi,PM}$ is constant across the EO comb devices, we predict a $1/L$ scaling of the modulation index $\beta_{PM} = \pi V_p / V_{\pi,PM}$ on the phase modulator system and the optical bandwidth. In other words, a longer am-

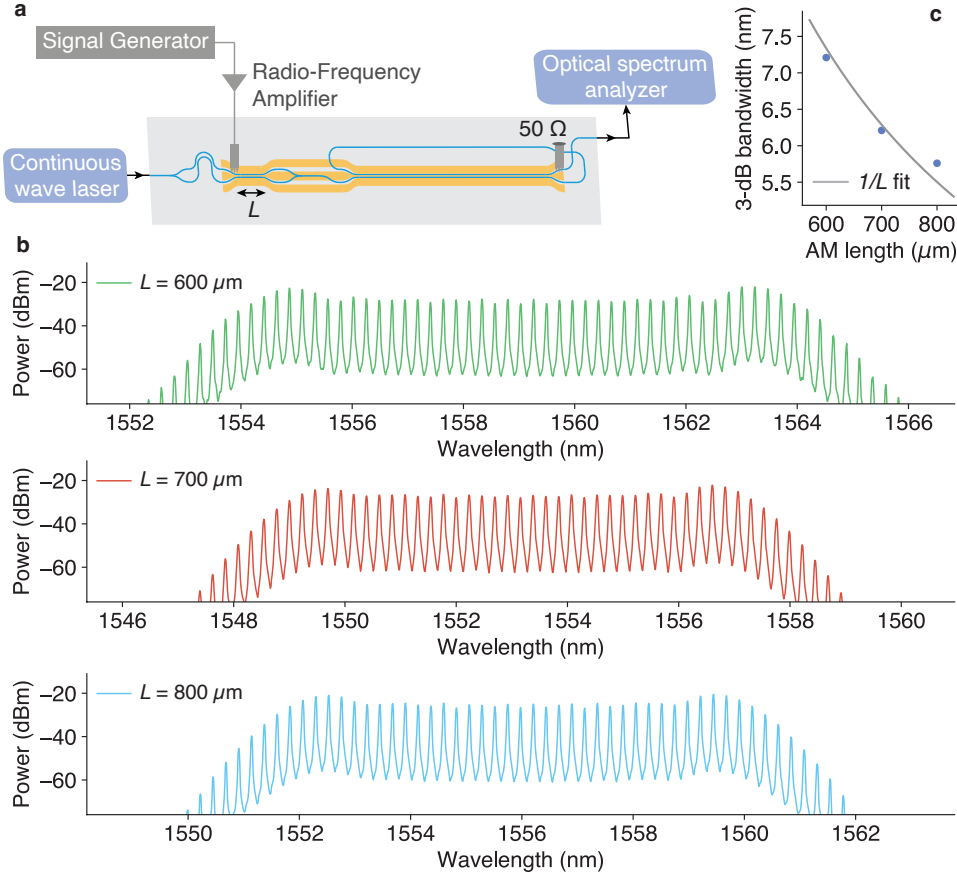


Figure 2.7: Optical spectrum characterization of the electro-optic comb generator chips. **a**, Chips with variable amplitude modulator length L (600, 700, and 800 μm) are tested. A tunable continuous wave laser is coupled into pulse generator chips with different MZI lengths. The generated spectrum is measured with an optical spectrum analyzer (OSA). **b**, 3-dB optical bandwidth plotted against amplitude modulator length, with the corresponding $1/L$ fit. **c**, Optical spectrum for the devices. For the 600 μm MZI length, spectral flatness of 3 dB is achieved over 31 comb lines, spanning 7.2 nm bandwidth. For the device with 700 μm length, spectral flatness of 2.2 dB is achieved over 27 comb lines, spanning 6.2 nm bandwidth. For the device with 800 μm MZI length, spectral flatness of 1.7 dB is achieved over 25 comb lines, spanning 5.7 nm bandwidth.

plitude modulator will require less microwave power to drive the device, resulting in a lower modulation strength on the phase modulation and a narrower optical bandwidth.

The optical spectra of the three devices are given in Figure 2.7b. The respective 3-dB optical bandwidth of the 600, 700, and 800 μm amplitude modulator chips are approximately 7.2 nm, 6.2 nm, and 5.7 nm. As predicted, the longer AMs require less microwave power to generate flat-top frequency combs, corresponding to a narrower optical bandwidth than their shorter counterparts with approximately $1/L$ scaling (Fig. 2.7c). We believe that the deviation from the fit can be explained by variation in microwave loss between the electrodes of different devices.

One outstanding challenge for EO comb sources discussed above is stabilization time: the microwave powers required to achieve large optical bandwidth and low pulse duration often result in significant heat load delivered to the chip, which can cause drifts and negatively impact fiber-chip coupling efficiency when performing traditional fiber-to-fiber measurements with XYZ positioners. To address this problem, we used photonic wire bonding (PWB) to attach fiber arrays to TFLN waveguides (Fig. 2.10a). Photonic wire bonding (PWB) is an emerging technology for photonic packaging that has been effectively utilized in silicon photonics and for III-V laser integration^{36,37,38}. PWBs are incredibly adaptable to optical mode profile and are a promising approach for combining devices and components across multiple materials into an integrated platform.

We photonic wire bond polarization-maintaining (PM) fiber arrays to the input and output waveguide facets of the TFLN chip with amplitude modulator length of 700 μm . Inverse taper couplers³² are used at the input and output of the device to improve coupling losses between the fiber array and LN waveguide. The PWBs are subsequently cured and cladded for robustness to atmosphere and the entire assembly is glued on a submount. Accounting for device insertion loss, we measure 5-dB/facet coupling loss

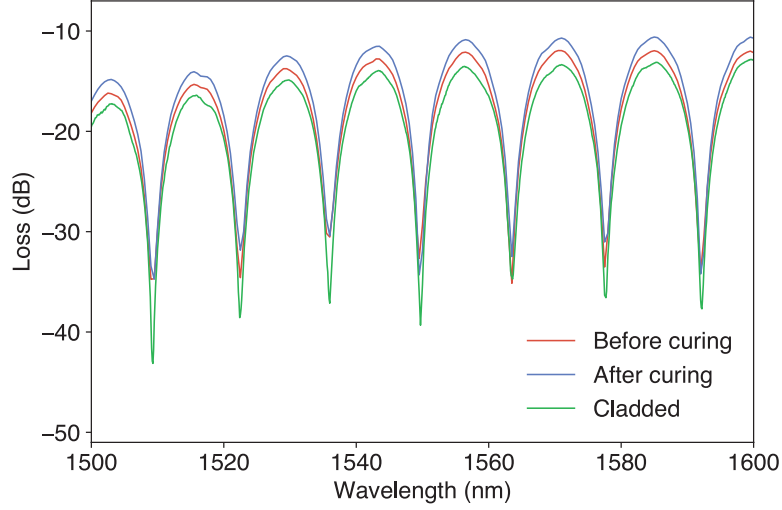


Figure 2.8: Characterization of optical losses in PWB process. Loss plotted against the wavelength during steps of the photonic wire bonding process, from before wire bonds are cured to after they are cladded. In the measurement, the loss can be improved further by adjusting the input polarization.

from the fiber into the LN waveguide. However, lower coupling losses are achievable. Figure 2.8 shows the measured passive transmission through the device at different stages of the photonic wire bonding process.

The measured assembly is pictured in Fig. 2.10b. The electrodes of the LN chip are externally probed to provide microwave signal to the modulators. Under 28.7 GHz operation, the pulse generation module gives a flat-top spectrum with <2-dB variation of 6.2 nm optical bandwidth, as shown previously in Fig. 2.7c, not affected by the packaging method.

As an additional step, we verify the system for ultrashort pulse generation using off-chip dispersion to compress the generated pulses. For ultrashort pulse generation, an additional consideration is made for the operation wavelength, illustrated in Fig. 2.9a. Because the phase between the amplitude and phase modulation is fixed by a common drive rather than set by a phase shifter, the selection of quadrature point is important

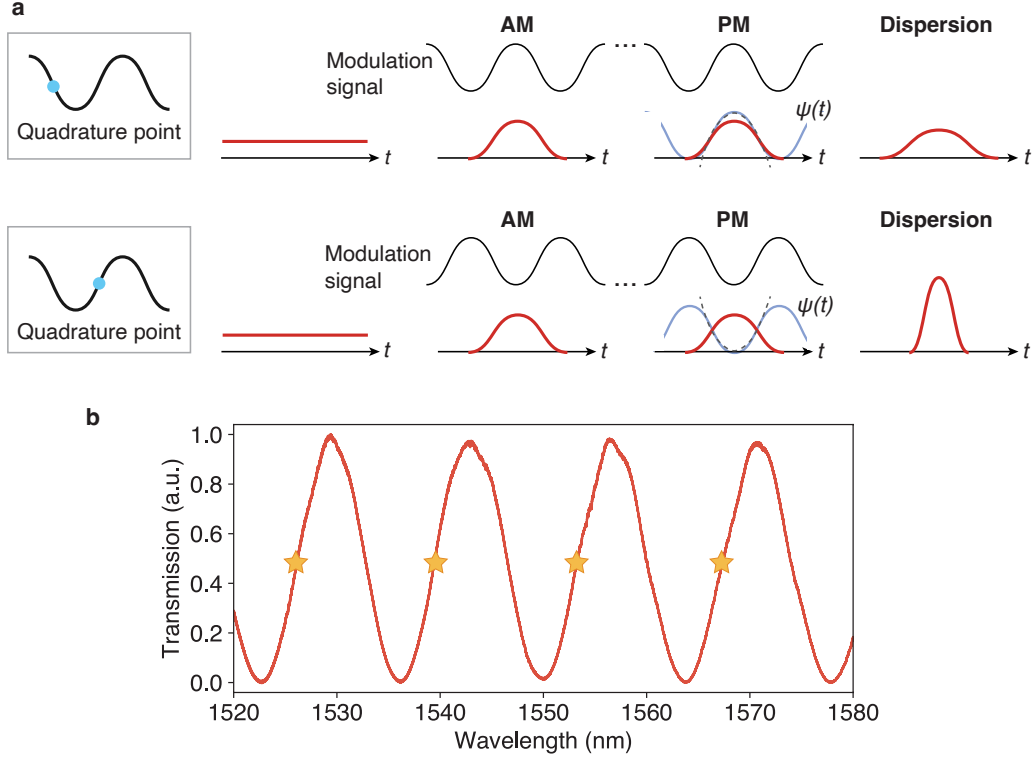


Figure 2.9: Wavelength operation consideration for single-drive device. **a**, Illustration showing the syncing of the pulse generated by the amplitude modulator with the modulation signal for quadrature points of the amplitude modulator. Because of the single drive operation, the phase modulation signal is automatically in phase with the generated pulse. We note that in the first case, the phase modulator will apply a negative quasi-quadratic phase to the pulse, whereas in the second, the phase modulator applies a positive quasi-quadratic phase. After sending through anomalous dispersion, the first case will translate to a broadening of the pulse (analogous to a concave lens) whereas the second will translate to pulse compression (analogous to a convex lens). **b**, Passive transmission of the photonic wire bonded device, showing the quadrature points of the unbalanced amplitude modulator. The points of operation for ultrashort pulse generation, where the phase modulation signal applies a positive quadratic phase on the generated pulse, are denoted by the stars.

and corresponds to the case where the applied quasi-quadratic phase is positive or negative with respect to the timing of the pulse. In turn, this condition determines whether pulse compression occurs under anomalous or normal dispersion. The passive transmission of the PWB device, showing the compatible wavelength operation points for SMF-28 (which has a slight anomalous dispersion), is given in Fig. 2.9b. We note that

this consideration need only be taken in the case of ultrashort pulse generation and not for flat-top EO frequency comb generation.

The output signal of the device is sent through 95 m of single-mode optical fiber (SMF-28), which compresses the pulses to their Fourier transform limit. The pulse duration is measured using an intensity autocorrelation system, which shows a final pulse width of 840 fs (Fig. 2.10c). This pulse width is in good agreement when comparing the time-bandwidth product against the frequency combs span compared against previous on-chip pulse generation results on TFLN shown earlier in this Chapter using the dual drive design².

Finally, we verify that photonic wire bonding drastically improves device stability when compared to unpackaged chips. To illustrate this, we perform a long term characterization of both PWB and a non-PWB device (specifically, with 800 μm amplitude modulator length) for comparison. We monitor the output power for both devices for the period of three hours after the microwave supply is turned on. As it can be seen, in the case of non-PWB device coupling quickly drifts away and it has to be reoptimized. We perform a manual coupling optimization whenever the output power drops by 30 dB, to ensure steady reading on the power meter. On the other hand, PWB device is stable and does not get out of alignment, even though it is driven at a higher microwave power than the fiber-coupled device.

The results of the measurement are shown in Fig. 2.10d. We see that while the coupling for the non-PWB device drifts substantially after the microwave power is turned on, requiring nearly two hours of hold time for stable operation, the PWB device demonstrates excellent operation stability over the entire measurement time. After the microwave power is turned on, the comb power of the PWB device varies < 0.3 dB over the entire measurement. We attribute the minor optical power variation to thermal effects,

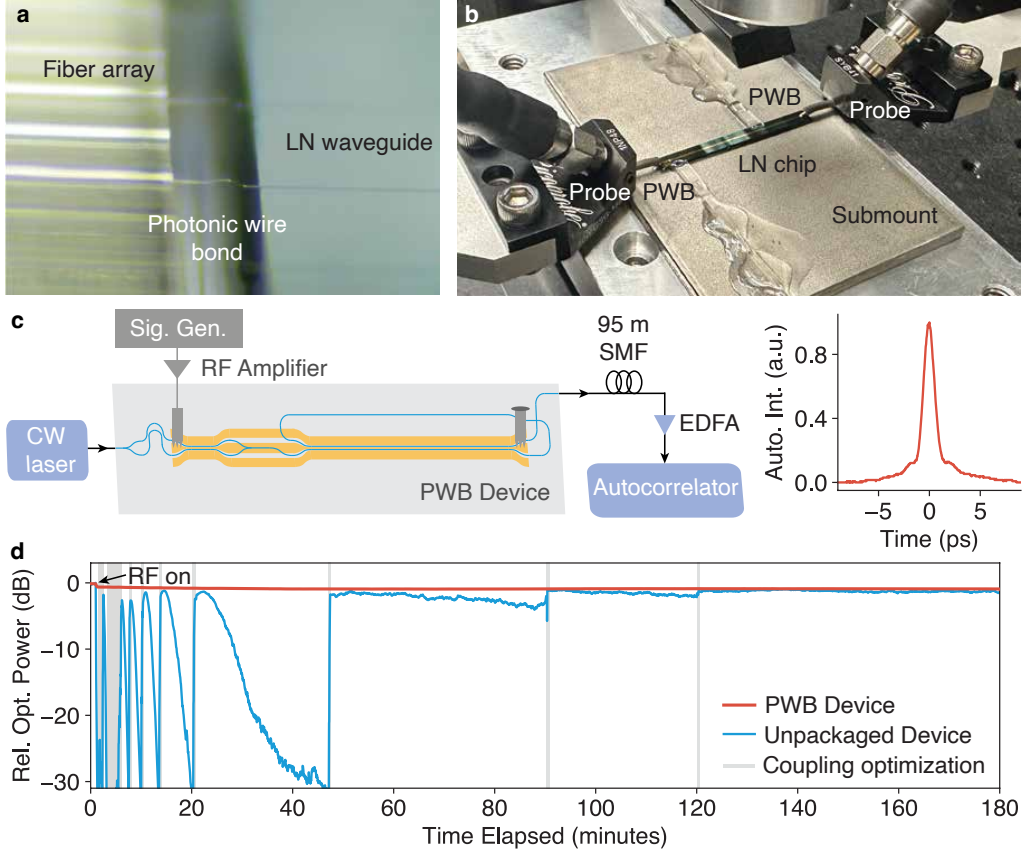


Figure 2.10: Photonic wire bonded pulse generation assembly. **a**, Image of photonic wire bond connecting the output of the fiber array to the input of the lithium niobate waveguide. **b**, Measurement image. The photonic wire bonded assembly sits on a submount, and GSG probes are touched down to the electrodes to drive the LN chip. **c**, Setup for the pulse width measurement and corresponding autocorrelator trace for the photonic wire bonded device. The pulse duration using an intensity autocorrelator after compression in 95 m of single-mode fiber. The final pulse duration is 840 fs. **d**, Stability measurement results conducted over three hours, comparing a photonic wire bonded device with an unpackaged device.

laser output power variation, and slight polarization drift.

2.4 CONCLUSION

In conclusion, we demonstrate a flat-top and frequency-agile EO comb generator on a LN photonic chip. Using a dual-drive design, we generate 526-fs pulses at a 30-GHz

repetition rate, directly pumped both with a chip-scale DFB laser and a tunable Santec laser. Using a single-drive design, we generate 840-fs pulses at 29-GHz repetition. The realization of the EO comb and ultrafast pulse generator is based on the construction of an EO time-lens system, which is, for the first time, realized on a miniaturized opto-electronic circuit since its proposal in 1960s. The combination of low-loss and tightly-confined optical waveguides and high EO efficiency tremendously reduces the footprint, microwave and optical power consumption, and complexity of the entire time-lens system, making it compatible with low-cost wafer-scale production. The photonic wire bonding method used to optically package the chip in the single-drive design enables ultra-stable operation at high microwave powers and would be compatible with future integration with chip-scale lasers.

While the coupling loss of the PWB interface was measured to be 5-dB/facet in our case, lower coupling losses are possible. We suspect that the coupling loss in our system is limited by the facet quality and vertical alignment of the photonic wire bond, which can be addressed in future iterations. Currently, vertical alignment of the PWB on the TFLN side is done using mode height simulations combined with reflection from the Si handle, which has errors due to variation in stack height. Using a mode imaging system would remove this uncertainty. Additionally, the loss can be reduced further with additional improvements on the LN coupler^{39,33}.

Our EO comb generator offers an excellent solution for frequency-agile operation on chip with tunable repetition rates and wavelengths, without any limitation from cavity resonances or the gain bandwidth present in microcombs and modelocked lasers. For the single-drive design, shorter pulse durations are achievable by shortening the relative size of the amplitude modulator with respect to the phase modulator system around the limitations of the available RF amplifiers. Additionally, full wavelength agility can be re-

alized on both systems through direct current (DC) tuning of the amplitude modulator quadrature through electro- or thermo-optics and the implementation of a balanced amplitude modulator geometry. Similarly, additional flexibility in the repetition rate can be achieved through the use of thermo-optics to tune the group velocity of the waveguide section in the loop-back design. However, the EO approach still requires Watts-level microwave power, which can be further reduced with implementation with low-loss electrodes and on-chip microwave resonators. Integration with an on-chip laser, either using flip-chip bonding with butt-coupling⁴⁰ or photonic wire bonding could help realize a fully integrated pulse source. Reducing the microwave losses using segmented electrodes can reduce the temporal focal length and increase the EO bandwidth for realizing even shorter pulses⁴¹.

One could also envision an entire nonlinear photonic system where the EO-based pulse source serves as the pump for octave-spanning frequency combs, optical parametric oscillation, and Raman-scattering microscopy. We address one of the potential applications of the pulse system in the next Chapter. The EO comb generator itself can serve as a multi-wavelength source for microwave photonics, telecommunication, and calibration of astronomical spectrographs. Finally, the electro-optic lensing could enable the temporal and spectral shaping of single photon^{42,43}, which could find applications in building the future quantum network⁴⁴.

3

Synchronous-pumped lithium niobate microresonators

3.1 INTRODUCTION

With the rapid development of integrated photonics, there has been significant study of on-chip frequency comb generation based on third-order nonlinearity in microresonators^{45,46} for applications in communications⁹, spectroscopy⁴⁷, optical clocks⁴⁸, and more⁴⁹. On-chip resonators not only offer much smaller form factor than bulk cavities, but also provide additional opportunities to control the generated output light by leveraging dispersion engineering in addition to phase matching. Kerr comb generation has been demonstrated on a wide range of material platforms, including silica⁵⁰, silicon nitride⁵¹, silicon carbide⁵², silicon^{53,54}, aluminum nitride⁵⁵, diamond⁵⁶, and lithium niobate^{57,58}. In most demonstrations, frequency combs are generated using continuous-wave (CW) laser light, requiring high-finesse cavities to achieve the high circulating powers needed for broadband spectrum generation. As a result, many on-chip Kerr combs have repetition rates ranging from 100s of GHz to THz, outside the range of most high-speed microwave electronics. Additionally, comb generation on crystalline material platforms such as lithium niobate usually suffers from strong Raman gain⁵⁹, requiring careful engineering of the free spectral range, pump wavelength, or coupling conditions to mitigate stimulated Raman scattering (SRS)^{60,61,58}.

Alternatively, these nonlinear resonators can be driven using pulsed laser light via synchronous pumping. Here, the free spectral range (FSR) of the cavity and the repetition rate of the pulse train must also be matched in addition to the pump wavelength to the cavity resonance. Synchronous pumping can lower threshold power for parametric oscillation due to high pulse peak power and has been shown to enable broadband on-chip frequency comb generation of much lower (10s of GHz) repetition rates^{62,63,64}. Driving with ultrashort pulses could also be an alternative means of mitigating stimu-

lated Raman scattering in microresonators, by simultaneously increasing the SRS threshold when the pulse duration is shorter than the phonon relaxation time⁶⁵ and decreasing the four-wave mixing (FWM) threshold by seeding the nonlinear process. However, an on-chip pulse-driven comb in which both the pulse generation and comb generation occur on the chip scale remains a large challenge. Unlike bulk oscillators, on-chip resonators have FSRs that are fixed by fabrication with relatively little tunability. Thus, for the realization of a fully chip-scale pulse-driven comb system, it becomes crucial to have an on-chip pulse generator with flexible center frequency and repetition rate. Electro-optic modulation is a well-known method for generating optical pulses from CW laser light with tunable frequency and repetition rate^{23,66}, and has been implemented with discrete-component modulators to generate tunable pulses for synchronous driving of integrated $\chi^{(2)}$ and $\chi^{(3)}$ resonators^{62,63,67,64,68}.

Thin-film lithium niobate (LN) is an attractive material platform boasting excellent electro-optic and nonlinear optic properties ($r_{33} = 30$ pm/V, $d_{33} = 27$ pm/V, $n_2 = 1.8 \times 10^{-19}$ m²/W)⁷. The periodic poling capability of the LN platform also make it appealing for quasi-phase matching of on-chip $\chi^{(2)}$ OPOs or cascaded $\chi^{(2)}$ process for higher effective $\chi^{(3)}$ nonlinearity. Due to these excellent material properties, broadband spectrum generation has been of particular interest on the thin-film LN platform. Previously, ultrashort on-chip pulse generation², CW- and synchronous-pumped periodically poled $\chi^{(2)}$ OPO^{68,69,70,71}, and Kerr comb generation via $\chi^{(3)}$ OPO^{57,58,60,72,73} has been demonstrated on TFLN. However, there is currently no work combining on-chip pulse generation with synchronous pumping of resonators using this or any other platform.

In this work, we demonstrate a broad 30-GHz Kerr comb source by leveraging $\chi^{(3)}$ nonlinearity of dispersion-engineered high- Q thin-film LN microresonator that is synchronously pumped using an electro-optic femtosecond pulse source realized via on-

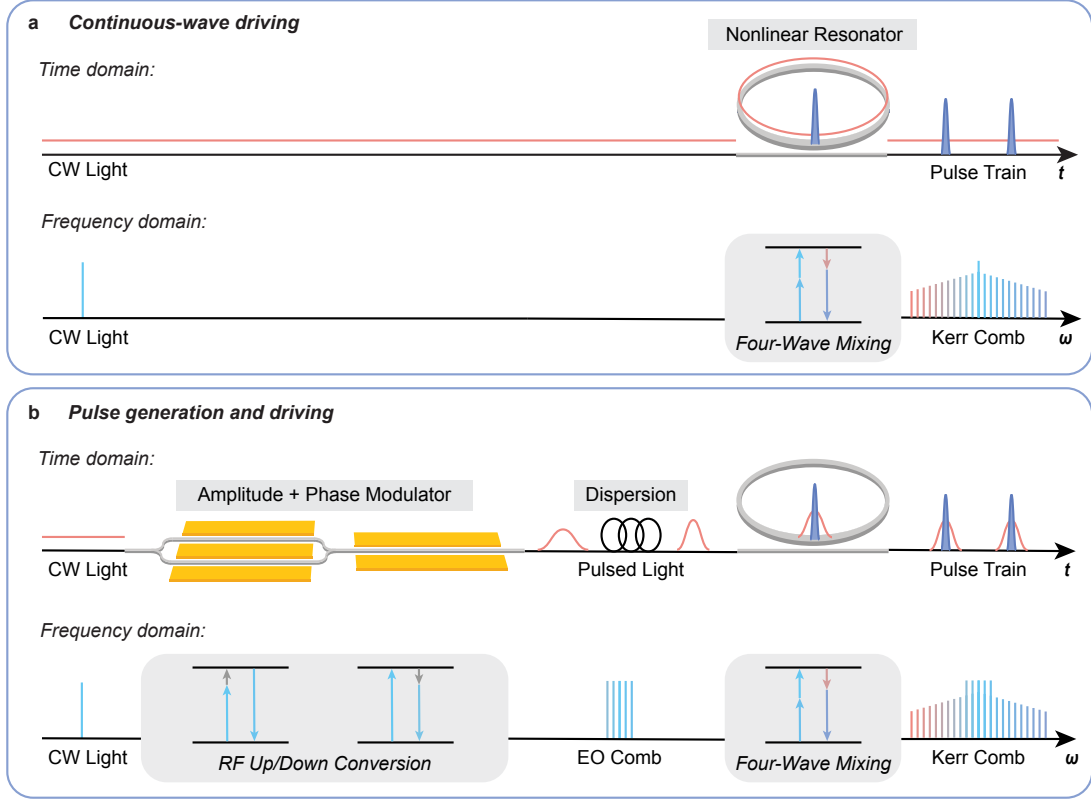


Figure 3.1: Visualization and concept of pulse and frequency comb generation. **a**, CW scheme in time and frequency domain; CW light is tuned to the resonant frequency of a microresonator and undergoes four-wave mixing process. The generated frequency comb pulse train has a repetition rate equal to the free spectral range of the microresonator. **b**, Pulse generation and synchronous pumping scheme in time and frequency domain; a pulse train is formed through amplitude and phase modulation of CW light and compressed in a dispersive medium before resonantly driving the microresonator.

chip electro-optic modulation on thin-film LN with pulse compression in optical fiber². The on-chip modulation not only reduces the footprint of our pulsed comb generation scheme, but also gives flexibility of the pulse repetition rate and center wavelength while maintaining small form factor. We find that by employing both synchronous pumping and a microresonator with normal dispersion, we see pump-to-comb efficiency higher than that of traditional CW-pumped Kerr comb generation schemes, which is consis-

tent with previous works focused on studying efficiency in synchronous-pumped and normally dispersive comb generation schemes^{63,74}. We also find that pulse-pumped operation significantly mitigates Raman scattering process that has plagued LN^{60,59} and other crystalline resonators⁶¹. Notably, stimulated Raman scattering is well mitigated without any engineering of the free spectral range, pump wavelength, or resonator coupling rates which have been employed in previous works^{60,61,58,75}. The final frequency comb spectrum spans 400 nm over 1400 comb lines (over 2/5 of an octave above the noise floor of our spectrum analyzer, -70 dBm in our case) with a pump-to-comb conversion efficiency of over 10%.

3.2 RESULTS

Our frequency comb is based on $\chi^{(3)}$ four-wave mixing process, which is the dominant process used in traditional microresonator Kerr comb sources^{45,46,17}. Fig. 3.1a illustrates the time- and frequency-domain behavior of comb generation under a typical CW-pumped scheme. In Fig. 3.1b, we show the comb generation scheme with synchronous pulsed pumping. Pulses are generated from CW light using an electro-optic time lens system through amplitude and phase modulation and dispersion, before being sent to the nonlinear resonator.

Our pulse generator chip is fabricated as described in Ref.² (Chapter 2) on a 600 nm X-cut LN substrate, consisting of one amplitude and one phase modulator with 2-cm long travelling-wave electrodes. The fiber-to-fiber insertion loss of this chip is 9 dB (3 dB/facet for each facet and 3 dB loss from device operation). The pulse generator chip features a recycled-PM design to lower the V_π of the phase modulator. As such, the optimal operating condition is achieved when the optical signal is in phase with the

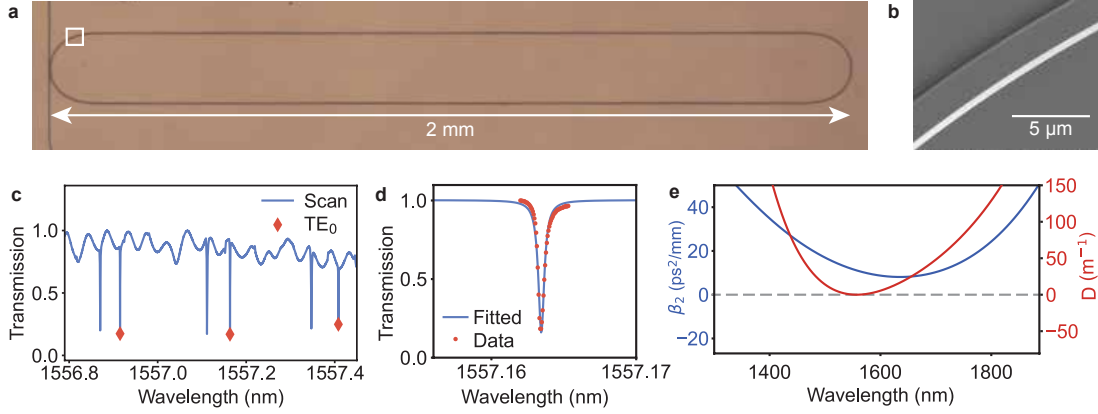


Figure 3.2: Device images and parameters. **a**, Optical microscope image of racetrack resonator showing 2 mm footprint size. **b**, Scanning electron microscope (SEM) image of waveguide sidewalls in the bending region of the racetrack resonator. **c**, Wide resonance scan (blue) showing FSR of 30.14 GHz for the fundamental TE mode (indicated in red) around 1557 nm. **d**, Resonance linewidth fitting of fundamental TE mode, showing loaded optical Q -factor of 3.2 million and linewidth of ~ 60 MHz. **e**, Dispersion profiles of fabricated geometry with pumping at the resonance frequency.

microwave drive and the phase modulator exhibits resonant behavior, hitting the lowest V_π every 2.8 GHz. One such resonant operating frequency occurs at 30.135 GHz. When operated at this RF frequency, the generated pulses reach maximum compression after passing through 59-m of single-mode fiber and achieve a pulse duration of 526 fs² (Chapter 2).

The resonator (Fig. 3.2a,b) is fabricated on Z-cut LN in order to avoid TE/TM mode crossing in the racetrack bends caused by material birefringence. The racetrack is designed so that the FSR of its fundamental TE mode is within the resonant operating range of the EO pulse generator, measured to be 30.14 GHz (Fig. 3.2c). After the optical layers are defined, the device then undergoes a 2-hour annealing process at atmospheric pressure with O₂ ambient, intended to decrease the thin film material absorption limit and increase the optical quality factors of our resonator device¹. Accordingly, the waveguides are designed to be wide and multimode (2 μm) so that the optical quality fac-

tor is not limited by fabrication-induced sidewall scattering. Importantly, the annealing process slightly changes the material dispersion, pulling the waveguide dispersion from the anomalous to the normal regime (Figure 3.3). The fundamental TE mode of the resonator is near-critically coupled and has a loaded optical linewidth of 60 MHz and optical Q -factor of 3.2 million (Fig. 3.2d). We note a near 5x improvement in the Q of this device compared to our previous 250 GHz Kerr comb devices⁷², made possible by this annealing process. The calculated dispersion profiles for our waveguide geometry are given in Fig. 3.2e. The insertion loss of this chip is 11 dB (5.5 dB/facet for each facet and negligible on-chip loss).

For the measurement (Fig. 3.5a), light is coupled between the two LN chips using lensed fiber and single-mode fiber (SMF-28). The total length of SMF between the two devices is carefully calibrated to reach maximum pulse compression. An erbium-doped fiber amplifier (EDFA) is used to amplify the pulse between the two chips. The center frequency of the pulse source is tuned to match the resonance frequency, and its repetition rate is fine-tuned to precisely match the FSR of the Kerr ring with 1 MHz accuracy.

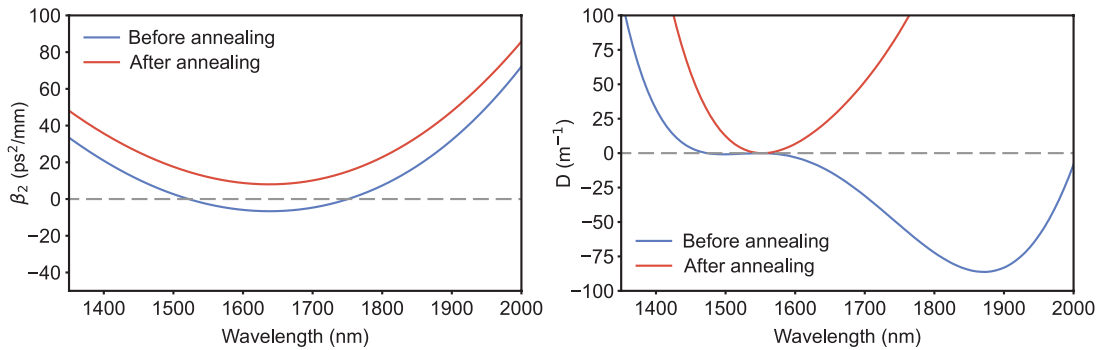


Figure 3.3: Effect of annealing on dispersion. Group velocity dispersion (left) and dispersion operator (right) curves of resonator waveguide geometry simulated using thin-film index data collected before and after high temperature furnace annealing. The annealing process pulls the dispersion from the anomalous into normal regime. Index measurements performed by J.A. Woollam.

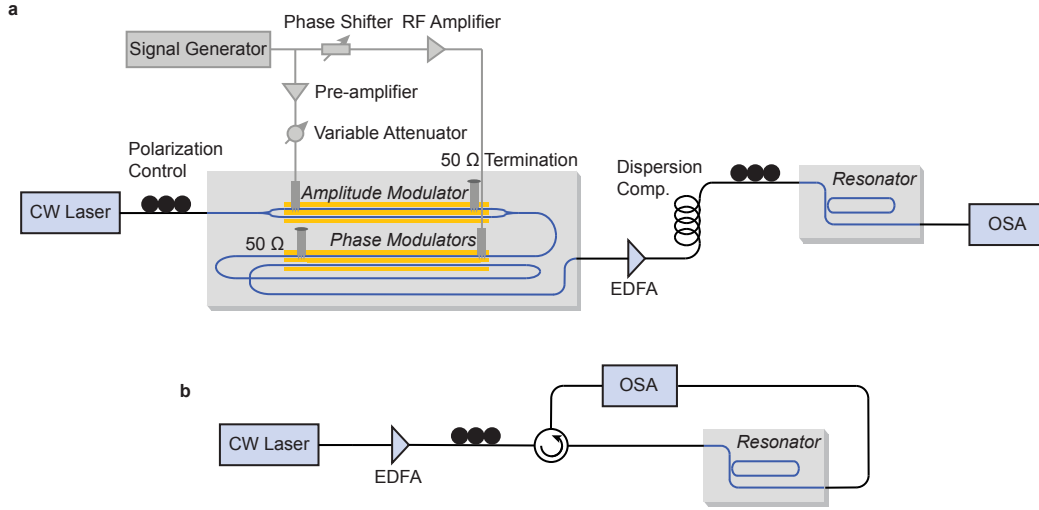


Figure 3.4: Full experimental setup. **a**, Setup schematic for pulse generation and driving across two LN photonic chips; **b**, Setup for testing spectrum under CW pumping. A circulator is used to monitor the spectrum in the backward direction. EDFA: erbium doped fiber amplifier; OSA: optical spectrum analyzer.

We discuss the sensitivity of the resulting spectrum to repetition rate detuning later in the Chapter.

For generation of the frequency comb spectrum, the EDFA gain is set to 12 dB. The gain is largely used to compensate for the chip-chip coupling loss (3 dB at the output of EO pulse generator, and 5.5 dB at the input of resonator) and supplies an additional 3.5 dB of gain to the system. The generated comb spectrum is given in Fig. 3.5b. The asymmetric profile of the frequency comb reflects the dispersion operator curve given in Fig. 3.2e, which biases the comb in the red direction with respect to the pump. The final frequency comb is achieved using ~ 12.5 mW of average on-chip power, corresponding to 0.8 W of peak power and 0.45 pJ pulse energy, in the bus waveguide, and spans $2/5$ of an octave over 1500 comb lines. The conversion efficiency of our frequency comb, which we define as the total power of the frequency comb divided by the average power in the bus waveguide, is measured to be $\sim 10\%$. The high efficiency is enabled by both

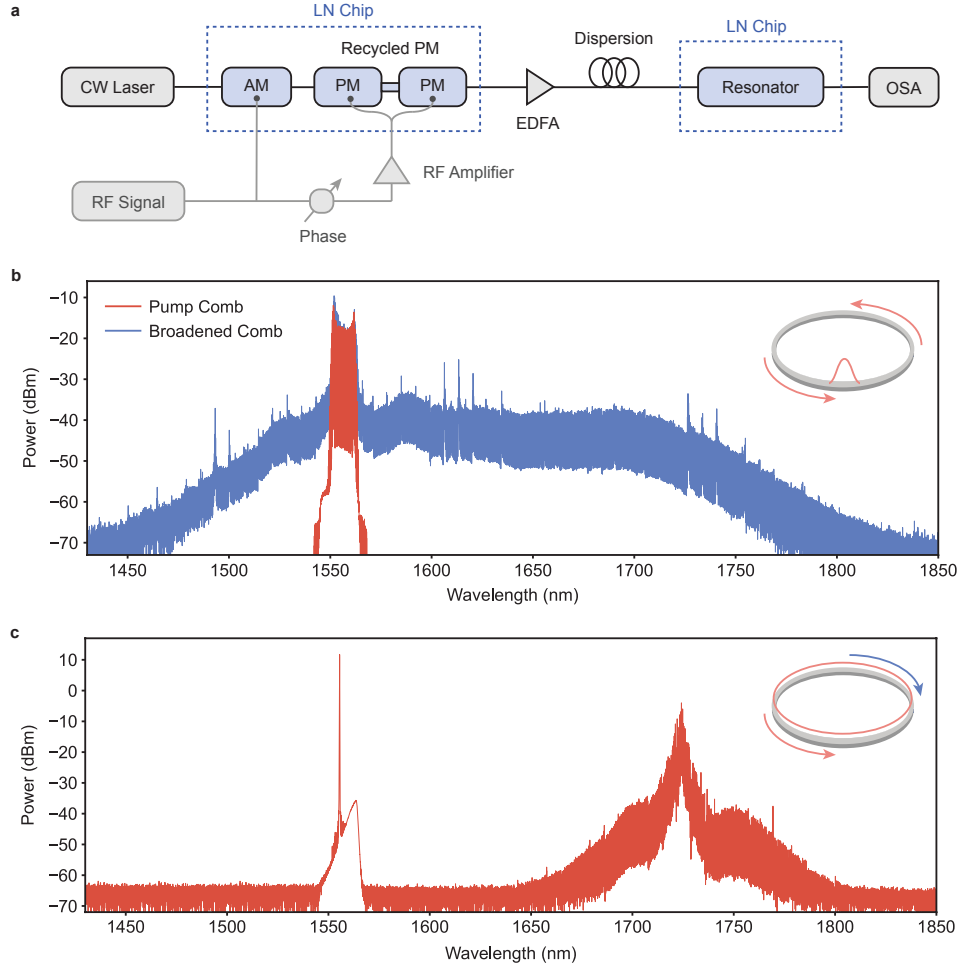


Figure 3.5: Frequency comb generation in pulse-driven vs. CW-driven case. **a**, Measurement scheme for on-chip pulse-pumped frequency comb. CW light is coupled onto the first LN chip for pulse generation. The outgoing pulse train is amplified using an EDFA and compressed in single mode fiber. The pulsed light is then coupled to the second LN chip for spectral broadening. **b**, Pulse-pumped spectrum, generated with 12.5 mW average on-chip power. The spectrum spans 2/5 octave. Red trace: electro-optic frequency comb produced by the pulse generator chip, measured before coupling into the resonator chip. Blue trace: frequency comb spectrum produced via pulse-pumping, measured at the resonator chip output. The power level of the red trace is adjusted to account for chip-chip coupling loss, so that the relative power of the pump comb and broadened comb match appropriately. Inset: visualization of light circulation in microresonator for synchronous driving. SRS is avoided and FWM co-propagates with the pump. **c**, CW-pumped spectrum, generated with 200 mW on-chip power, monitored in the backward direction. A strong Raman peak is observed at 1720 nm. Inset: visualization of light circulation. SRS threshold is lower than that for FWM, and so both SRS and subsequent comb lines generated through FWM counter-propagate with the pump.

synchronous driving as well as normally dispersive operation⁶³. To date, this is the highest reported pump-to-comb conversion efficiency of Kerr comb on thin-film LN. Key features of the comb, including bandwidth, center wavelength, and repetition rate, are compared against other LN Kerr combs in Table 3.1.

Importantly, synchronous driving overcomes strong stimulated Raman scattering (SRS) in the material, another limitation in LN-based frequency combs and $\chi^{(3)}$ OPOs based on crystalline materials⁵⁹. SRS has been observed in both X and Z crystal cuts of thin-film lithium niobate and extensively studied along both crystal orientations in X-cut LN⁵⁹, which finds that light generated through SRS in LN scatters in the backward direction to the optical pump direction. In our pulsed scheme, we excite the resonator synchronously with an EO-generated comb, thus seeding and lowering the power threshold of the four-wave mixing process. This favors the Kerr comb generation in the forward direction over strong Raman scattering followed by Kerr comb generation in the backward direction. To confirm this experimentally, the LN resonator is also excited using a high-power CW source. Fig. 3.5c shows the generated comb spectrum at 200 mW of on-chip power in the CW driving case, monitored in the backward direction. We see a strong peak in the comb spectrum at ~ 1725 nm, corresponding to a Raman frequency shift of 625 cm^{-1} and the resonance of the $E(\text{TO})_9$ phonon branch^{76,77,78,79,80}. Moreover, the microcomb that forms around the peak at 1725 nm is significantly stronger than the comb lines around the pump, similar to previously observed Raman frequency combs in LN and further supporting the observance of Raman scattering under this generation scheme. To our knowledge, this is the first demonstration of Raman suppression through pulsed pumping.

The effect of repetition rate detuning on the spectrum is studied from 0 to 3 MHz detuning at 1 MHz steps, shown in Figure 3.6. The effect on the spectrum is visible at 2

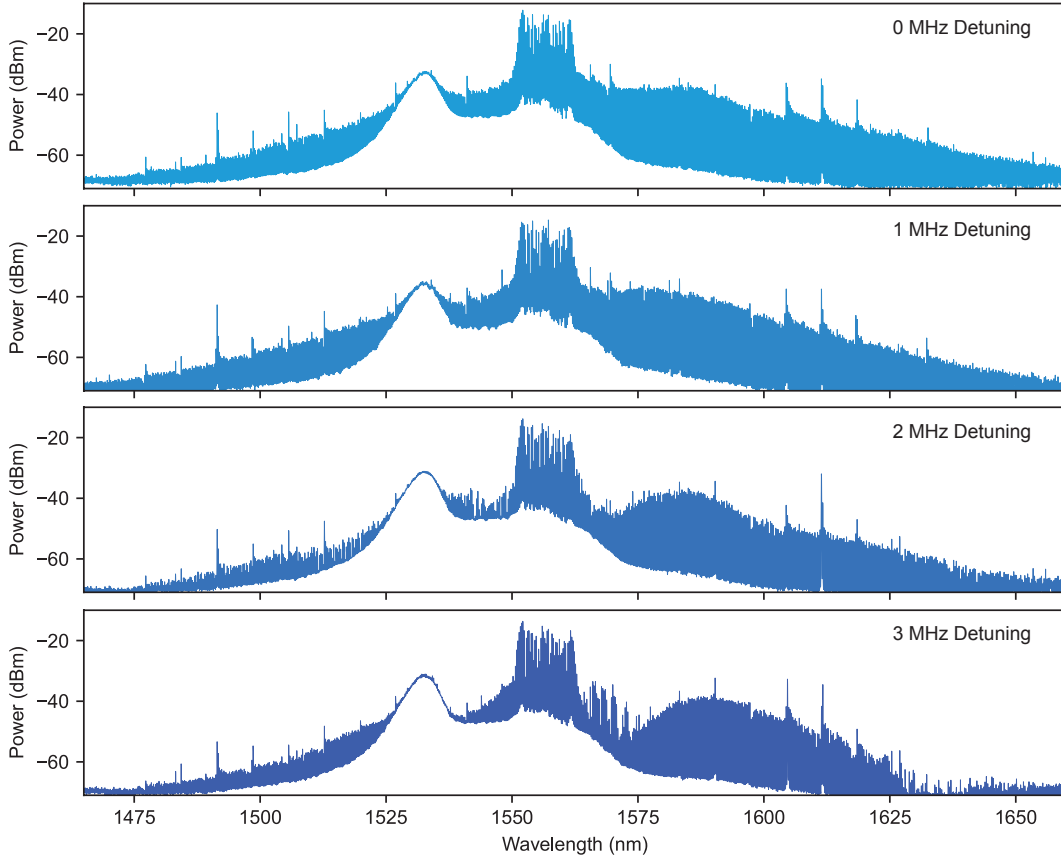


Figure 3.6: Optical spectrum as a function of repetition rate detuning. Broadened comb spectra collected at 0-3 MHz RF detuning of the pulse source repetition rate from the Kerr resonator's free spectral range with constant average pump power. The spectrum is noticeably affected for >2 MHz detuning.

MHz detuning and becomes significant at 3 MHz detuning, at which the stability of the spectrum is also greatly sacrificed. We note that the hump in the OSA floor around 1530 nm is due to unfiltered amplified spontaneous emission (ASE) noise from the EDFA source used between the setups. From these results, we conclude that 1 MHz repetition rate sensitivity is sufficient for our scheme, although detuning may more clearly affect broadband spectrum at higher optical pump powers.

Finally, we observe the broadening behavior of our system at low and varying opti-

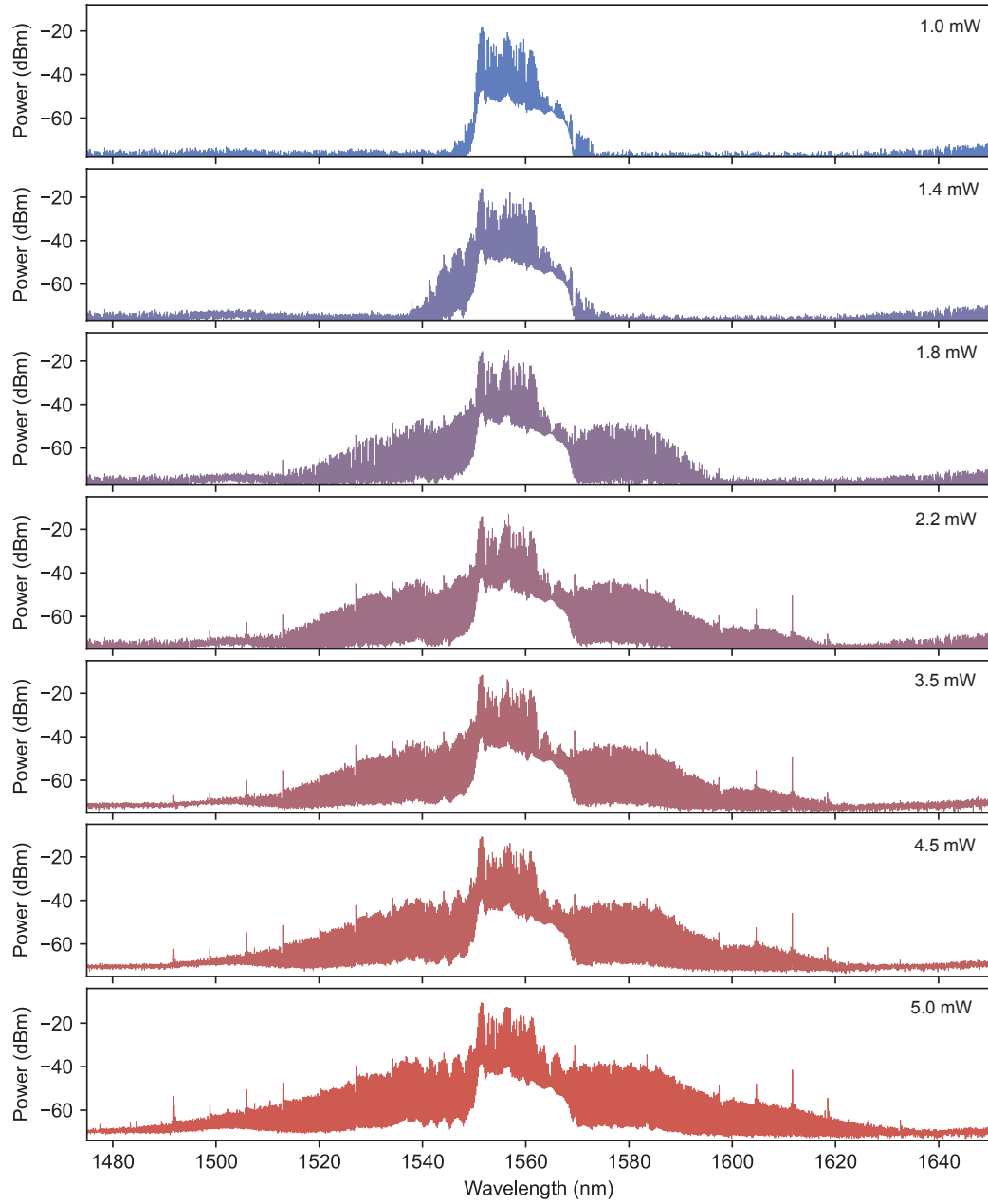


Figure 3.7: Pulse-pumped spectrum for different optical pump powers. Generated optical spectra for average on-chip powers ranging from 1.0 to 5.0 mW. The power is adjusted by tuning the current on the EDFA. As expected, we see broader comb spectrum for increasing optical power.

cal powers by tuning the EDFA gain to change the on-chip power of the resonator chip, which allows for control of the power without any effect of the optical dispersion between the chips. The output spectrum from the resonator is measured for a range of average on-chip optical powers from 1.0 to 5.0 mW (corresponding to 64 mW to 320 mW of peak power and 36 fJ to 0.18 pJ of pulse energy). As optical power increases, the spectrum broadens around the pump EO comb due to self-phase modulation, as shown in Fig. 3.7. We note that the characteristics of the spectra in Fig. 3.6 and Fig. 3.7 differ slightly from the main spectrum given in Fig. 3.5b. This is posited to be due to aging of the resonator sample over long time periods between measurements, which cannot be completely recovered through reannealing of the sample.

3.3 CONCLUSION

In conclusion, we show a synchronously driven LN resonator and resonant Kerr broadening of an electro-optic optical pulse source on thin-film lithium niobate, generating a 30-GHz comb spanning 400 nm in bandwidth. Additionally, the seeded four-wave mixing process of the pulsed-pumping scheme greatly reduces the threshold power for comb generation and bypasses the Raman scattering process commonly observed in thin-film LN comb generation. The mitigation of stimulated Raman scattering allows for generation of broadband LN optical frequency combs at lower repetition rates. For applications requiring high conversion efficiencies, pump-to-comb conversion efficiency can be improved further through operating the resonator in the overcoupled regime⁶³ or through the introduction of an auxiliary resonator⁸¹. Careful study of the interplay between stimulated Raman scattering and soliton formation under pulsed pumping could be explored further for even broader bandwidth combs in the normally disper-

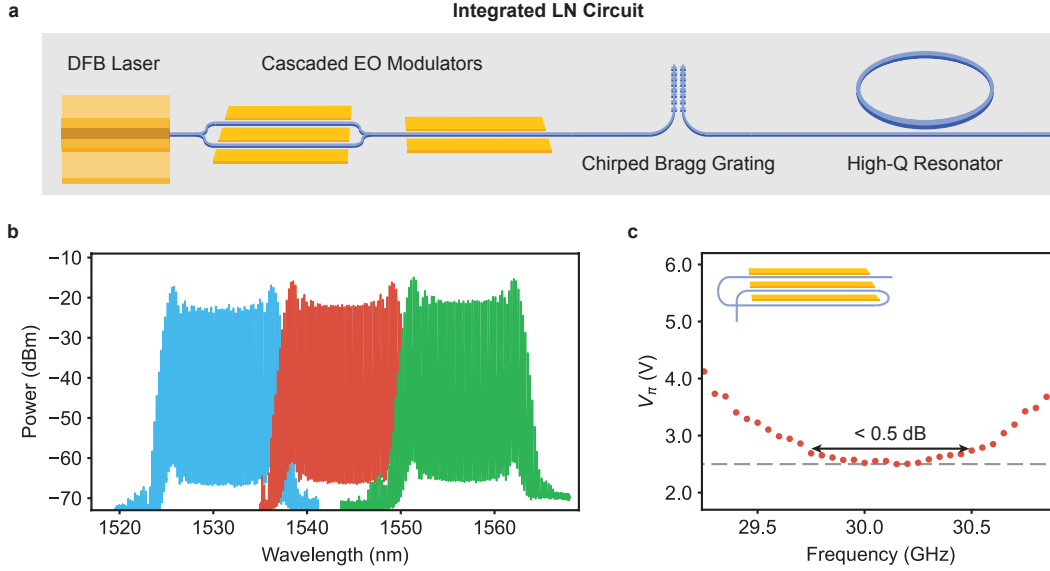


Figure 3.8: Outlook of synchronously driven resonators on thin-film LN. **a**, Schematic for integrated on-chip pulse-driven comb generation, which includes integrated on-chip DFB laser source, amplitude and phase modulators for pulse generation, on-chip dispersion management, and dispersion engineered $\chi^{(2)}$ or $\chi^{(3)}$ resonators. For future integrated schemes, fabrication developments can be made on modulator side (Z-cut modulators) or on the resonator side (dispersion engineering with respect to material birefringence) for compatibility of integration. **b**, Pulse generation at different center wavelengths. The pulse source has flexible wavelength tunability and can be precisely matched to resonator frequency without large need for thermal tuning. **c**, V_π of the recycled phase modulator (layout shown in inset) utilized in the pulse generator design. The waveguide wraps around before passing through modulation again, allowing the phase modulator to be driven with a single GSG electrode and microwave source but imparting a periodic resonance condition on the V_π . The V_π is measured around the resonance condition 30.135 GHz (2.5 V). Within the range that we can fabricate reproducibly, indicated by black arrow, there is < 0.5 dB change in the V_π and minor effect on the pulse generation efficiency.

sive regime^{82,83}.

We believe that the thin-film LN platform is a promising host for on-chip nonlinear optics beyond what has been demonstrated in this work. The poling capability of LN makes it excellent candidate for $\chi^{(2)}$ pulsed optical parametric oscillators, and $\chi^{(3)}$ Kerr combs operating in the anomalous dispersion regime for octave spanning comb generation are also a candidate for synchronous driving. While our current measurement was

done across two separate chips, we look forward to future compatibility of this scheme for full integration (Fig. 3.8a). This would not only greatly reduce the footprint of the comb generator, but also allow us to avoid on- and off-chip coupling losses between the two chips. Combined with higher pump-power lasers, this loss reduction would also eliminate the need for an intra-system amplifier. Platform compatibility between the two chips should be possible with careful design and fabrication development. For example, the electro-optic components can be made on Z-cut LN with development of high efficiency Z-cut LN modulators⁸⁴. On the other hand, the resonator chip could be fabricated on X-cut LN through careful engineering of the racetrack bends to avoid mode crossing caused by material birefringence⁸⁵. If the dispersion of the wire bond is taken into account, photonic wire bonding could also bridge these devices.

While fabrication across two chips allows flexibility to sweep a range of device sizes to precisely match the optimal repetition rate of the pulse generator, the on-chip pulsed resonator scheme should be robust to fabrication should the two components be combined. The center wavelength of the pulse source is flexible (Fig. 3.8b), as the recycled phase modulator is not optically resonant and the quadrature point of the amplitude modulator could be tuned with DC bias (electro-optics) or heaters (thermo-optics). Additionally, we conservatively estimate that we can target the resonance FSR within 300 MHz of the intended design given fabrication variation and error. We find that within this RF frequency range, there is negligible ($\sim 1\%$) change in the V_π of the recycled phase modulator (Fig. 3.8c). Moreover, the resonant frequency of the recycling phase modulator could be tuned beyond this point using thermo-optic heaters on the loop-back waveguides to change the group refractive index of the optical path between phase modulators.

An outstanding bottleneck to a fully integrated scheme not demonstrated in this

work is the need for on-chip dispersion compensation to compress the pulses generated by the modulator system. One approach to on-chip dispersion is through the use of a long waveguide with large engineered dispersion. However, because of the waveguide geometries usually needed to achieve high dispersion, this approach is limited largely by the propagation loss in these waveguides, currently much higher than the state-of-the-art TFLN propagation loss². Waveguide gratings are a more commonly used method for dispersion compensation on integrated photonics platforms and can achieve large dispersion factors over a small propagation distance^{86,87}; they have been implemented on the TFLN platform with low insertion loss and small footprint². Because these gratings operate in the reflection mode, most implementations of on-chip dispersion gratings require optical circulators or beam splitters. Recently, circulator-free dispersion schemes based on coupled waveguide gratings⁸⁸ (used to illustrate a potential approach to on-chip dispersion in Fig. 3.8a) and mode de-multiplexers⁸⁹ have been proposed and demonstrated. We discuss some results and provide an outlook for on-chip dispersion in Chapter 6.

Recently, in addition to pulse generation, compression, and nonlinear broadening, efforts in integrating high power lasers have been demonstrated on thin-film lithium niobate⁴⁰. We look forward to a fully integrated scheme that combines not only the pulse generator and resonator, but also on-chip pulse compression and laser integration. We envision a scheme that combines all these components together, which could be possible with further fabrication development, improvements in loss, and larger scale processing of lithium niobate components.

Reference	Repetition rate (GHz)	Center wavelength (nm)	On-chip power (mW)	Bandwidth (octave)	Pump scheme	Dispersion	Mitigated SRS
This work	30	1557	12.5	0.4	Pulsed	Normal	Yes
Ref. ⁷²	250	1550	300	0.65	CW	Anomalous	No
Ref. ⁵⁸	335	1550	240	0.8	CW	Anomalous	Yes
Ref. ⁶⁰	200	2000	90	0.3 [†]	CW	Anomalous	Yes
Ref. ⁵⁷	200	1550	33	0.25 [†]	CW	Anomalous	Yes
Ref. ⁷³	200	1550	600	1.15	CW	Anomalous	Yes
Ref. ⁷⁵	45	1550	396	0.35 [†]	CW	Anomalous	Yes
	20	1550	282	0.17 [†]	CW	Anomalous	Yes

Table 3.1: Performance comparison of our pulse-pumped racetrack with other LN Kerr combs [†]: extracted.

4

High performance
electro-optic-based optical isolators

4.1 INTRODUCTION

Recent decades have witnessed the rapid advancement of integrated photonic technology for applications including optical communications^{90,9}, microwave photonics⁹¹, computing^{92,93}, optical atomic clock⁹⁴, quantum information processing⁹⁵, light detection and ranging⁹⁶ and sensing^{97,98}. In parallel, Hertz-linewidth semiconductor lasers have been recently demonstrated on chip⁹⁹, showing promise as photonic engines for future large-scale photonic processors. The urgent need to combine the narrow-linewidth semiconductor laser with the functional photonic modules demands a high-performance reliable integrated optical isolator.

Various approaches for realizing integrated on-chip optical isolators have been demonstrated, including magneto-optics^{100,101,102,103,104}, electro-optics^{105,106,107,108,109,110,111}, acousto-optics^{112,113,114} as well as optical nonlinearities^{115,116,117}. Magneto-optical isolators feature robust operation and broad bandwidth but face challenges of material integration or interfacing with atomic systems which require optical readout and are highly sensitive to magnetic interference¹¹⁸. Nonlinear optical approaches either require additional optical pump lasers and filtering^{115,119} to remove the pump or show power-dependent isolation^{116,117}. Recent demonstrations based on electro-optics and acousto-optics require narrowband optical cavities^{111,113,114} to enhance non-reciprocity, engineered phase-matching of two optical modes^{113,114} or hybrid integration of piezoelectric material and silicon photonics^{112,114}. These requirements limit the operational bandwidth and system scalability. Electro-optic (EO) waveguide-based device has immense potential to overcome these limitations. However, previous demonstrations based on silicon or III-V EO modulators^{108,109,110} have suffered from high insertion loss and low isolation ratio due to the tradeoff between phase modulation and absorption. There-

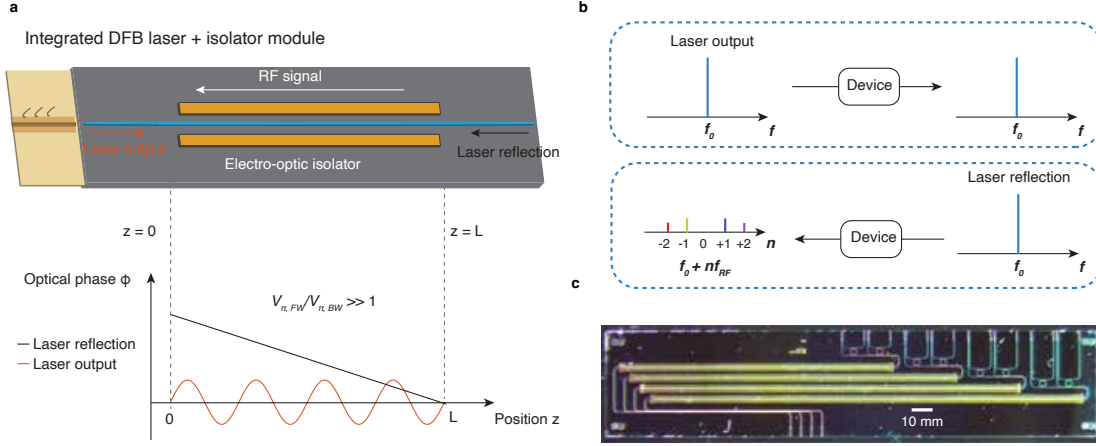


Figure 4.1: Concept of an integrated electro-optic isolator. a, The schematic of an integrated laser system consisting of a III-V-based distributed feedback laser (DFB) followed by an electro-optic isolator on thin film lithium niobate (LN). The microwave signal is counter-propagating with respect to the DFB laser output. Under single-tone travelling-wave-based microwave drive at f_{RF} , the accumulated optical phase of the laser output after passing a phase modulator of length L is equal to zero when $f_{RF} = mc/(2Ln_{opt})$ where m is an integer number and c and n_{opt} are light velocity and optical group index, respectively. Any reflected laser light which propagates in the backward direction will undergo efficient phase modulation, which results in $V_{\pi,FW}/V_{\pi,BW} \gg 1$. FW: forward; BW: backward. b, The illustration of the travelling-wave EO isolator in frequency domain. The laser output at frequency f_0 is fully transmitted since there is no added phase modulation. All the energy of the reflection signal at frequency f_0 is distributed to the optical sidebands at frequency equal to $f_0 + nf_{RF}$, where n is an integer number. The energy at frequency f_0 can be fully depleted when the modulation depth $\beta \sim 0.765\pi$ satisfying $J_0(\beta) = 0$. c, The optical micrograph of the EO isolator chip on thin film LN, which consists of four individual devices with varying modulation length L of 1.25, 1.5, 1.75 and 2 cm.

fore, the realization of a high performance integrated isolator with low insertion loss, high isolation ratio, low power consumption, broad bandwidth, and high fabrication tolerance remains challenging.

Thin film (TF) lithium niobate (LN) is well positioned to address this challenge, as it simultaneously supports phase-only modulation and low propagation loss. Recent advancement of the integrated LN platform has enabled ultra-low-loss travelling-wave EO modulators operating at CMOS-compatible voltages and high EO bandwidth⁶. Other excellent photonic properties⁷, including wide transparency window, large second or-

der and Kerr nonlinearity and piezoelectric coefficient, makes LN a highly compelling choice for realizing future coherent optical systems.

Here, we demonstrate an integrated isolator based on a travelling-wave-based phase modulator realized in TFLN, and use it to protect a distributed feedback (DFB) laser edge-coupled to the TFLN chip (Figure 4.1). The phase modulator is driven with a single-tone sinusoidal microwave field, at the frequency of f_{RF} , propagating in the direction opposite from the optical signal. The total phase accumulation $\phi(f_{RF}, t)$ is given by the expression $\frac{V_{pk}}{V_{\pi L}} \pi \int_0^L [e^{-\alpha z/2} \cos [2\pi f_{RF}(\frac{z}{c} n_{opt} \pm \frac{z}{c} n_{RF} - t)]] dz$, where V_{pk} , V_{π} , α , n_{opt} , n_{RF} , and c are peak voltage, half-wave voltage, microwave power attenuation constant, optical group index, microwave phase index and light velocity in vacuum, respectively. + and - are chosen to indicate situations when the optical and microwave fields are counter-propagating and co-propagating, respectively. Since the laser output is counter-propagating with the microwave signal, the phase accumulation is a periodic function of the propagating distance in the modulator. Assuming $\alpha = 0$, the phase accumulation equals zero when the microwave driving frequency $f_{RF} = mc/(2Ln_{opt})$ where m is an integer number. In this case, the laser light is not modulated and is fully transmitted in the forward direction. On the other hand, reflected light travelling in the backward direction co-propagates with the microwave field, resulting in a linearly increasing phase accumulation for any given t . The reflected light at the input frequency f_0 can be fully depleted when the modulation depth $\beta \sim 0.765\pi$, satisfying $J_0(\beta) = 0$ where J_0 is the 0^{th} order Bessel function of the first kind (see Appendix Figure A.1). In this case, all the optical power is distributed to frequencies equal to $f_0 + nf_{RF}$, where n is an integer (Fig. 4.1b).

Figure 4.1c shows an optical micrograph of the LN chip with modulator lengths of 1.25 cm to 2 cm at a step size of 0.25 cm. The chip is fabricated on a 600-nm X-

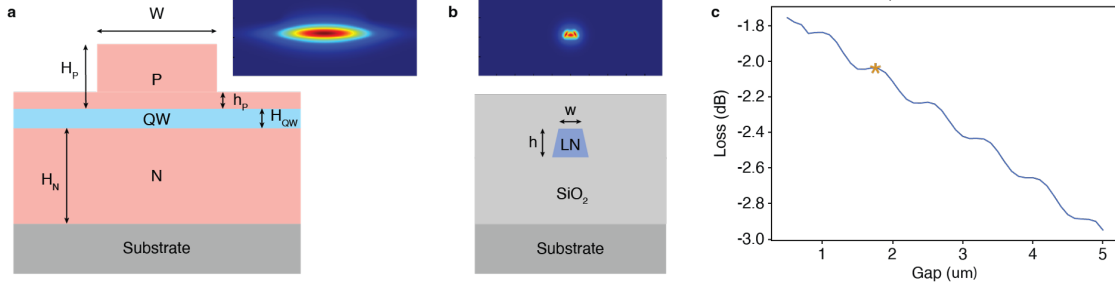


Figure 4.2: Coupling between DFB laser and LN chip. **a**, Quantum well waveguide structure and (inset) simulated output mode of the III-V DFB laser: $W = 5 \mu\text{m}$, $H_P = 2.5 \mu\text{m}$, $h_p = 500 \text{ nm}$, $H_{QW} = 450 \text{ nm}$, $H_N = 4 \mu\text{m}$. **b**, Structure of the LN mode converter at the coupling facet: $h = 300 \text{ nm}$, $w = 250 \text{ nm}$. **c**, Simulation of the coupling efficiency between the DFB laser mode and LN facet. An anti-reflective coating is added to the laser structure, as in experiment, to minimize reflection between the laser and the air gap. The minimum coupling gap achievable in our experiment is $\sim 2 \mu\text{m}$, corresponding to about 2-dB coupling loss (indicated by the star). This loss is further verified experimentally on other LN chips.

cut LN wafer with an etch depth of 320 nm. The group velocity of the light matches the phase velocity of the microwave ($n_{opt} = n_{RF} = 2.26$) to ensure the efficient phase modulation at microwave driving frequencies at 10s of GHz. The chip has both input and output bilayer taper couplers³² for a low-loss interface with DFB laser (2-dB/facet insertion loss, see Figure 4.2) and lensed fiber (3-dB/facet insertion loss). Our device utilizes a simple optical structure of a single modulated waveguide and operates with a single-frequency microwave drive on the condition that the microwave power is tuned to the pump depletion point.

4.2 RESULTS

We show that the integrated phase modulator allows for high performance optical isolation at the laser frequency. A detailed measurement scheme describing the methods for all the results in this chapter is given in Figure 4.3. First, we characterize the isolation ratio and optical bandwidth using a tunable continuous-wave laser and a phase modulator

of 1.75 cm in length. The isolation ratio is defined as the ratio of optical power propagating in the forward and backward direction at the laser frequency f_0 . Figure 4.4a plots the optical spectrum in both forward and backward directions using $f_{RF} = 26.5$ GHz ($m = 7$). In the forward direction, the laser light (wavelength of 1553.2 nm) is nearly completely transmitted, with an on-chip insertion loss of 0.5 dB. In the backward direction, however, most of the light is coupled to sidebands, at a frequency spacing equal to harmonics of the f_{RF} ($= 26.5$ GHz). Next, we demonstrate the ability of our isolator to operate at different microwave frequencies, f_{RF} of 26.5 and 10.4 GHz (Figure 4.4b,c). At 26.5 GHz, an isolation ratio of 48 dB is achieved at the pump wavelength, using 21 dBm of microwave power, corresponding to $\beta \sim 0.765\pi$ with a $V_\pi = 4.7$ V. A microwave power stability of $< 1\%$ is required for achieving > 44 dB isolation ratio. Since the microwave power scales as $P \sim (0.0765 V_\pi)^2$, power reductions are possible by reducing V_π . At 10.4 GHz, the isolation ratio is estimated to be > 44 dB (here, the measurement is

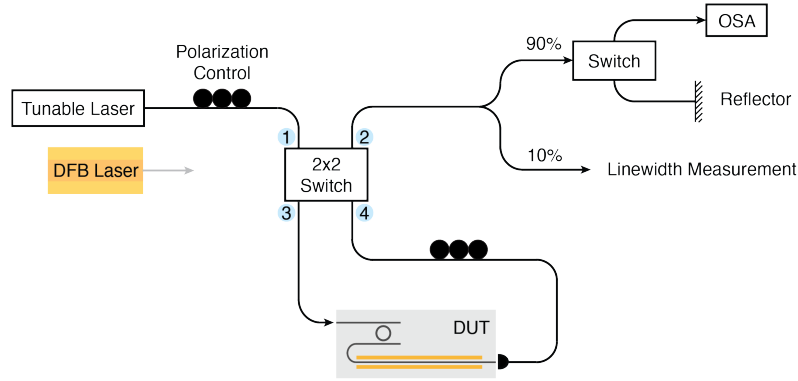


Figure 4.3: Full measurement setup for the LN electro-optic isolator. Forward and backward spectra of the isolator are monitored using a tunable laser and 2x2 switch (forward: 1 $\rightarrow$ 3, backward: 1 $\rightarrow$ 4) to couple light onto the chip. Polarization controllers are used in both directions to couple light into the transverse-electric (TE) waveguide mode. Spectra are measured on an OSA (optical spectrum analyzer). For III-V laser operation, the DFB laser is butt-coupled directly to the isolator chip, foregoing the switch. The output is then split between a retroreflector and linewidth measurement to monitor the behavior of the DFB laser with and without isolation. The DUT (device under test) includes both devices with and without the add-drop filter.

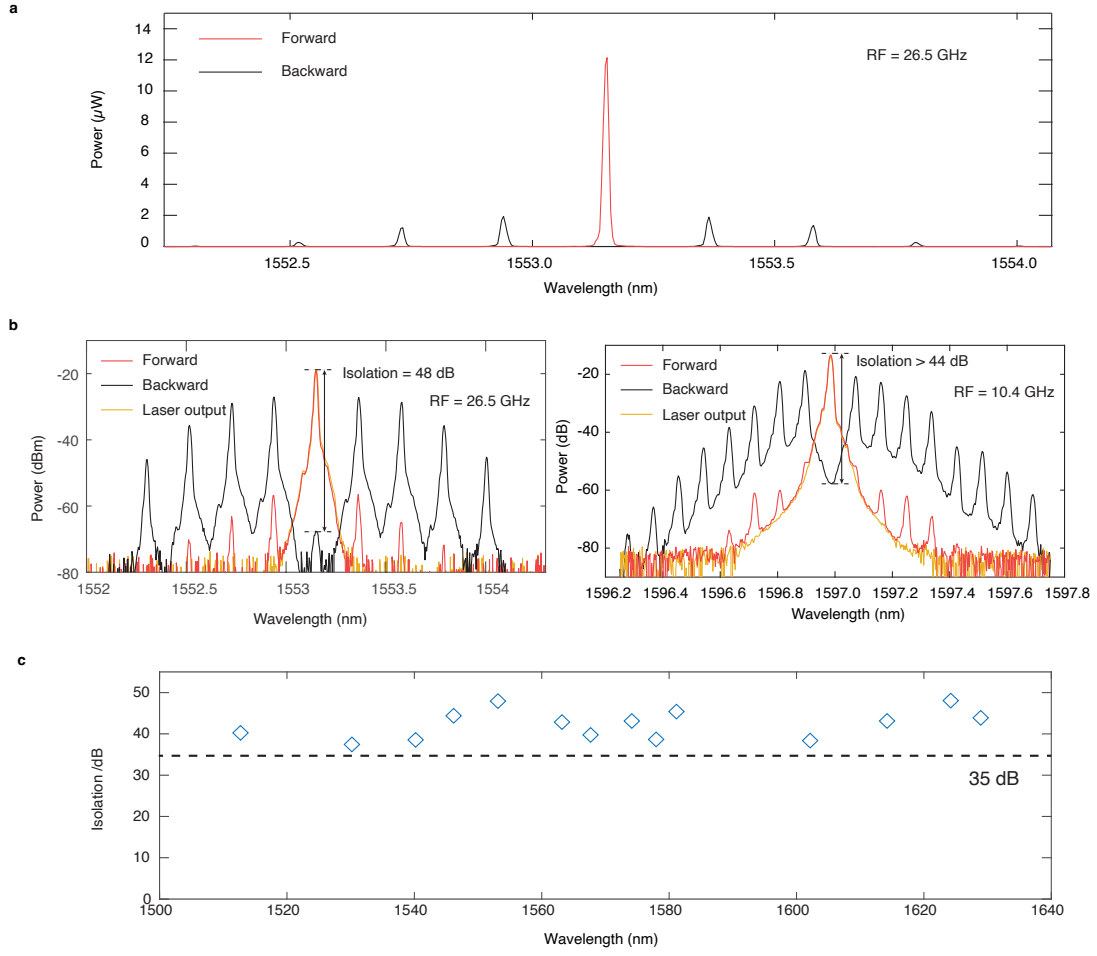


Figure 4.4: Characterization of isolation and optical bandwidth of the electro-optic isolator. **a**, The optical spectrum of the forward and backward optical signal when the microwave signal is propagating in the backward direction at a frequency of 26.5 GHz and a power of 21 dBm (125 mW). **b**, The optical spectra (power in dB scale) at two different microwave driving frequencies of 26.5 GHz (left) and 10.4 GHz (right), respectively. The achieved isolation at the input laser frequency is 48 dB at 26.5 GHz driving, and > 44 dB at 10.4 GHz driving (instrumental resolution limited), respectively. We have observed sidebands with > 37 dB lower power than the pump line in the forward signal in both cases. On-chip insertion loss is 0.5 dB. The yellow curve (laser output) is sitting on top of the red curve (forward signal). **c**, The isolation ratio as a function of optical wavelength. The isolation ratio remains larger than 37 dB across a tuning range of 1510 – 1630 nm.

limited by the instrumental resolution). At both frequencies, the isolation is likely limited by the purity of the light polarization, as there is excellent agreement of the isolation

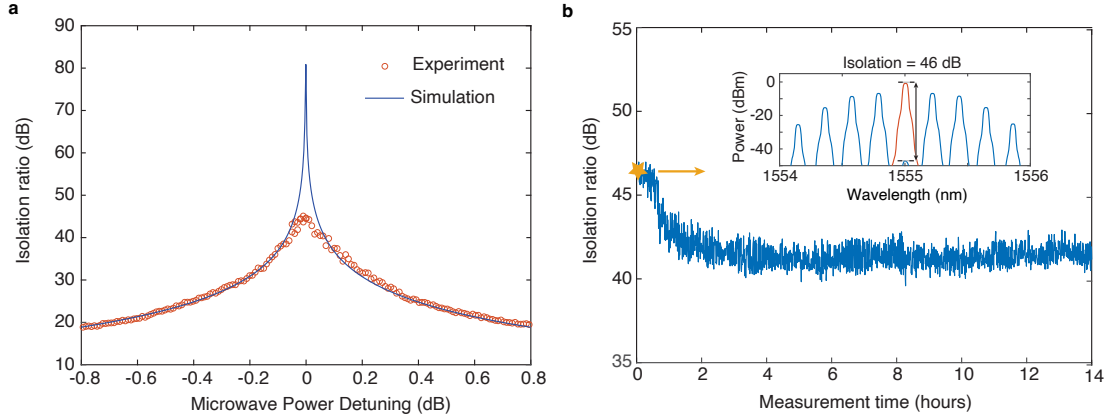


Figure 4.5: Stability measurement of the isolation ratio. **a**, The measured isolation ratio as a function of microwave power. The simulation is performed using the fundamental Bessel function $J_0(\beta)$ (see Appendix A), where the microwave power is proportional to β^2 . The highest simulated isolation ratio is limited by the finite sampling of simulation points. The experimental data shows excellent agreement with the simulation when the microwave power is detuned more than 0.05 dBm ($\sim 1.1\%$) from the optimal power. The maximum isolation ratio is likely limited by polarization purity of the input light. **b**, The measured isolation ratio over 14 hours at a time interval of 30 seconds. At the beginning of the measurement, the isolation of 46 dB is achieved at an optical wavelength of 1555 nm and similar microwave setting to that in Fig. 4.4a and b (inset: the optical spectrum). To assess the effects of environmental conditions, such as temperature, air fluctuation, and acoustic noise, the isolation ratio is monitored as a function of time while keeping all the operation conditions the same. The isolation gradually decreases to 41 dB after 2 hours and stabilizes to a mean value of 41.3 dB with a standard deviation of 0.56 dB over a 12-hour period. The main reason for drop in isolation is the polarization variation over time, which is confirmed by the fact that the isolation ratio can be adjusted back to 46 dB after manually adjusting the polarization controller in the setup.

ratio with simulation for microwave power detuning of >0.05 dB (Figure 4.5a), while the measured isolation ratio becomes increasingly sensitive to input polarization close to the optimal microwave power. There is likely some residual leakage of the transverse-magnetic (TM) mode into the isolator device. This is consistent with the fact that most commercial stress-induced birefringence-based polarization paddles state a polarization extinction ratio of 30 dB.

The ability to achieve large isolation using different RF frequencies offers flexibility in the system design: for example, it allows sidebands to be placed off-resonance with the laser cavity, or outside the laser instability range, thus ensuring stable laser opera-

tion. We note that the forward light exhibits first sidebands that are 37 dB below the pump, which is attributed to the microwave power attenuation along the electrode ($\alpha = 0.7 \text{ dB/cm/GHz}^{0.5}$) and the facet reflection. These sidebands cause minimal loss and intensity modulation to the laser signal and could be filtered out. We further tested the optical bandwidth of our device by tuning the laser wavelength and repeating the isolation ratio measurement from 1510 to 1630 nm. As shown in Fig. 4.4dc, the optical isolation remains $>37 \text{ dB}$ across the entire tuning range. We note that this bandwidth is tested using a single-frequency laser and should not apply for broadband light source input. In principle, the EO isolator could also be used with directly modulated laser or laser followed by an EO modulator as long as the modulation frequency is smaller than the microwave frequency f_{RF} used to drive the isolator, to avoid spectral overlap between the input and the backward sidebands¹²⁰. Finally, we tested the stability of the isolation ratio at 1555 nm and achieved an isolation ratio of $41.3 \pm 0.56 \text{ dB}$ over 12 hours, shown in Figure 4.3b. Notably, the few-dB decrease in the isolation ratio is attributed to polarization variation over time, as after the stability measurement, the maximum isolation ratio can be achieved again simply by adjusting the polarization control paddles in the setup. This problem is expected to be less pronounced for fully packaged or integrated systems.

In cases where the total power reflected to the laser needs to be minimized, an add-drop ring-resonator filter, transmitting at laser wavelength and featuring free spectral range (FSR) larger than f_{RF} , can be combined with the phase modulator, as shown in Fig. 4.6a (inset). Here, the laser light first enters the ring at Port 1 and exits through Port 2, going through the phase modulator without experiencing modulation. The reflected light goes back through the phase modulator, generating sidebands. The sidebands are off-resonance with the filter, do not couple to it, and are instead directed towards Port 3.

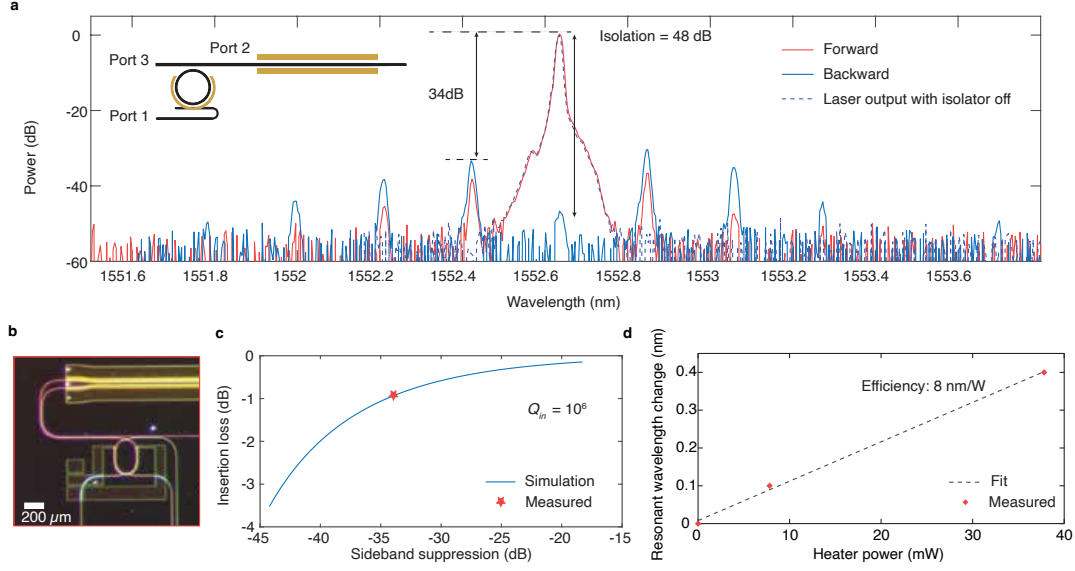


Figure 4.6: Integration of a tunable add-drop ring filter on one thin film lithium niobate chip. **a**, The optical spectra of the forward and backward signals. Laser output spectrum is taken when the microwave drive is turned off. The inset shows an add-drop-ring filter with an integrated heater, followed by a travelling-wave EO isolator. In the forward direction, the laser inputs at Port 1, travels to Port 2 and exits from the phase modulator. In the backward direction, the laser inputs from the modulator, travels to Port 2 and exits from Port 1. With the ring filter, the sideband power is 34 dB lower (0.04%) than the pump power while the isolation ratio remains 48 dB. **b**, The optical micrograph of the ring filter with an integrated NiCr heater. **c**, The insertion loss of the micro-ring filter as a function of sideband suppression at an intrinsic quality factor (Q_m) of 1×10^6 . The sideband suppression is defined as the power ratio between the first sideband and the pump. We measured an insertion loss of 0.9 dB along with the 34-dB sideband suppression at 26.5 GHz. **d**, Characterization of the integrated heater. The resonance wavelength is recorded as a function of heater power. We can tune the cavity resonance over one free spectral range of 1.0 nm with 125 mW heater power, based on a fitted tuning efficiency of 8 nm/W.

The reflected power can be easily suppressed as the sidebands are multiple f_{RF} away from the laser frequency. The FSR and the intrinsic Q factor of the microring resonator are 120 GHz and 1×10^6 , respectively. Figure 4.6a shows the measured optical spectra, where the highest sideband power is suppressed and is 34 dB below the laser power and the 48-dB isolation ratio remains unchanged. The suppression can be further increased using a higher Q factor microresonator and a higher f_{RF} . The total backward-propagating power is 30 dB lower than the forward-propagating power. Furthermore, an integrated NiCr

heater is fabricated with a measured tuning efficiency of 8 nm/W (about 125-mW heat power to tune one FSR of 120 GHz), as shown in Fig. 4.6b and d. Thus, the device can be used to isolate a single-frequency laser by tuning the heater within the broad spectral range shown in Fig. 4.4d. Figure 4.6d shows the relationship between the insertion loss induced by the filter and the sideband suppression ratio based on the intrinsic Q factor. We measured an insertion loss of 0.9 dB and suppression ratio of 34 dB, which is in good agreement with the simulation.

Finally, we verify the performance of our optical isolator using a complete III-V-LN laser module, as shown in Fig. 4.1a. We construct the system by edge-coupling a prefabricated InP DFB laser with the LN isolator chip (Fig. 4.7a). The coupling between the III-V and LN components is carefully engineered via mode-matching, which leads to a measured 2-dB loss at the interface (Fig. 4.2). To assess the effectiveness of the isolator in protecting the DFB laser, we send the output of the circuit through a 90:10 beam-splitter. The 90% port gets sent to a retroreflector which can be turned on/off to act as unwanted facet reflections at high laser power operation (see Figure 4.3). We estimate the reflected power is 0.4% of the laser power based on the measured insertion loss, taking into the account the losses at each facet and at the beam splitter. The 10% port is sent to a self-heterodyne beatnote measurement system to monitor the DFB linewidth at different operating conditions (Fig. 4.7b). Fig. 4.7c shows the linewidth measurement at different laser operating conditions. We see that when reflection is introduced with the isolator turned off, multimode oscillation is observed with additional peaks in the spectrum spaced ~ 6.5 MHz from the center peak. This frequency spacing corresponds to the FSR of the cavity formed by the retroreflector and DFB laser, which includes the LN chip and optical fiber. When the LN isolator is turned on, the linewidth of the laser is restored, showing that our optical isolator successfully protects the single mode oper-

ation of laser from reflection. Additionally, we measured the optical spectra when the reflection is turned on and off (isolator is kept on) and observed similar spectral features. This indicates that the attenuated reflected sidebands do not affect the laser emission efficiency at the fundamental mode. We posit that there is additional noise from facet reflections on the LN chip, adding further noise to the measurement, which we believe explains the asymmetric spacing of the linewidth peaks. To our knowledge, this is the first demonstration of an on-chip laser and isolator module for phase stability protection, which clearly indicates the practicality of our isolator approach.

4.3 CONCLUSION

In conclusion, we demonstrate an integrated electro-optic modulator-based optical isolator on thin film lithium niobate, and verify its ability to protect the phase stability of an edge-coupled on-chip laser. Our device features 48 dB isolation ratio at the laser frequency, 30-dB power isolation, 0.5 dB on-chip insertion loss (1.5 dB with the ring filter) and a tunable optical operating wavelength across C and L bands, using 21-dBm microwave power. To the best of our knowledge, this is the largest optical isolation and the lowest on-chip insertion loss ever demonstrated for an integrated optical isolator. The non-resonant EO device is compatible with a wide range of laser wavelengths. The isolator only requires a counter-propagating single-tone microwave signal and is independent of input laser power or phase. We overcome the limitations of high insertion loss and low isolation contrast which typically exist in a non-resonant system due to lack of Purcell enhancement factor in the light-matter interaction^{113,114}. Our approach significantly benefits from the combination of low insertion loss and high EO efficiency of the integrated LN platform and offers the highly competitive performance across all

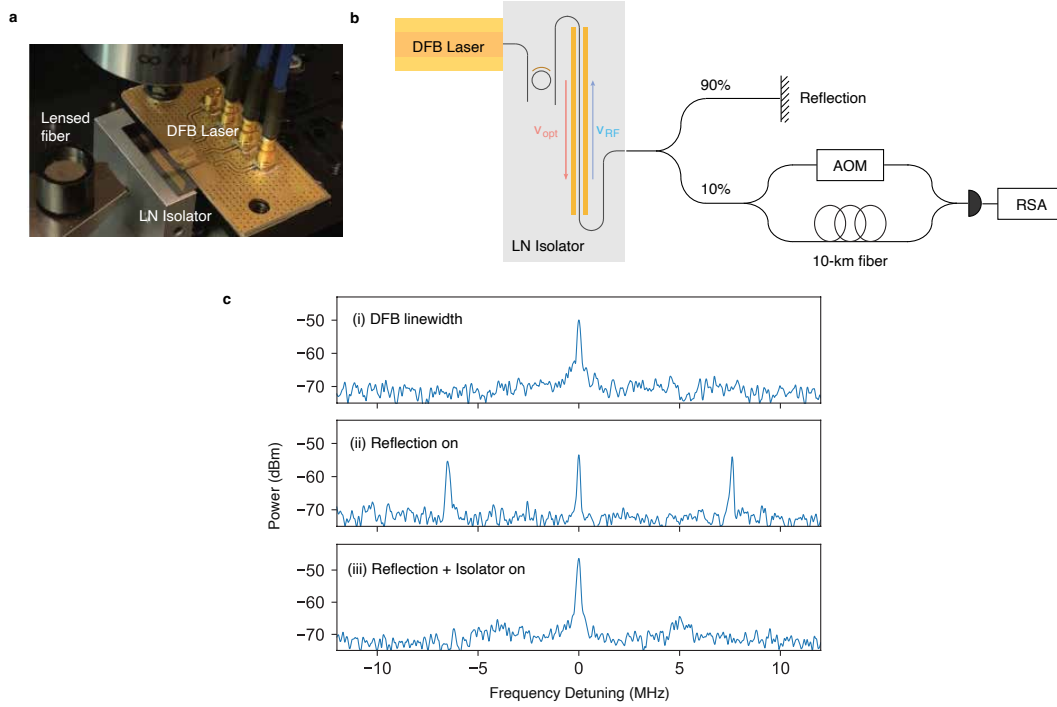


Figure 4.7: Characterization of the DFB laser and isolator module. **a**, Photograph of measurement system. An InP DFB laser is butt-coupled to the LN isolator chip. The laser system output is collected with a lensed fiber. **b**, Measurement setup. The output of the laser system is split between a reflection port and a self-heterodyne linewidth measurement system to monitor the DFB linewidth. AOM: acousto-optic modulator, RSA: RF spectrum analyzer. **c**, Linewidth measurements are given for (i) with isolator turned on and reflection turned off, (ii) with isolator turned off and reflection turned on, (iii) with both isolator and reflection turned on. RSA resolution bandwidth: 20 kHz.

metrics compared to other on-chip approaches (see Table 4.1). We envision further reduction of both the on-chip modulator and filter insertion loss to less than 0.1 dB based on the state-of-art linear loss of 3 dB/m on thin film LN¹⁰³ and of laser-LN chip coupling loss to <2 dB through implementation of an anti-reflection coating on the LN side, direct heterogeneous or homogeneous integration of the laser and LN device to reduce the coupling gap, or photonic wire bonding. Further improvement of power isolation to 60 dB can be achieved by cascading another tunable ring-based filter. Broadband

optical isolation can be achieved based on interference effect by splitting into multiple modulation paths¹¹⁰. Microwave power consumption can be reduced to 30 mW by reducing the half-wave voltage by half via implementation of a double-pass phase modulator design² and further down to 13 mW by utilizing high performance electrode design⁴¹ with 0.25 dB/cm/GHz^{0.5}. Finally, the entire laser module could be integrated on the same substrate via hybrid integration between of a semiconductor laser, the LN isolator and an on-chip stable microwave generator^{40,121}. Such a fully integrated, coherent on-chip laser system will be indispensable for applications from optical communication to time-frequency transfer.

Platform	Device	Isolation ratio (dB)	Insertion loss (dB)	Power (dBm)	Wavelength (nm)	Tested with on-chip laser	Ref.
LN (Fig. 4.4)	EO; waveguide	48	0.5	21	1553 Tunable	No	This work
LN (Fig. 4.6)	EO; waveguide + ring filter	48	1.4	21	1552 Tunable	No	This work
LN (Fig. 4.7)	EO; waveguide + ring filter	42	2.9	21	1556	Yes	This work
Ce:YIG/Si	MO; resonator; WB	32	2.3	10	1555	No	Ref. ¹⁰¹
Ce:YIG/Si	MO; resonator; WB	32	1.1	15	1550	No	Ref. ¹⁰²
Ce:YIG/SiN	MO; resonator; PLD	28	1	NA	1570	No	Ref. ¹⁰⁰
Ce:YIG/Si	MO; waveguide; PLD	30	5	NA	1585	No	Ref. ¹⁰³
AlN/Si	AO; waveguide	12	0.6	20.8	1523	No	Ref. ¹¹⁴
LN	AO; resonator	12.75	1.13	29	1550	No	Ref. ¹¹³
		39.31	0.65		1550		
AlN/SiN	AO; resonator	9.3	0.8	25	1550	No	Ref. ¹¹²
		41	1.8	33	1550		
LN	$\chi^{(2)}$; waveguide	18	6.6/1.4*	5 [†]	1575	No	Ref. ¹¹⁵
Si	Kerr; waveguide	4	NA	NA	1582	No	Ref. ¹¹⁹
SiN	Kerr; resonator	23	4.6	20 ^{†^}	1550	Yes	Ref. ¹¹⁷
		17	1.3				
Silica	Kerr; resonator	24	5	21 ^{†^}	1550	No	Ref. ¹¹⁶
InP	EO; waveguide	11	2.3	26	1580	No	Ref. ¹⁰⁹
Si	EO; waveguide	3	11.1	24	1556	No	Ref. ¹¹⁰
Si	EO; resonator	13	18	-3	1550	No	Ref. ¹⁰⁷
GaAs/AlGaAs	EO; waveguide	30	8	18	1550	No	Ref. ¹⁰⁸
Si	EO; waveguide	3	70	25	1555	No	Ref. ¹⁰⁵

Table 4.1: Performance comparison with other on-chip isolator approaches. EO: electro-optic; MO: magneto-optic; AO: acousto-optic; WB: wafer bonding; PLD: pulsed laser deposition. *: including 3.7 dB loss per facet; †: optical power; †^: power dependent; ^: only backward transmission tested.

5

Extended cavity diode lasers on thin-film lithium niobate

5.1 INTRODUCTION

In the past decade, the thin-film lithium niobate (TFLN) platform has positioned itself as a promising alternative to silicon on insulator, boasting many material properties that other platforms lack: its large electro-optic coefficient has made it an excellent host for efficient high-speed, wide-bandwidth modulators; its large second order nonlinearity and ability to be periodically poled allows for efficient nonlinear three-wave mixing processes; its wide transparency window allows for low-loss propagation over a wide range of wavelengths. Recently, a slew of TFLN components have been demonstrated, verifying it as an excellent platform for large-scale photonic systems⁷. However, a key element in the TFLN platform to drive all of the on-chip components is missing: a tunable, narrow-linewidth and high-power laser that removes the need for an off-chip, bulk-optic source.

Co-integration of lasers with TFLN will result in more compact, energy efficient and higher performance on-chip systems. As a result, there has been growing interest in the integration of on-chip lasers on TFLN. Crucially, the gain component needed for realizing these laser cavities is missing from the TFLN platform. While there exist efforts in erbium-doped lithium niobate solutions for supplying native gain to the platform⁸, most recent efforts involve the integration of III-V direct bandgap semiconductor materials with TFLN, following naturally from hybrid system lasers on other photonic platforms^{122,123}.

A laser is realized through the implementation of active gain medium in a frequency-selective optical cavity, which can be formed directly in the III-V waveguide with reflection gratings or with photonic circuits on another platform. Different approaches to realizing these laser cavities generally show a trade-off of various performance metrics,

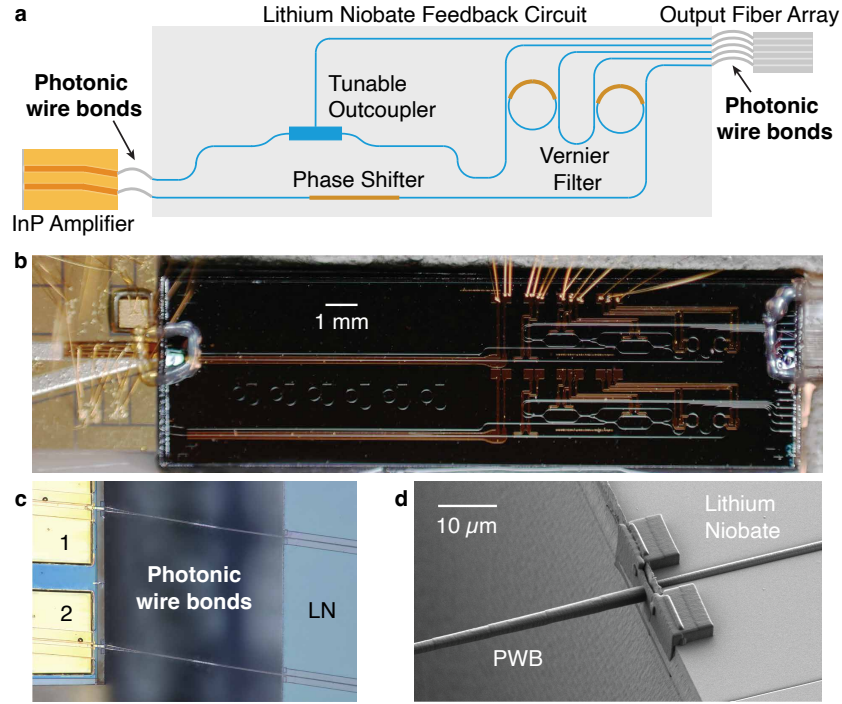


Figure 5.1: Photonic wire bonded TFLN-based ECL. **a**, Schematic showing the design of the laser, which comprises of a dual channel InP amplifier, an LN feedback circuit and a fiber array, with all components integrated by photonic wire bonding. **b**, Fully integrated laser with photonic and electrical wire bonds. **c**, Microscope image of the InP-LN interface, with photonic wire bonds and waveguides angled to mitigate Fresnel reflections. **d**, Scanning electron microscope image of a photonic wire bond to a TFLN waveguide.

including power and efficiency, linewidth, and tunability. III-V distributed feedback (DFB) lasers have a shorter cavity length and mature fabrication, usually translating to higher optical power output and slope efficiency due to lower overall loss in the laser cavity if the laser-waveguide coupling efficiency is high. However, because the cavity is defined fully on the III-V chip, the wavelength agility is usually fairly limited and tunable only within a few nm range using temperature control. Moreover, the effective cavity length is relatively short, limiting the laser linewidth¹²⁴. Self-injection locking can further reduce the linewidth for DFB lasers in low-loss photonic platforms¹²⁵. Extended cavity lasers (ECL) form one mirror of the laser cavity using a highly reflective coating

on one end of the gain waveguide and the other mirror using a photonic feedback circuit. These lasers benefit from the longer laser cavity formed by the photonic circuit and also active tuning of the photonic components, which can lead to lower linewidth and wavelength agility, respectively. However, because the on-chip propagation loss and coupling loss between the two platforms are directly translated into the laser cavity, there are a larger number of loss mechanisms which can increase the laser linewidth. Therefore, implementation of fully integrated ECLs can be challenging.

The growing need for high-performance on-chip lasers for the platform has motivated many TFLN-laser integration activities, including heterogeneous^{126,127} and hybrid-integration approaches^{128,40,129,130,131}. However, all current demonstrations lack the combination of high performance with scalability. Heterogeneous integration requires complex, wafer scale fabrication strategies and has yet to show high performance devices. Hybrid strategies have shown promising performance using stage-coupled components but have yet to be adopted at the wafer scale due to stringent alignment requirements.

An emerging technology that could address scaling issues without compromising performance is photonic wire bonding (PWB), which enables three-dimensional and low-loss optical connections between different photonic chips via polymer waveguides¹³². In contrast to traditional hybrid and heterogeneous integration strategies, photonic wire bonding boasts excellent optical power handling capabilities and allows for nearly unlimited scalability. Because the PWB formation process can take into account the relative position and geometry of different facets, it can overcome large misalignment and mode mismatch between a whole range of components, including but not limited to amplifiers, waveguides, and fibers. In contrast to traditional hybrid and heterogeneous integration strategies, photonic wire bonding boasts excellent optical power handling capabilities and allows for nearly unlimited scalability. The flexibility of this technology

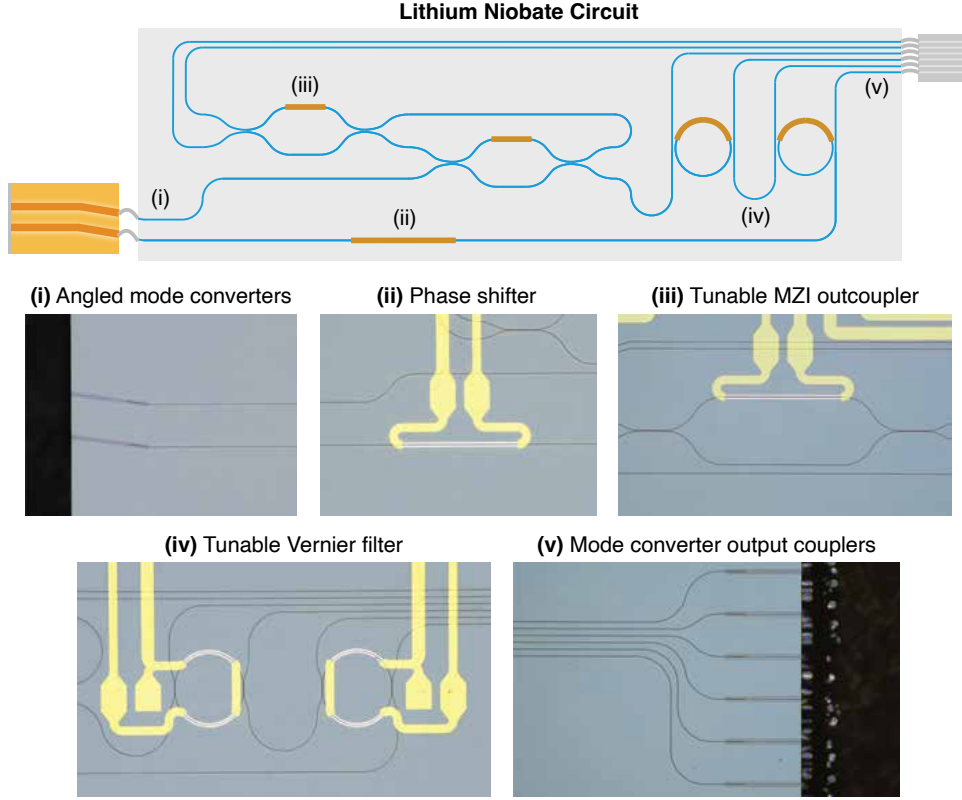


Figure 5.2: Full design schematic of the extended cavity diode laser. The lithium niobate circuit consists of a Vernier filter and two MZI couplers to maximize the output power. Ti-Pt heaters are included to tune the phase, on each ring to align the Vernier filter, and on the MZI couplers. All of the waveguides extend to the end of the LN chip and are photonic wire bonded to individual fibers in the grating array. The top waveguide is the laser output, and the bottom waveguide is used to characterize the coupling loss. (i)-(v) shows microscope images of each of these components taken before the packaging process.

could enable the integration of more complex systems and functionalities with much higher throughput, overcoming challenges of previous hybrid and heterogeneous integration strategies. Previous work has already heralded the upcoming strength of PWB combined with silicon photonics, showing its applicability for high confinement platforms^{37,133}. However, the TFLN platform, with triple the Si material bandgap, enables PWB to tailor to a much wider range of applications at a lower film loss.

Here, we use photonic wire bonding as an integration technique and realize an ex-

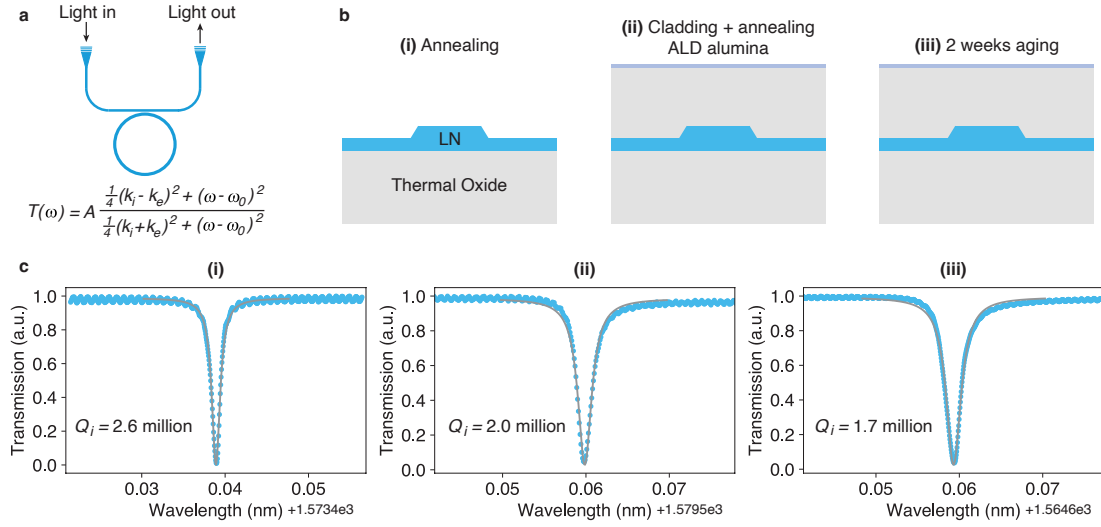


Figure 5.3: Loss characterization of the TFLN circuit at different points in the fabrication process. **a**, Measurement method for characterizing diagnostic ring resonator devices. Grating coupled rings are probed at each stage of fabrication and the resonances are fitted to the transmission function $T(\omega)$. A : background signal; k_i : intrinsic loss rate; k_e : extrinsic loss rate; ω_0 : resonance frequency. **b**, Stages of the fabrication process during which the intrinsic Q is fitted. (i) The chips first undergo an annealing process as in Ref.¹. (ii) The devices are cladded with PECVD silicon dioxide, annealed again, and a 20 nm layer of aluminum oxide is deposited via atomic layer deposition (ALD) to prevent oxide degradation. (iii) Finally, the devices are measured again after 2 weeks of aging in ambient environment. **c**, Sample transmission spectra taken at points (i)-(iii) with fitted Q_i . We see a drop in the intrinsic Q after cladding, which is likely due to a combination of weaker confinement and low-quality oxide. After 2 weeks, the intrinsic Q drops further to 1.7 million, indicating that the aluminum oxide layer has slowed down oxide degradation but not completely eliminated it.

tended cavity laser that integrates indium phosphide (InP) amplifiers with a lithium niobate feedback circuit (Fig. 5.1a and b, Fig. 5.2). Scaling up the power is realized by photonic wire bonding two InP amplifier waveguides to a single cavity. The LN waveguide circuit provides the cavity extension and frequency selection in the laser using a Vernier filter consisting of two tunable, micro-ring resonators¹²³. The rings, couplers, and cavity phase can be actively tuned using thermo-optic heaters.

5.2 RESULTS

The LN circuit is fabricated on 600 nm X-cut lithium niobate on insulator. To ensure low optical propagation loss in our laser cavity, the devices are annealed in O₂ ambient after the dry etching process and again after cladding with PECVD oxide. Additionally, to mitigate aging of the photonic circuit after integration with PWB and maintain low-loss operation, a thin layer of aluminum oxide is deposited using atomic layer deposition (ALD), intended to act as a dense barrier for the more porous oxide (Fig. 5.3b). Ti-Pt thermo-optic heaters are deposited with Ti-Au contacts, and the waveguide facets are defined using photolithography and etched with reactive ion and Bosch etching.

Figure 5.4 shows the characterization data of the InP amplifiers, labeled 1 and 2 as depicted in Fig. 5.1c. The power from amplified spontaneous emission (ASE) is measured using an integrating sphere. Both of the amplifiers are tested up to 800 mA of current. At 500 mA on the amplifier, the gain curve is measured using an optical spectrum analyzer (OSA), which shows the peak gain and gain bandwidth for each amplifier. Amplifier 1 has a peak gain at 1512 nm, a 3-dB gain bandwidth of 40 nm, and 10-dB gain bandwidth of 69 nm. Amplifier 2 has a peak gain at 1518 nm, a 3-dB gain bandwidth of 35 nm, and a 10-dB gain bandwidth of 75 nm. We note a minor deviation in the peak gain wavelength between the two amplifier waveguides.

The InP chip, LN chip, and output fiber array are mounted a common carrier, each vertically aligned within the PWB tolerance. To mitigate any Fresnel reflections at the InP-LN interface, the waveguides in both chips and the photonic wire bond here are angled (Fig. 5.1c). After the photonic wire bonds are written (Fig. 5.1d), for endured stability and reduced loss, each interface is cladded with a UV-cured epoxy. The single-pass PWB loss between amplifier and LN is 4 dB. The PWB loss from the output wave-

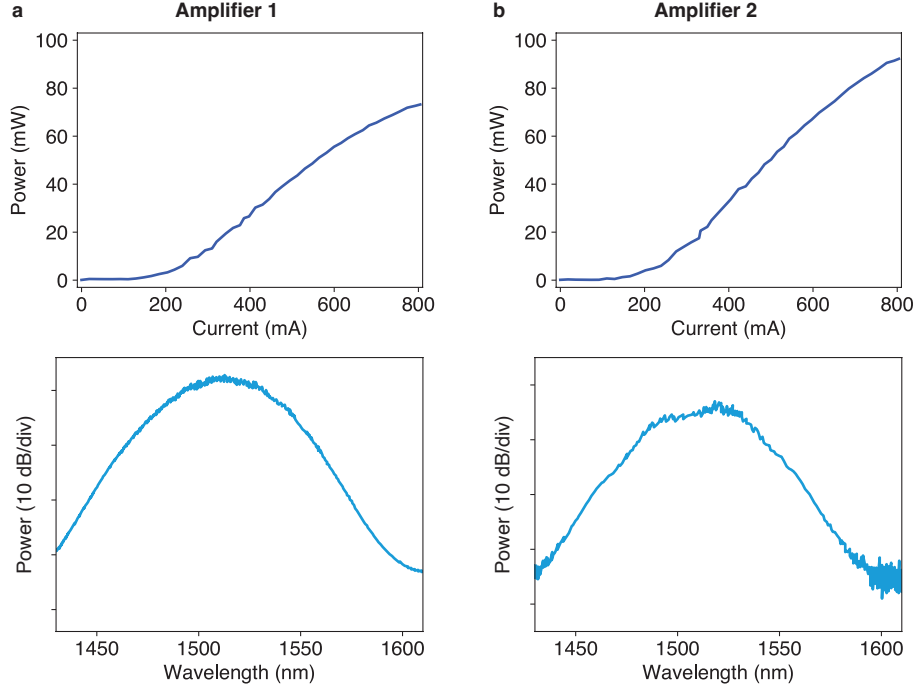


Figure 5.4: Characterization of the InP gain waveguides before photonic wire bonding. **a**, *LI* curve measured using an integrating sphere and gain curve measured on an optical spectrum analyzer for amplifier 1 as labeled in Fig. 5.1c. At the maximum current of 800 mA, the total ASE power of the amplifier reaches 73 mW. The peak gain for the amplifier is at 1512 nm. **b**, *LI* and gain curve for amplifier 2 as labeled in Fig. 5.1c. At the maximum current of 800 mA, the total ASE power of the amplifier reaches 92 mW. The peak gain for the amplifier is at 1518 nm.

uide to fiber is 3 dB. The LN-fiber loss is acquired from a passive device with identical couplers and photonic wire bonds on each side. The 4 dB loss at the InP-LN interface was determined by finding the total loss from the gain waveguide (measured before photonic wire bonding) to the output fiber (using the circuit drop port) and subtracting the known losses. The propagation loss on the LN circuit is low at 5.8 dB/m, obtained from the measured transmission spectra of single ring resonators (Fig. 5.3), fitted using the transmission equation in Ref. ¹³⁴. Gold wire bonds provide the electrical connection to the fully packaged laser, and a Peltier and on-chip thermistor are used for temperature

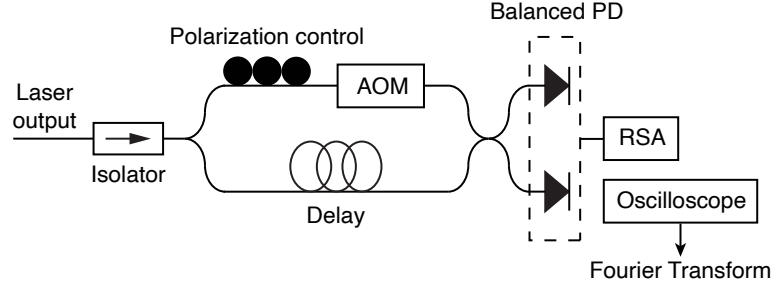


Figure 5.5: Delayed self heterodyne set-up for measuring the linewidth of our integrated laser. The output of the laser is sent through an optical isolator before being sent into measurement setup. The signal is split into two arms; one arm is modulated at 160 MHz and the other arm has a fixed delay length. The combined signal is measured using a balanced photodetector and optimized using a polarization controller. For the frequency noise measurement, the delay is set to 19.6 m and the time trace is measured using a digital oscilloscope and Fourier transformed. For the Voigt fit for the integrated linewidth, the delay is adjusted to 20 km and the beat is detected on an RSA. AOM: acousto-optic modulator; PD: photodetector; RSA: RF spectrum analyzer.

control (Fig. 5.1d). A full and detailed schematic of the laser design and LN circuit with microscope images is given in Figure 5.2.

With the laser in operation, we find a laser threshold at 100 mA and an approximately linear increase of output power with current (Fig. 5.6a). To explore power up-scaling, the amplifiers were driven with different currents, 200 mA and 800 mA respectively, giving a maximum 78 mW of on-chip power. We note a discontinuous jump in the LI curve at this point; we discuss in the chapter the reproducibility of these higher optical powers which currently limits the collection of a full LI curve. Optical spectra show that, at all currents reported here, the laser oscillates single longitudinal mode with a high side mode suppression ratio up to 61 dB (Fig. 5.6b). The demonstrated wavelength tuning range is 40.6 nm, centered around the gain maximum at 1520 nm, which is beyond the 3-dB gain bandwidth of the amplifiers.

The linewidth of our laser is characterized using a delayed self heterodyne set-up (Fig. 5.5), using existing methods¹³⁵, with a 19.6 m length delay between both arms.

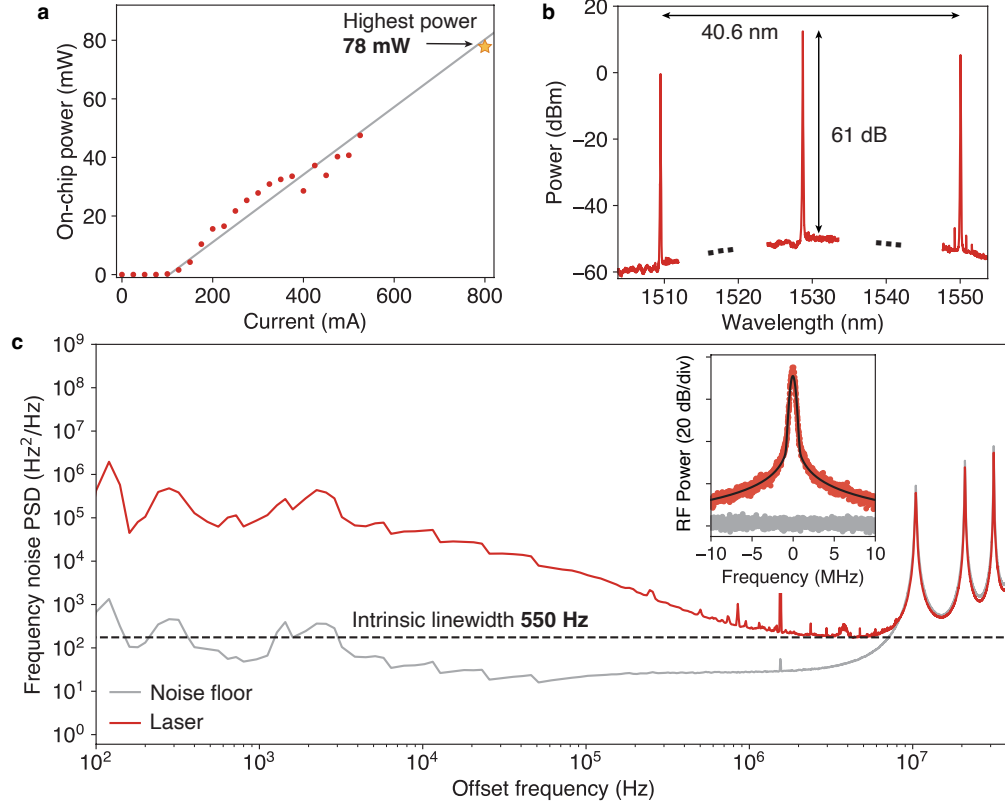


Figure 5.6: Laser performance. **a**, On-chip optical power as a function of current applied to each amplifier. Residual variations to the linear trend are caused by longitudinal mode hops. The highest power of 78 mW corresponds to 200 mA to amplifier 1 and 800 mA to amplifier 2. **b**, Optical spectrum showing single frequency operation with up to 61 dB side mode suppression at the peak gain wavelength. The laser is tunable over 40.6 nm (shaded region) beyond the 3-dB gain bandwidth. **c**, Frequency noise power spectral density plot acquired using delayed self-heterodyne measurement with a 19.6 m delay, showing an ultra-narrow 550 Hz intrinsic linewidth. Inset: beat note recorded in the frequency domain confirming the low intrinsic linewidth.

The light from both arms is combined on a balanced photodiode, and the beat recorded on an oscilloscope and consequently analyzed. The measured frequency noise power spectral density, computed from five 100-ms long oscilloscope traces of the heterodyne beat note, is shown in Fig. 5.6c. The frequency noise flattens at the quantum-limited noise, at a level that corresponds to an ultra-narrow intrinsic linewidth of 550 Hz. The

narrow linewidth is confirmed by measuring the beat note in the frequency domain using a long delay of 20 km and performing a fit to the wing of the lineshape.

5.3 CONCLUSION

In conclusion, we demonstrate a high-power and ultra-narrow-linewidth extended cavity laser in TFLN integrated with photonic wire bonding, addressing a previously missing component of the thin-film lithium niobate platform. To our knowledge, this laser shows the lowest intrinsic linewidth of any laser demonstrated using TFLN. We attribute this mainly to the low-loss and long extended cavity, which has an effective length of 24 cm. The highly frequency selective Vernier design allows the laser wavelength to be tuned over double the range of what has been reported before using stage coupling. Finally, the maximum on-chip output power exceeds existing TFLN extended cavity lasers by 1-2 orders of magnitude^{128,126}, in part due to the power scaling of the dual amplifier design.

We did notice that the laser performance had degraded with time, seen as a lower maximum power. We attribute this to gradual moisture diffusion into the top oxide cladding, which is deposited using PECVD, leading to increased absorption and optical loss. This claim is supported by diagnostic loss measurements taken during the fabrication process using grating coupled ring resonators (Fig. 5.3a). A standard fabrication flow will involve annealing after etching the LN ridge waveguides, then cladding with PECVD oxide and subsequent annealing¹. However, previous fabrication iterations showed that the low quality oxide degraded considerably over the course of a week, reducing the intrinsic quality factor by an order of magnitude. To combat the oxide degradation, an aluminum oxide sealing layer, was deposited on the top oxide by ALD

(Fig. 5.3b). We found from transmission measurements that this ALD layer slows the degradation down but does not eliminate it fully (Fig. 5.3c). We believe that higher optical powers could be achieved reproducibly with the implementation of higher density oxides and hermetically sealed packaging techniques.

Increasing the PWB link transmission and reducing the LN propagation loss to <3 dB/m^{34,1} would lower the intra-cavity loss and improve the performance further. In further experiments, the PWB showed no degradation with up to 1 W of optical power in the bond. Photonic wire bonding to tapered amplifiers could open the path towards optical powers in the Watt-level regime. In addition to completing the TFLN platform with a tunable, narrow-linewidth and high-power laser, our work verifies photonic wire bonding as a scalable solution for performance and integration. The power scaling through photonic wire bonding can be used to enable different nonlinear processes, while a combination with fast modulation can be useful for microwave photonic applications. For example, future work could involve integration of these lasers with other on-chip functionalities of lithium niobate, such as high performance electro-optic modulators and on-chip detectors¹³⁶ to realize a fully integrated link.

6

Conclusions and outlook

6.1 SUMMARY AND CONCLUSION

This thesis presented recent work on the thin-film lithium niobate on the generation of frequency comb sources and the integration of hybrid lasers on the platform. Both are important for future system-level demonstrations on the platform.

I first showed that by leveraging the high electro-optic efficiency and low propagation loss of the platform combined with novel waveguide structures enabled by integrated photonics, we can generate flat-top electro-optic frequency combs on TFLN, with pulses compressible to ultrashort durations using off-chip dispersion. Notably, in contrast to the realization of such systems using bulk modulators, the high modulation index and device performance is achieved without the need of cascading multiple phase modulators after each other, dramatically reducing the power consumption and complexity of the system. Additionally, further iterations of the system using single-drive electrode designs simplify the system operation further, requiring just one microwave source and amplifier and no splitters or phase shifters. In both cases, the use of off-chip fiber as a dispersive medium compresses a ~ 30 -GHz pulse train to sub-ps pulse duration.

Next, I demonstrate an application of the ultrashort pulse source by using as a source to synchronously pump a high- Q racetrack resonator fabricated on Z-cut TFLN. By using single-mode fiber between the two chips to compress the pulse, the source can be broadened with 12.5 mW of average on-chip power to a spectrum spanning over 1400 comb lines with 30-GHz repetition rate. Importantly, the use of pulsed pumping is demonstrated as a way of mitigating strong stimulated Raman scattering in these resonators and generating smooth spectra with low repetition rates in comparison to other Kerr combs.

From here, the thesis transitions towards considerations for integration of III-V gain and on-chip lasers onto the TFLN platform. Looking forward to future integrated on-chip lasers embedded into photonic systems, optical isolators are crucial for protecting the laser cavity against system reflections, which could cause the laser to lock to the reflection and multimode oscillation behavior. Using a travelling-wave phase modulator whose microwave signal travels against the direction of light propagation, I show an optical isolator with 48-dB isolation ratio and low insertion loss of 0.5 dB, outperforming existing on-chip isolator solutions. Additionally, the implementation of an add-drop ring filter brings the total sideband suppression of the isolator down to 34 dB with only an additional 0.9 dB loss in the entire isolator system. The effectiveness of the isolator is verified using a butt-coupled DFB laser.

Finally, I present our work on realizing extended cavity diode lasers on thin-film lithium niobate. Using photonic wire bonding to integrate the InP gain waveguides with our TFLN circuit, we take advantage of the flexibility of PWB and are able to utilize two gain sections in our laser. Our laser shows single-mode lasing behavior with excellent side mode suppression of 61 dB. Additionally, we confirm an ultra-low intrinsic linewidth of 550 Hz. We drove our lasers up to 525 mA current on both diodes for 50 mW of on-chip power. Additionally, we achieved 78 mW on-chip by setting the gain waveguides at 800 mA and 200 mA. The optical power, tunability, and linewidth of our laser outperforms previous ECL demonstrations on TFLN.

6.2 OUTLOOK

A natural outlook for the work presented in this thesis is towards the possibility of combining some or all of its components together. The laser and isolator exist on compatible

platforms and could be combined and embedded into any photonic system on TFLN. With these two components in combination with the electro-optic pulse generation system, the chip output could be pigtailed or photonic wire bonded to the correct length of fiber for dispersion and used as a (mostly) chip-scale femtosecond pulse source.

Additionally, all components of the thesis could be combined for the realization of a fully integrated and broadband comb system, similar to the outlook presented in Fig. 3.8 from Chapter 3. The success of such a system relies on the successful replacement of all off-chip dispersion and amplifiers with on-chip components. Amplification can be foregone under the assumption that the component-to-component loss would be reduced in an integrated demonstration combined with higher power pump lasers. The 78 mW on-chip power realized from our laser in Chapter 5 is sufficient if component-to-component coupling losses are reduced to 3-4 dB from their current 8.5 dB. (Our spectrum in Chapter 3 is generated with 11 dBm optical power. Adding the 3 dB of device insertion loss from the EO time lens system and 1.5 dB insertion loss from the isolator, this translates to 3.5 dB difference in power at the input of the time lens assuming ~ 80 mW or 19 dBm of optical power.) These losses are achievable considering current results in photonic wire bonding (3 dB/link) and could be improved further with optimized processes and couplers.

A larger missing component which would also benefit in the case of an integrated ultra-fast pulse source is low-loss on-chip dispersion management. On-chip dispersion has been implemented in conjunction with the amplitude and phase modulator system in Chapter 2 for pulse compression using both dispersion engineered waveguides and a chirped Bragg grating (Figure 6.1). While dispersion engineered waveguides such as the ones shown in Fig. 6.1a can significantly increase the dispersion parameter $|D|$ in comparison to standard waveguides or single-mode optical fiber (Fig. 6.1b), the required

length to realize full compression of the 30-GHz pulse train is still significant at nearly 50 cm in length (0.49 m). With the state-of-the-art losses on TFLN ridge waveguides³⁴, this length would equate to 1.5 dB of loss. However, the geometry of the waveguides to achieve reasonable dispersion is less confined and the optical mode sees more of the waveguide sidewall, leading to much higher losses of 1.5 dB/cm² or a whopping 75 dB of loss for full pulse compression. With the current losses, the length of the dispersive waveguide is limited and the pulse can only be compressed to 8 ps (Fig. 6.1c). Stronger phase modulation in the time lens system would impart a larger quadratic phase profile to the pulse, requiring less dispersion to compress the pulse to the Fourier transform limit and allowing the realization of even shorter pulses. Implementing low-loss segmented electrodes⁴¹ could reduce the microwave loss by over a factor of 2, reducing the required waveguide length to 25 cm. However, the major limitation of this approach is still in the propagation loss.

A more viable solution to on-chip dispersion management is the use of a chirped waveguide grating (Fig. 6.1d) where the grating period is linearly chirped over a length L , giving a group delay $2n_g L/c$ over the reflection bandwidth (Fig. 6.1e). Such chirped gratings have been implemented for compact dispersion management in Si and SiN systems. In comparison to the long dispersive waveguide, the use of a chirped grating is significantly more compact. The grating used to compress the 30-GHz pulses from Chapter 2 is 1.75 mm in length with extremely low propagation loss of 0.033 dB/mm.

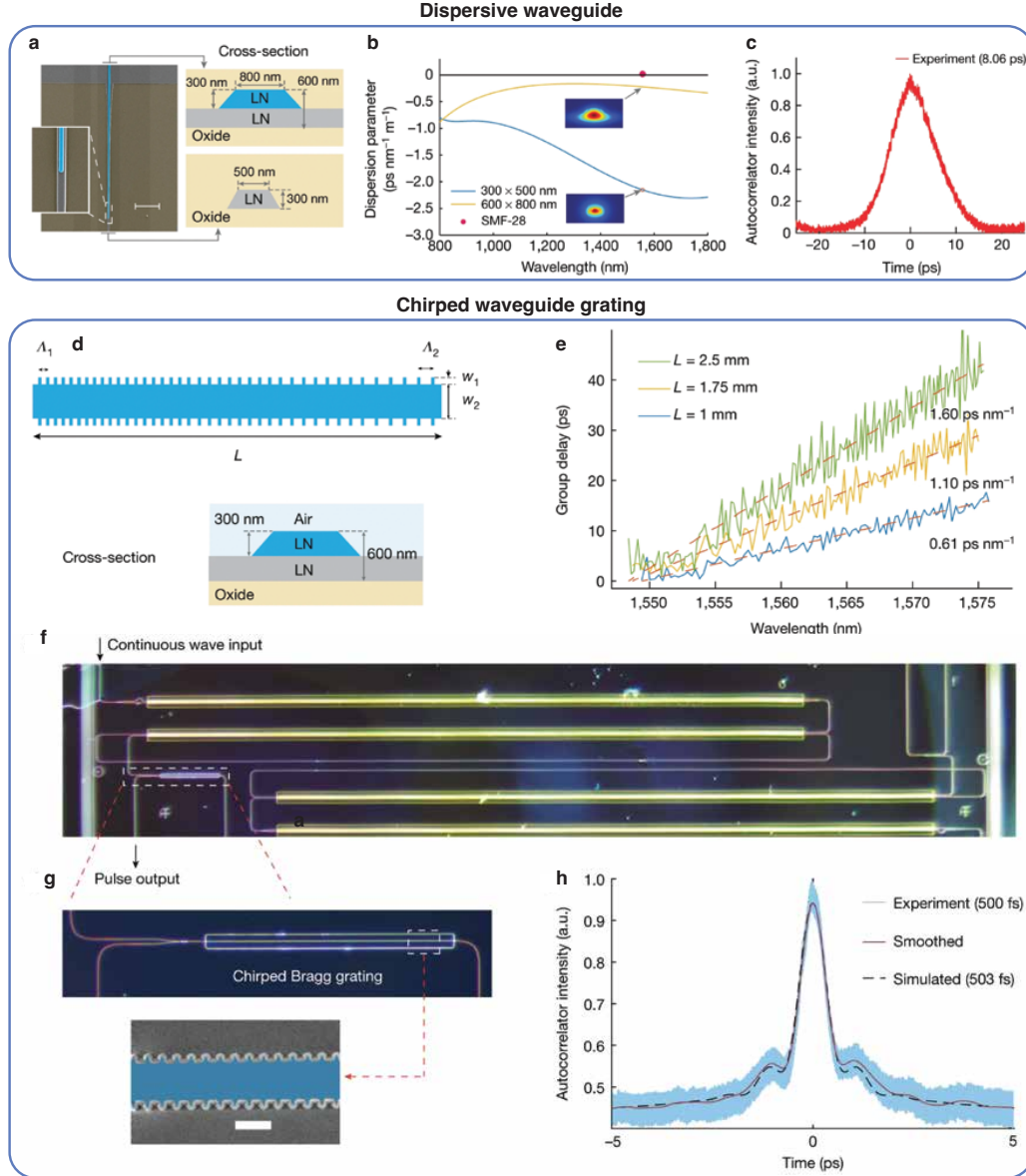


Figure 6.1: Solutions for on-chip dispersion compensation, adapted from Ref. ². Dispersive waveguide approach. **a**, SEM micrograph of the transition taper to the compression waveguide. The partially etched waveguide with a top width of 800 nm is adiabatically tapered to a top width of 300 nm with a $1\ \mu\text{m}$ slab width. The fully etched slab waveguide is then tapered to a final width of 500 nm. Scale bar, $20\ \mu\text{m}$. **b**, The dispersion parameter D of the integrated lithium niobate waveguides. The compression waveguide is dispersion-engineered to have a large dispersion value of $-2.15\ \text{ps nm}^{-1}\text{m}^{-1}$ at 1557 nm. Its absolute value is a factor of 10 and 120 larger than the $|D|$ of the partially etched waveguide and single-mode fiber. The total required length to reach full pulse compression at 30 GHz is 0.49 m. **c**, On-chip pulse compression using a 9.56-cm dispersive waveguide down to 8.06 ps. **Chirped waveguide grating approach.** **d**, The grating period is linearly chirped from $\Lambda_1 = 406.5\ \text{nm}$ to $\Lambda_2 = 414.5\ \text{nm}$ along the grating length L . The Bragg grating device is air-cladded with a fin width w_1 of 200 nm and a waveguide width w_2 of $1.1\ \mu\text{m}$. **e**, Group delay dispersion of three different grating lengths, extracted from the wavelength-dependent fringe periods in its corresponding reflection spectrum. Dashed lines represent fits. **f**, Optical image of the TFLN chip, which consists of the time lens system from Chapter 2 followed by a Y-splitter and a chirped Bragg grating waveguide. **g**, Magnified view of the chirped Bragg grating. A window is opened to remove the cladding (SEM graph, bottom). Scale bar, $1\ \mu\text{m}$. **h**, Autocorrelator trace showing the pulse is fully compressed to its Fourier transform limit of 500 fs using a 1.75-mm integrated lithium niobate Bragg grating.

The chirped grating structure was successfully implemented in the time lens system by opening a window in the cladding and wet etching with hydrofluoric acid for the uncladded grating structure (Fig. 6.1fg). Additionally, the grating successfully compresses the pulses generated by the modulator system to ~ 500 fs pulse duration (Fig. 6.1h). An issue with such gratings, however, is that they operate in the reflection mode, necessitating circulators or beamsplitters for implementation. On the fabricated chip, the use of a Y-splitter along with the grating imposes an inherent 6 dB of additional loss in the system. Currently, circulator-free grating approaches are being explored for on-chip dispersion on other platforms and could be implemented on TFLN in the future. Because on-chip solutions have a relatively fixed dispersion value, a single-electrode time lens scheme could be a good candidate for full integration, as the modulation index of the phase modulator is determined in the design based on its relative length to the amplitude modulator.

A few outstanding issues need to be addressed for viable integration. Most work presented in this thesis was performed on what we call the “standard stack” in our group, which is a 300 nm deep ridge waveguide on 600 nm X-cut LN (300 nm remaining slab) with 4.7 μm buried oxide layer. The isolator chip features a 320 nm etch depth (see Appendix B) but can be adapted easily to be compatible with the pulse generation and laser circuits. On the other hand, the racetrack resonator presented in Chapter 3 was fabricated on 600 nm Z-cut TFLN to avoid mode crossing from material birefringence. Therefore, single-platform implementation of the existing system is not possible without significant development on the resonator side for compatibility with the X-cut fabrication flow. Likely, mode engineering of the resonator bends or linear resonators is required. A more likely solution is photonic wire bonding, which can be low loss with proper design of the Z-cut facet coupler.

The long-term stability and packaging of these systems is also important. In recent years, to achieve high quality factors ($Q_i > 10^6$) and reduce photorefractive drift, furnace annealing has been utilized^{1,137}, posited to reduce surface damage and minimize carrier charge sites in the material. However, in ambient conditions, the effects of annealing degrade over time. To recover performance, the devices can be re-annealed, but the annealing process is not compatible with packaged devices or with metallization. In this thesis, the devices that required ultra-low propagation loss and high quality factor, the uncladded resonator from Chapter 3 and the laser cavity circuit from Chapter 5 went through the annealing process and experienced degradation in performance over the course of 2 weeks to 1 month. For the resonator device, the Q -factor could be nearly fully recovered through cleaning and re-annealing. However, after metallization and packaging, the laser device could not be reannealed and was only heated below the compliance temperature of the photonic wire bonds (80°C), which does not fully recovered performance. Future packaged integrated systems would also need to be robust to environment and could not undergo such destructive post-processing.

Improvements on the LN film quality and fabrication process could improve the pre-annealing loss for most non-quantum applications³⁴. We believe that a major source of our existing loss stems from low-quality and porous oxides. The use of higher quality oxides could also improve the waveguide loss and reduce the effects of aging. For the uncladded devices for nonlinear optics, different cladding-compatible geometries could be explored if the high Q -factor can be maintained and preserved through the cladding process. Additionally, the effectiveness of packaging on the long-term performance of different device configurations and geometries has yet to be explored.



Electro-optic modulation

A.1 BASIC FORMALISM OF THE ELECTRO-OPTIC EFFECT

The linear electro-optic effect for a material with Pockels tensor r_{ijk} is described by the relationship of the inverse permittivity tensor to the applied electric field. This is given

by¹³⁸

$$\Delta \left(\frac{1}{n^2} \right)_{ij} = \sum_k r_{ijk} E_k \quad (\text{A.1})$$

where indices 1, 2, 3 correspond to the x, y, z coordinates with respect to the material's crystal orientation. Due the symmetry rule $r_{ijk} = r_{jik}$, the equation can be reduced to a first order matrix equation using notation $r_{ijk} \equiv r_{bk}$, where

$$r_{bk} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \\ r_{41} & r_{42} & r_{43} \\ r_{51} & r_{52} & r_{53} \\ r_{61} & r_{62} & r_{63} \end{bmatrix} \quad \text{where } b = \begin{cases} 1 & \text{for } ij = 11 \\ 2 & \text{for } ij = 22 \\ 3 & \text{for } ij = 33 \\ 4 & \text{for } ij = 23, 32 \\ 5 & \text{for } ij = 13, 31 \\ 6 & \text{for } ij = 12, 21 \end{cases} \quad (\text{A.2})$$

The nonzero entries of the Pockels tensor are defined by each material's properties and crystal symmetry group. Lithium niobate belongs to the $3m$ crystal group, with a measured electro-optic tensor of⁷

$$r_{bk, LN} = \begin{bmatrix} 0 & -3.4 & 8.6 \\ 0 & 3.4 & 8.6 \\ 0 & 0 & 30.8 \\ 0 & 28 & 0 \\ 28 & 0 & 0 \\ 3.4 & 0 & 0 \end{bmatrix} \text{ pm/V} \quad (\text{A.3})$$

Notably, lithium niobate's r_{33} component (30.8 pm/V) is its largest, which corresponds to a change in the refractive index along the z crystal axis when an external electric field is applied along the z axis. Historically this has been the most utilized electro-optic tensor component in both bulk and thin-film LN modulators^{5,6}. Therefore, when selecting crystal cut of TFLN for EO devices, x -cut wafers are preferred, as it allows the patterning of planar electrodes for applying electric field along the extraordinary crystal axis.

A.2 PRINCIPLE OF ELECTRO-OPTIC PHASE MODULATION

Consider a simple electro-optic traveling wave phase modulator. The optical field of continuous-wave light $E(t) = E_0 e^{j\omega_o t}$ with optical frequency ω_o modulated at a single tone microwave frequency ω_m can be described by

$$E(t) = E_0 e^{j\omega_o t} e^{j\beta \sin(\omega_m t)} \quad (\text{A.4})$$

where $\beta = \pi V_p / V_\pi$ is the modulation depth, which gives the ratio of the peak drive voltage to the half wave voltage. By applying the Jacobi-Anger expansion, the expression for the optical field can be rewritten in terms of Bessel functions

$$E(t) = E_0 e^{j\omega_o t} \left[\sum_{n=0}^{\infty} J_n(\beta) e^{jn\omega_m t} + \sum_{n=1}^{\infty} (-1)^n J_n(\beta) e^{-jn\omega_m t} \right] \quad (\text{A.5})$$

where J_n is the n th Bessel function of the first kind. The Bessel functions J_n from $n = 0$ to $n = 4$ are plotted in Figure A.1. Taking the Fourier transform to the frequency

domain, we have

$$E(\omega) = E_0 \sum_0^{\infty} J_n(\beta) \delta(\omega - n\omega_m - \omega_o) + E_0 \sum_1^{\infty} J_n(\beta) \delta(\omega + n\omega_m - \omega_o) \quad (\text{A.6})$$

which describes the power of the spectral elements at $\omega_o + n\omega_m$ as a function of the input continuous-wave optical power, where m is an integer and $\omega_o + n\omega_m$ is the frequency of the m th order modulation sideband.

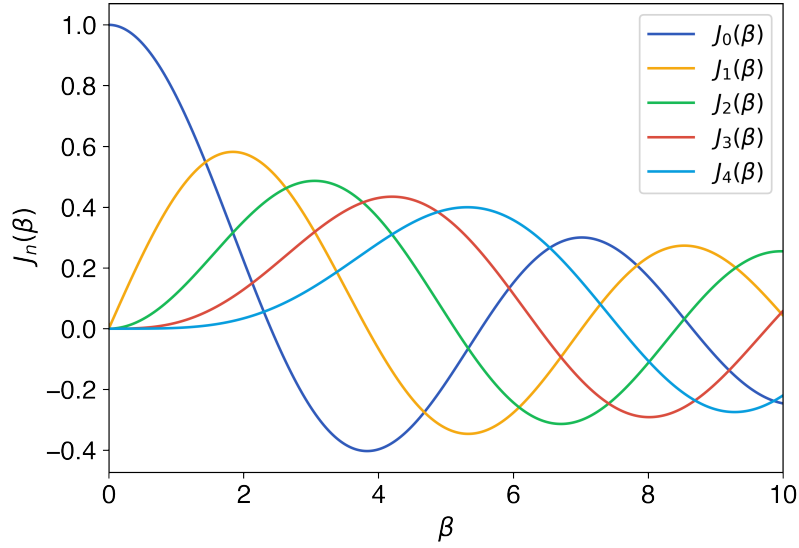


Figure A.1: Plot of the Bessel functions of the first kind J_n for indices $n = 0, 1, 2, 3, 4$.

A.3 V_π MEASUREMENT

For characterization measurement of phase modulators driven at frequencies higher than available oscilloscope bandwidth, the frequency domain expansion in equation A.6 can be used in conjunction with the optical spectrum to extract the half wave volt-

age V_π . The strength of the pump frequency can be described as a function of the peak drive voltage by

$$E(V_p) = E_0 J_0 \left(\pi \frac{V_p}{V_\pi} \right) \quad (\text{A.7})$$

We can therefore extract the half wave voltage at a given drive frequency by noting the power at the pump for a known modulation strength V_p , where

$$\frac{P}{P_0} = \left| J_0 \left(\pi \frac{V_p}{V_\pi} \right) \right|^2 \quad (\text{A.8})$$

This method is used to extract the modulation frequency-dependent half wave voltage of both of the cascaded phase modulator systems described in Chapter 2. In the measurement, the amplitude modulator is kept passive.

B

Fabrication methods

B.1 DEVICE FABRICATION FOR CHAPTER 2

Devices are fabricated on a thin-film X-cut lithium niobate on insulator wafer (NanoLN) with a film thickness of 600 nm and an oxide insulator thickness of 2 μm . Electron-beam lithography (EBL) is used to define the optical waveguide with hydrogen silsesquioxane (HSQ) resist, which is then partially etched by Ar⁺ based reactive ion etching with an

etch depth of 300 nm. The thickness of the remaining slab is 300 nm. The device is then cladded with silicon dioxide via plasma-enhanced chemical vapor deposition (PECVD). Photolithography and hydrofluoric acid wet etching is used to remove the oxide cladding in the regions of the spot size converter. A second EBL and reactive ion etching (RIE) process is used to define the final input taper and the compression waveguide on the slab layer, which is followed by PECVD oxide cladding. Microwave electrodes are patterned using both EBL and photolithography, and gold metal contacts are deposited using e-beam evaporation and lift-off processes. The facets of the chip are etched by deep RIE for end-fire optical coupling.

The width of partially etched optical waveguides is 1.5 μm , except for the width of the Y-splitter waveguides, which is tapered to 0.8 μm to avoid coupling to higher-order spatial modes. The cladding thickness is 820 nm. The dimension of the spot size converter at both facets is 300×250 nm for off-chip coupling. The coupling loss to the lensed fiber (mode size of 2 μm diameter) and distributed feedback laser chip is measured to be 3 dB and 2 dB, respectively. The propagation loss on chip is measured to be 3.2-3.6 dB, and ~ 0.3 dB/cm on the partially etched waveguides given a total length of 10 cm. The microwave strip has a metal thickness of 1.6 μm , an electrode gap of 5 μm and a signal and ground pad width of 35 μm and 290 μm , respectively.

B.2 DEVICE FABRICATION FOR CHAPTER 3

The X-cut pulse generator chips used in this chapter are taken from Chapter 2 and their fabrication methods are described above. The resonator device is fabricated on thin-film Z-cut lithium niobate on insulator wafer (also NanoLN) with 600 nm film thickness and 2 μm buried oxide thickness. The optical layer is defined using electron-beam

lithography (EBL) with hydrogen silsesquioxane resist, which is then partially etched by Ar⁺-based reactive ion etching. The Z-cut device is partially etched by 485 nm with a remaining slab of 115 nm. The chip undergoes 2-hours of high temperature furnace annealing in O₂ ambient. The facets of the Z-cut LN chip are cleaved to ensure a smooth facet to minimize coupling loss.

B.3 DEVICE FABRICATION FOR CHAPTER 4

The device fabrication methods in this work mirror that of Chapter 2 almost exactly. In this work, however, the etch depth of the devices is 320 nm, with 280 nm of slab remaining.

B.4 DEVICE FABRICATION FOR CHAPTER 5

Devices are fabricated on a thin-film X-cut lithium niobate on insulator wafer (NanoLN) with a film thickness of 600 nm and an oxide insulator thickness of 4.7 μ m. Electron-beam lithography (EBL) is used to define the optical waveguide with mAN resist, which is then partially etched by Ar⁺ based reactive ion etching with an etch depth of 300 nm. The thickness of the remaining slab is 300 nm. The device is then annealed at 520°C for 2-hours in O₂ ambient.

For patterning the second layer of the spot size converter, a second EBL layer is performed using ZEP resist and etched again using Ar⁺. The device is cladded with PECVD silicon dioxide and annealed a second time. Immediately after annealing, a 20 nm layer of aluminum oxide is deposited using atomic layer deposition (ALD).

Ti-Pt thermo-optic heaters are defined using photolithography and deposited with electron beam evaporation. The thickness of the Ti adhesion layer is 10 nm and the

thickness of the Pt layer is 25 nm. A second round of photolithography is used to deposit 300 nm thick gold contact pads with a 10 nm Ti adhesion layer. The facets of the chip are etched by deep RIE for end-fire optical coupling.

Bibliography

- [1] A. Shams-Ansari, G. Huang, L. He, Z. Li, J. Holzgrafe, M. Jankowski, M. Churraev, P. Kharel, R. Cheng, D. Zhu, *et al.*, *Apl Photonics* **7** (2022).
- [2] M. Yu, D. Barton III, R. Cheng, C. Reimer, P. Kharel, L. He, L. Shao, D. Zhu, Y. Hu, H. R. Grant, *et al.*, *Nature* **612**, 252 (2022).
- [3] G. T. Reed, G. Mashanovich, F. Y. Gardes, and D. Thomson, *Nature photonics* **4**, 518 (2010).
- [4] D. A. B. Miller, *Journal of Lightwave Technology* **35**, 346 (2017).
- [5] E. L. Wooten, K. M. Kissa, A. Yi-Yan, E. J. Murphy, D. A. Lafaw, P. F. Hallemeier, D. Maack, D. V. Attanasio, D. J. Fritz, G. J. McBrien, *et al.*, *IEEE Journal of selected topics in Quantum Electronics* **6**, 69 (2000).
- [6] C. Wang, M. Zhang, X. Chen, M. Bertrand, A. Shams-Ansari, S. Chandrasekhar, P. Winzer, and M. Lončar, *Nature* **562**, 101 (2018).
- [7] D. Zhu, L. Shao, M. Yu, R. Cheng, B. Desiatov, C. Xin, Y. Hu, J. Holzgrafe, S. Ghosh, A. Shams-Ansari, *et al.*, *Advances in Optics and Photonics* **13**, 242 (2021).
- [8] Z. Chen, Q. Xu, K. Zhang, W.-H. Wong, D.-L. Zhang, E. Y.-B. Pun, and C. Wang, *Optics Letters* **46**, 1161 (2021).
- [9] L. Lundberg, M. Mazur, A. Mirani, B. Foo, J. Schröder, V. Torres-Company, M. Karlsson, and P. A. Andrekson, *Nature communications* **11**, 201 (2020).
- [10] A. D. Ludlow, M. M. Boyd, J. Ye, E. Peik, and P. O. Schmidt, *Reviews of Modern Physics* **87**, 637 (2015).
- [11] J. Lee, Y.-J. Kim, K. Lee, S. Lee, and S.-W. Kim, *Nature Photonics* **4**, 716 (2010).
- [12] Y. Na, C.-G. Jeon, C. Ahn, M. Hyun, D. Kwon, J. Shin, and J. Kim, *Nature Photonics* **14**, 355 (2020).

- [13] K. J. Mohler, B. J. Bohn, M. Yan, G. Mélen, T. W. Hänsch, and N. Picqué, *Optics letters* **42**, 318 (2017).
- [14] M. P. Moreno and S. S. Vianna, *JOSA B* **28**, 2066 (2011).
- [15] J. Ye and S. T. Cundiff, *Femtosecond optical frequency comb: principle, operation and applications* (Springer Science & Business Media, 2005).
- [16] M. Jankowski, C. Langrock, B. Desiatov, A. Marandi, C. Wang, M. Zhang, C. R. Phillips, M. Lončar, and M. Fejer, *Optica* **7**, 40 (2020).
- [17] A. L. Gaeta, M. Lipson, and T. J. Kippenberg, *nature photonics* **13**, 158 (2019).
- [18] P.-H. Wang, J. A. Jaramillo-Villegas, Y. Xuan, X. Xue, C. Bao, D. E. Leaird, M. Qi, and A. M. Weiner, *Optics express* **24**, 10890 (2016).
- [19] M. Zhang, B. Buscaino, C. Wang, A. Shams-Ansari, C. Reimer, R. Zhu, J. M. Kahn, and M. Lončar, *Nature* **568**, 373 (2019).
- [20] M. L. Davenport, S. Liu, and J. E. Bowers, *Photonics Research* **6**, 468 (2018).
- [21] K. Van Gasse, S. Uvin, V. Moskalenko, S. Latkowski, G. Roelkens, E. Bente, *et al.*, *IEEE Photonics Technology Letters* **31**, 1870 (2019).
- [22] T. Kobayashi, T. Sueta, Y. Cho, and Y. Matsuo, *Applied Physics Letters* **21**, 341 (1972).
- [23] K. Beha, D. C. Cole, P. Del’Haye, A. Coillet, S. A. Diddams, and S. B. Papp, *Optica* **4**, 406 (2017).
- [24] J. Giordmaine, M. Duguay, and J. Hansen, *IEEE Journal of Quantum Electronics* **4**, 252 (1968).
- [25] V. Torres-Company and A. M. Weiner, *Laser & Photonics Reviews* **8**, 368 (2014).
- [26] A. J. Metcalf, V. Torres-Company, D. E. Leaird, and A. M. Weiner, *IEEE Journal of Selected Topics in Quantum Electronics* **19**, 231 (2013).
- [27] T. Yamamoto, T. Komukai, K. Suzuki, and A. Takada, *Journal of lightwave technology* **27**, 4297 (2009).
- [28] B. H. Kolner, in *Broadband Optical Modulators* (CRC Press, 2016) pp. 448–475.
- [29] R. Salem, M. A. Foster, and A. L. Gaeta, *Advances in Optics and Photonics* **5**, 274 (2013).

- [30] V. Torres-Company, J. Lancis, and P. Andrés, Optics letters **33**, 1822 (2008).
- [31] D. R. Carlson, D. D. Hickstein, W. Zhang, A. J. Metcalf, F. Quinlan, S. A. Diddams, and S. B. Papp, Science **361**, 1358 (2018).
- [32] L. He, M. Zhang, A. Shams-Ansari, R. Zhu, C. Wang, and L. Marko, Optics Letters **44**, 2314 (2019).
- [33] P. Ying, H. Tan, J. Zhang, M. He, M. Xu, X. Liu, R. Ge, Y. Zhu, C. Liu, and X. Cai, Optics letters **46**, 1478 (2021).
- [34] M. Zhang, C. Wang, R. Cheng, A. Shams-Ansari, and M. Lončar, Optica **4**, 1536 (2017).
- [35] J. M. Dudley, G. Genty, and S. Coen, Reviews of modern physics **78**, 1135 (2006).
- [36] N. Lindenmann, S. Dottermusch, M. L. Goedecke, T. Hoose, M. R. Billah, T. P. Onanuga, A. Hofmann, W. Freude, and C. Koos, Journal of Lightwave Technology **33**, 755 (2015).
- [37] Y. Xu, P. Maier, M. Blaicher, P.-I. Dietrich, P. Marin-Palomo, W. Hartmann, Y. Bao, H. Peng, M. R. Billah, S. Singer, *et al.*, Scientific Reports **11**, 16426 (2021).
- [38] M. R. Billah, M. Blaicher, T. Hoose, P.-I. Dietrich, P. Marin-Palomo, N. Lindenmann, A. Nesic, A. Hofmann, U. Troppenz, M. Moehrle, *et al.*, Optica **5**, 876 (2018).
- [39] C. Hu, A. Pan, T. Li, X. Wang, Y. Liu, S. Tao, C. Zeng, and J. Xia, Optics Express **29**, 5397 (2021).
- [40] A. Shams-Ansari, D. Renaud, R. Cheng, L. Shao, L. He, D. Zhu, M. Yu, H. R. Grant, L. Johansson, M. Zhang, *et al.*, Optica **9**, 408 (2022).
- [41] P. Kharel, C. Reimer, K. Luke, L. He, and M. Zhang, Optica **8**, 357 (2021).
- [42] M. Karpiński, M. Jachura, L. J. Wright, and B. J. Smith, Nature Photonics **11**, 53 (2017).
- [43] S. Mittal, V. V. Orre, A. Restelli, R. Salem, E. A. Goldschmidt, and M. Hafezi, Physical Review A **96**, 043807 (2017).
- [44] H. J. Kimble, Nature **453**, 1023 (2008).
- [45] T. J. Kippenberg, R. Holzwarth, and S. A. Diddams, science **332**, 555 (2011).

- [46] T. J. Kippenberg, A. L. Gaeta, M. Lipson, and M. L. Gorodetsky, *Science* **361**, eaan8083 (2018).
- [47] T. Ideguchi, S. Holzner, B. Bernhardt, G. Guelachvili, N. Picqué, and T. W. Hänsch, *Nature* **502**, 355 (2013).
- [48] S. B. Papp, K. Beha, P. Del’Haye, F. Quinlan, H. Lee, K. J. Vahala, and S. A. Diddams, *Optica* **1**, 10 (2014).
- [49] T. Fortier and E. Baumann, *Communications Physics* **2**, 153 (2019).
- [50] P. Del’Haye, A. Schliesser, O. Arcizet, T. Wilken, R. Holzwarth, and T. J. Kippenberg, *Nature* **450**, 1214 (2007).
- [51] Y. Okawachi, K. Saha, J. S. Levy, Y. H. Wen, M. Lipson, and A. L. Gaeta, *Optics letters* **36**, 3398 (2011).
- [52] L. Cai, J. Li, R. Wang, and Q. Li, *Photonics Research* **10**, 870 (2022).
- [53] M. Yu, Y. Okawachi, A. G. Griffith, N. Picqué, M. Lipson, and A. L. Gaeta, *Nature communications* **9**, 1869 (2018).
- [54] A. G. Griffith, R. K. Lau, J. Cardenas, Y. Okawachi, A. Mohanty, R. Fain, Y. H. D. Lee, M. Yu, C. T. Phare, C. B. Poitras, *et al.*, *Nature communications* **6**, 6299 (2015).
- [55] H. Jung and H. X. Tang, *Nanophotonics* **5**, 263 (2016).
- [56] B. Hausmann, I. Bulu, V. Venkataraman, P. Deotare, and M. Lončar, *Nature Photonics* **8**, 369 (2014).
- [57] Y. He, Q.-F. Yang, J. Ling, R. Luo, H. Liang, M. Li, B. Shen, H. Wang, K. Vahala, and Q. Lin, *Optica* **6**, 1138 (2019).
- [58] Z. Gong, X. Liu, Y. Xu, and H. X. Tang, *Optica* **7**, 1275 (2020).
- [59] M. Yu, Y. Okawachi, R. Cheng, C. Wang, M. Zhang, A. L. Gaeta, and M. Lončar, *Light: Science & Applications* **9**, 9 (2020).
- [60] Z. Gong, X. Liu, Y. Xu, M. Xu, J. B. Surya, J. Lu, A. Bruch, C. Zou, and H. X. Tang, *Optics letters* **44**, 3182 (2019).
- [61] Y. Okawachi, M. Yu, V. Venkataraman, P. M. Latawiec, A. G. Griffith, M. Lipson, M. Lončar, and A. L. Gaeta, *Optics Letters* **42**, 2786 (2017).
- [62] M. H. Anderson, R. Bouchand, J. Liu, W. Weng, E. Obrzud, T. Herr, and T. J. Kippenberg, *Optica* **8**, 771 (2021).

- [63] J. Li, C. Bao, Q.-X. Ji, H. Wang, L. Wu, S. Leifer, C. Beichman, and K. Vahala, *Optica* **9**, 231 (2022).
- [64] M. H. Anderson, W. Weng, G. Lihachev, A. Tikan, J. Liu, and T. J. Kippenberg, *Nature communications* **13**, 4764 (2022).
- [65] A. Berry and D. Hanna, *Optical and Quantum Electronics* **15**, 367 (1983).
- [66] T. Kobayashi, H. Yao, K. Amano, Y. Fukushima, A. Morimoto, and T. Sueta, *IEEE journal of quantum electronics* **24**, 382 (1988).
- [67] E. Obrzud, S. Lecomte, and T. Herr, *Nature Photonics* **11**, 600 (2017).
- [68] S. J. Herr, V. Brasch, J. Szabados, E. Obrzud, Y. Jia, S. Lecomte, K. Buse, I. Breunig, and T. Herr, *Optics letters* **43**, 5745 (2018).
- [69] J. Lu, A. Al Sayem, Z. Gong, J. B. Surya, C.-L. Zou, and H. X. Tang, *Optica* **8**, 539 (2021).
- [70] T. P. McKenna, H. S. Stokowski, V. Ansari, J. Mishra, M. Jankowski, C. J. Sarabalis, J. F. Herrmann, C. Langrock, M. M. Fejer, and A. H. Safavi-Naeini, *Nature Communications* **13**, 4532 (2022).
- [71] L. Ledezma, A. Roy, L. Costa, R. Sekine, R. Gray, Q. Guo, R. Nehra, R. M. Briggs, and A. Marandi, *Science Advances* **9**, eadf9711 (2023).
- [72] C. Wang, M. Zhang, M. Yu, R. Zhu, H. Hu, and M. Loncar, *Nature communications* **10**, 978 (2019).
- [73] Y. He, R. Lopez-Rios, Q. Yang, J. Ling, M. Li, K. Vahala, and Q. Lin, *CLEO, STu2G 1* (2021).
- [74] X. Xue, P.-H. Wang, Y. Xuan, M. Qi, and A. M. Weiner, *Laser & Photonics Reviews* **11**, 1600276 (2017).
- [75] Y. He, R. Lopez-Rios, U. A. Javid, J. Ling, M. Li, S. Xue, K. Vahala, and Q. Lin, *Nature Communications* **14**, 3467 (2023).
- [76] V. Caciuc, A. Postnikov, and G. Borstel, *Physical Review B* **61**, 8806 (2000).
- [77] S. Sanna, S. Neufeld, M. Rüsing, G. Berth, A. Zrenner, and W. Schmidt, *Physical Review B* **91**, 224302 (2015).
- [78] A. Ridah, P. Bourson, M. Fontana, and G. Malovichko, *Journal of Physics: Condensed Matter* **9**, 9687 (1997).

- [79] Y. Repelin, E. Husson, F. Bennani, and C. Proust, *Journal of Physics and Chemistry of Solids* **60**, 819 (1999).
- [80] I. Kaminow and W. Johnston Jr, *Physical Review* **160**, 519 (1967).
- [81] Ó. B. Helgason, M. Girardi, Z. Ye, F. Lei, J. Schröder, and V. Torres-Company, *Nature Photonics* **17**, 992 (2023).
- [82] Z. Li, Y. Xu, S. Shamilov, X. Wen, W. Wang, X. Wei, Z. Yang, S. Coen, S. G. Murdoch, and M. Erkintalo, *Nature Photonics* , 1 (2023).
- [83] Y. Xu, A. Sharples, J. Fatome, S. Coen, M. Erkintalo, and S. G. Murdoch, *Optics Letters* **46**, 512 (2021).
- [84] M. Zhang, C. Wang, P. Kharel, D. Zhu, and M. Lončar, *Optica* **8**, 652 (2021).
- [85] Y. Hu, M. Yu, N. Sinclair, D. Zhu, R. Cheng, C. Wang, and M. Lončar, *Nature Communications* **13**, 6293 (2022).
- [86] Z. Du, C. Xiang, T. Fu, M. Chen, S. Yang, J. E. Bowers, and H. Chen, *APL Photonics* **5** (2020).
- [87] C. Klitis, M. Sorel, and M. J. Strain, *Micromachines* **10**, 569 (2019).
- [88] J. W. Choi, E. Sahin, B.-U. Sohn, G. F. Chen, D. K. Ng, A. M. Agarwal, L. C. Kimerling, and D. T. Tan, *Light: Science & Applications* **10**, 130 (2021).
- [89] S. Liu, C. Zhang, H. Cao, S. Hong, Z. Yu, and D. Dai, *ACS Photonics* **10**, 1654 (2023).
- [90] H. Shu, L. Chang, Y. Tao, B. Shen, W. Xie, M. Jin, A. Netherton, Z. Tao, X. Zhang, R. Chen, *et al.*, *Nature* **605**, 457 (2022).
- [91] D. Marpaung, J. Yao, and J. Capmany, *Nature photonics* **13**, 80 (2019).
- [92] X. Xu, M. Tan, B. Corcoran, J. Wu, A. Boes, T. G. Nguyen, S. T. Chu, B. E. Little, D. G. Hicks, R. Morandotti, *et al.*, *Nature* **589**, 44 (2021).
- [93] A. Sludds, S. Bandyopadhyay, Z. Chen, Z. Zhong, J. Cochrane, L. Bernstein, D. Bunandar, P. B. Dixon, S. A. Hamilton, M. Streshinsky, *et al.*, *Science* **378**, 270 (2022).
- [94] Z. L. Newman, V. Maurice, T. Drake, J. R. Stone, T. C. Briles, D. T. Spencer, C. Fredrick, Q. Li, D. Westly, B. R. Ilic, *et al.*, *Optica* **6**, 680 (2019).
- [95] J. Wang, F. Sciarrino, A. Laing, and M. G. Thompson, *Nature Photonics* **14**, 273 (2020).

- [96] P. Trocha, M. Karpov, D. Ganin, M. H. Pfeiffer, A. Kordts, S. Wolf, J. Krockenberger, P. Marin-Palomo, C. Weimann, S. Randel, *et al.*, Science **359**, 887 (2018).
- [97] A. Shams-Ansari, M. Yu, Z. Chen, C. Reimer, M. Zhang, N. Picqué, and M. Lončar, Communications Physics **5**, 88 (2022).
- [98] M.-G. Suh, Q.-F. Yang, K. Y. Yang, X. Yi, and K. J. Vahala, Science **354**, 600 (2016).
- [99] W. Jin, Q.-F. Yang, L. Chang, B. Shen, H. Wang, M. A. Leal, L. Wu, M. Gao, A. Feshali, M. Paniccia, *et al.*, Nature Photonics **15**, 346 (2021).
- [100] W. Yan, Y. Yang, S. Liu, Y. Zhang, S. Xia, T. Kang, W. Yang, J. Qin, L. Deng, and L. Bi, Optica **7**, 1555 (2020).
- [101] D. Huang, P. Pintus, C. Zhang, Y. Shoji, T. Mizumoto, and J. E. Bowers, IEEE Journal of Selected Topics in Quantum Electronics **22**, 271 (2016).
- [102] P. Pintus, D. Huang, C. Zhang, Y. Shoji, T. Mizumoto, and J. E. Bowers, Journal of Lightwave Technology **35**, 1429 (2017).
- [103] Y. Zhang, Q. Du, C. Wang, T. Fakhru, S. Liu, L. Deng, D. Huang, P. Pintus, J. Bowers, C. A. Ross, *et al.*, Optica **6**, 473 (2019).
- [104] L. Bi, J. Hu, P. Jiang, D. H. Kim, G. F. Dionne, L. C. Kimerling, and C. Ross, Nature Photonics **5**, 758 (2011).
- [105] H. Lira, Z. Yu, S. Fan, and M. Lipson, Physical review letters **109**, 033901 (2012).
- [106] L. D. Tzuang, K. Fang, P. Nussenzveig, S. Fan, and M. Lipson, Nature photonics **8**, 701 (2014).
- [107] N. Dostart, H. Gevorgyan, D. Onural, and M. A. Popović, Optics Letters **46**, 460 (2021).
- [108] S. Bhandare, S. K. Ibrahim, D. Sandel, H. Zhang, F. Wust, and R. Noé, IEEE Journal of selected topics in quantum electronics **11**, 417 (2005).
- [109] C. R. Doerr, N. Dupuis, and L. Zhang, Optics letters **36**, 4293 (2011).
- [110] C. Doerr, L. Chen, and D. Vermeulen, Optics express **22**, 4493 (2014).
- [111] J. F. Herrmann, V. Ansari, J. Wang, J. D. Witmer, S. Fan, and A. H. Safavi-Naeini, Nature Photonics **16**, 603 (2022).
- [112] H. Tian, J. Liu, A. Siddharth, R. N. Wang, T. Blésin, J. He, T. J. Kippenberg, and S. A. Bhave, Nature Photonics **15**, 828 (2021).

- [113] D. B. Sohn, O. E. Örsel, and G. Bahl, *Nature Photonics* **15**, 822 (2021).
- [114] E. A. Kittlaus, W. M. Jones, P. T. Rakich, N. T. Otterstrom, R. E. Muller, and M. Rais-Zadeh, *Nature Photonics* **15**, 43 (2021).
- [115] K. Abdelsalam, T. Li, J. B. Khurgin, and S. Fathpour, *Optica* **7**, 209 (2020).
- [116] L. Del Bino, J. M. Silver, M. T. Woodley, S. L. Stebbings, X. Zhao, and P. Del’Haye, *Optica* **5**, 279 (2018).
- [117] A. D. White, G. H. Ahn, K. V. Gasse, K. Y. Yang, L. Chang, J. E. Bowers, and J. Vučković, *Nature Photonics* **17**, 143 (2023).
- [118] J. Kitching, *Applied Physics Reviews* **5** (2018).
- [119] K. Saha, Y. Okawachi, O. Kuzucu, M. Menard, M. Lipson, and A. L. Gaeta, in *CLEO: QELS_Fundamental Science* (Optica Publishing Group, 2013) pp. QF1D-2.
- [120] I. A. Williamson, M. Minkov, A. Dutt, J. Wang, A. Y. Song, and S. Fan, *Proceedings of the IEEE* **108**, 1759 (2020).
- [121] C. O. de Beeck, F. M. Mayor, S. Cuyvers, S. Poelman, J. F. Herrmann, O. Atalar, T. P. McKenna, B. Haq, W. Jiang, J. D. Witmer, *et al.*, *Optica* **8**, 1288 (2021).
- [122] D. Liang and J. E. Bowers, *Nature photonics* **4**, 511 (2010).
- [123] K.-J. Boller, A. van Rees, Y. Fan, J. Mak, R. E. Lammerink, C. A. Franken, P. J. van der Slot, D. A. Marpaung, C. Fallnich, J. P. Epping, *et al.*, *Photonics* **7**, 4 (2020).
- [124] A. L. Schawlow and C. H. Townes, *Physical review* **112**, 1940 (1958).
- [125] R. Lang, *IEEE Journal of Quantum Electronics* **18**, 976 (1982).
- [126] C. Op de Beeck, F. M. Mayor, S. Cuyvers, S. Poelman, J. F. Herrmann, O. Atalar, T. P. McKenna, B. Haq, W. Jiang, J. D. Witmer, *et al.*, *Optica* **8**, 1288 (2021).
- [127] T. J. Morin, J. Peters, M. Li, J. Guo, Y. Wan, C. Xiang, and J. E. Bowers, *Optics Letters* **49**, 1197 (2024).
- [128] M. Li, L. Chang, L. Wu, J. Staffa, J. Ling, U. A. Javid, S. Xue, Y. He, R. Lopez-Rios, T. J. Morin, *et al.*, *Nature communications* **13**, 5344 (2022).
- [129] V. Snigirev, A. Riedhauser, G. Lihachev, M. Churaev, J. Riemensberger, R. N. Wang, A. Siddharth, G. Huang, C. Möhl, Y. Popoff, *et al.*, *Nature* **615**, 411 (2023).

- [130] M. Han, J. Li, H. Yu, D. Li, R. Li, and J. Liu, Opt. Express **32**, 5632 (2024).
- [131] J. Ling, J. Staffa, H. Wang, B. Shen, L. Chang, U. A. Javid, L. Wu, Z. Yuan, R. Lopez-Rios, M. Li, *et al.*, Laser & Photonics Reviews **17**, 2200663 (2023).
- [132] N. Lindenmann *et al.*, Opt. Express **20**, 17667 (2012).
- [133] S. J. Chowdhury, K. Wickremasinghe, S. M. Grist, H. Zou, M. Mitchell, M. A. Al-Qadasi, B. Lin, D. Birdi, S. Smythe, S. Shekhar, *et al.*, Optics Express **32**, 3085 (2024).
- [134] M. Aspelmeyer, T.J. Kippenberg, and F. Marquardt, Reviews of Modern Physics **86**, 1391 (2014).
- [135] T. Okoshi, K. Kikuchi, and A. Nakayama, Electronics letters **16**, 630 (1980).
- [136] X. Guo, L. Shao, L. He, K. Luke, J. Morgan, K. Sun, J. Gao, T.-C. Tzu, Y. Shen, D. Chen, *et al.*, Photonics Research **10**, 1338 (2022).
- [137] Y. Xu, M. Shen, J. Lu, J. B. Surya, A. Al Sayem, and H. X. Tang, Optics Express **29**, 5497 (2021).
- [138] R. W. Boyd, *Nonlinear Optics*, 3rd ed. (Academic Press, Boston, 2008)